Università di Pisa
Department of Physics
Master's Degree in Physics

Anomalous Transport in Soaring Flights

Collective Strategies and Intermittent Search Models

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Abstract

Soaring birds and human pilots stay aloft by extracting energy from atmospheric updrafts, producing long, intermittent trajectories that alternate climbs within thermals with quasi-ballistic glides. Such motion is a candidate for *anomalous transport*, that is, a departure from ordinary Brownian diffusion. This thesis characterizes the transport statistics of real soaring flights from a large empirical dataset and seeks a minimal stochastic transport model able to reproduce those statistics. As the work develops it will also examine how flying in competition changes the motion relative to solo flights.

This document reports the current state of the project. A first dataset has been acquired reproducibly—the archive of the FFVL *Coupe Fédérale de Distance*, in its paraglider and hang-glider disciplines—and Chapter 2 describes its acquisition, its coverage and the pre-processing that turns the raw logs into analysis-ready trajectories.

Chapter 3 measures the transport of the retained ensemble without segmenting a flight. Displacement from take-off turns out to measure the geometry of the launch site rather than the motion, and is retired as a source of exponents in favour of a within-flight estimator that annihilates any polynomial trend. Over 60 s to 2,000 s that estimator gives $\alpha = 2.02 \pm 0.10$ for paragliders and 1.94 ± 0.13 for hang gliders — strongly super-diffusive, and moved by no stratification the archive allows by more than its own uncertainty. Beyond the second moment, the spectrum of moments is straight, so the motion is monofractal and is *not* a Lévy walk; the propagator is nonetheless not Gaussian; and no leg duration is distinguished, which is a constraint on any segmentation attempted next. Chapter 4 sets out that segmentation and the modelling it enables, and is a plan rather than a result.

Contents

Chapter 1

Introduction

1.1 Motivation

Gliders, paragliders, and soaring birds carry no continuous source of propulsion; to remain airborne they extract energy from air currents in the atmosphere, primarily from *thermals*, the buoyant updrafts that form as sunlight heats the ground. A cross-country flight thus proceeds as a repeating cycle of three phases. In the *transition* phase the flyer glides along a nearly straight path toward the next source of lift. If none is reached at a safe altitude, a *search* phase follows, during which the flyer probes the surrounding air mass to locate lift. Once lift is found, a *climb* phase begins, in which the flyer circles within the rising air to regain altitude. The cycle alternates long, directed displacements with episodes of slow, localized motion.

Intermittent and directionally persistent motion of this kind is one of the settings in which *anomalous transport* arises. Let $\mathbf{r}(t)$ denote the horizontal position of a flight, measured from take-off. Transport is anomalous when the mean-squared displacement grows as a non-trivial power of time, $\langle|\mathbf{r}(t)|^2\rangle \sim t^\alpha$ with $\alpha \neq 1$, as opposed to the linear growth ($\alpha = 1$) of ordinary Brownian diffusion. The continuous-time random walk (CTRW) is one of the standard modelling frameworks for such regimes [1]: the walker waits a random time τ, then makes a random displacement $\mathbf{x}$, and the joint density $\psi(\mathbf{x}, \tau)$ of the two fixes the transport. When the two are independent, $\psi(\mathbf{x}, \tau) = \lambda(\mathbf{x})\, w(\tau)$, a heavy-tailed w makes the walk subdiffusive, while a heavy-tailed λ makes it a Lévy flight, in which an arbitrarily long displacement is made instantaneously and the second moment diverges. The Lévy walk is not a separate framework but the opposite, fully coupled case, which relates displacement to duration through a constant speed v,

$$\psi(\mathbf{x}, \tau) \propto \delta(|\mathbf{x}| - v\tau)\, w(\tau), \tag{1.1}$$

with the direction drawn uniformly, so that each waiting time is spent travelling ballistically rather than standing still [2]. Anomalous transport of this kind has been reported in random-search strategies [3], in disordered media [4], and in turbulent flows [5].

Both are renewal processes: each step draws a fresh pair $(\mathbf{x}, \tau)$ independently of every step before it, so the walk retains no memory beyond its current position. Correlated CTRWs relax that assumption. Correlating successive jumps alone already moves the exponent—it can turn the walk superdiffusive—and separates the ensemble average from the time average [6]; imposing power-law correlated noise on the jump lengths and the waiting times together gives a process carrying scale-free displacements, scale-free waiting times, and the long-range memory of fractional Brownian motion at once [7].

The soaring cycle has three parts rather than two: a directed glide, then a search phase, then a localized climb. The glide is a leg flown at nearly constant speed—the coupling of Eq. (1.1), and the only feature of it the cycle inherits, since the uniform direction drawn alongside that coupling is itself one of the things to be tested. The search and the climb that end each leg are episodes of slow, localized motion, closer to the waits of an uncoupled CTRW; whether they can be collapsed into a single waiting phase is a modelling decision the data have to support (Sec. 4.4).

Whether the resulting sequence of legs is renewal is left open until it can be measured. It can fail in four ways, each its own measurement: successive glide *directions*, successive glide *lengths* and successive *waiting times* may each be correlated, and a waiting time may be correlated with the *length of the glide beside it*. Two of the four have a mechanism behind them. A glide heading is set by the route flown, the pilot's strategy, and the state of the air mass, wind and terrain, none of which changes on the timescale of one glide. The wait–length coupling has no sign fixable in advance: what limits a glide is the altitude available at its start, and a waiting time is search and climb together, of which only the climb converts waiting into height while the search consumes it—so what a pooled waiting time shows depends on how the two mix, one more reason the collapse has to be argued from the data. These are reasons to look, not results: none of the four is assumed here, and since each relates consecutive legs, and a leg exists only once a flight has been cut into phases, none can be tested before the segmentation of Chapter 4.

The model this thesis works towards is the minimal one consistent with whichever of the four survive; its working hypothesis is a persistent random walk in continuous time—legs at finite speed, waiting times between them, correlated leg directions—and that persistence stands or falls on the measurement. Chapter 3 comes first because it needs no segmentation. It reads four things off the unsegmented trajectory: the exponent at which displacement grows with the time lag; whether moments of every order share that one exponent, as a self-similar process requires; how long a heading survives, read from the decay of the velocity autocorrelation; and whether any duration over which a flight holds its heading is distinguished, or whether those durations carry the power-law tail a Lévy walk needs. Together they fix the class of transport. What that chapter cannot do is reach the four correlations, and the reason is what each of them is indexed by. The four compare one leg with the next, so their index is the leg number, and it exists only once the flight has been cut. Every statistic in Chapter 3 is indexed by a time lag instead: it compares the trajectory at two instants a fixed interval apart, without knowing - if any - either instant falls in. A glide is approximately straight, so a heading held across a lag shorter than one glide is persistence *within* a leg, and a velocity autocorrelation that stays positive at those lags reports that and not whether the direction of one glide predicts the direction of the next; at longer lags the two are mixed rather than separated. The chapter can therefore show that the heading has memory without showing that the memory is of the leg-to-leg kind the model needs, and the other three correlations, which compare the lengths and the waiting times of legs that have not been delimited, it cannot reach at all. Chapter 4 cuts the flight, and the model is built on what both return. Whether the trajectories of real soaring flights belong to an anomalous regime, and of which type, is the question addressed in this thesis.

Soaring has previously been studied mainly as a control problem—how to centre and exploit a thermal efficiently [8]. The present work is statistical instead: it asks which transport properties emerge at the level of the ensemble of observed trajectories. Its direct precedent is the study of Vilpellet et al. [9], who analysed a large set of human soaring flights and found a horizontal transport that is sub-ballistic and approximately independent of aircraft type (Hurst exponent $H \approx 0.88$, i.e. $\langle|\mathbf{r}(t)|^2\rangle \sim t^{2H}$, between diffusive $H = 1/2$ and ballistic $H = 1$, so that a value $1/2 < H < 1$ marks superdiffusive transport). That study also introduced a phase-resolved analysis based on the transition–search–climb cycle. This thesis builds on it: it enlarges the dataset, revisits the cleaning and filtering that precede any analysis, and re-examines both the phase segmentation and the transport analysis.

1.2 Objectives

The thesis pursues the following objectives:

1. to assemble a large, reproducible, and documented dataset of real soaring flights;
2. to clean and filter the raw GPS logs into analysis-ready trajectories, and to define, measure and analyze suitable transport observables exploited for testing anomalous transport;
3. to define a segmentation strategy for the transition–search–climb cycle and characterize the motion statistics within each phase;
4. to build a minimal stochastic model able to reproduce the empirical statistics;
5. to extend the dataset to competition flights and examine how flying in competition changes the motion relative to solo flights.

A first dataset—the archive of the *Coupe Fédérale de Distance* (CFD) of the French free-flight federation (FFVL), covering paraglider and hang-glider flights—has been assembled and verified and is described in Chapter 2; data collection continues and will be extended to further sources, in particular sailplanes and competition flights (Sec. 2.1).The dataset (Secs. 2.2–2.5) and its cleaning-and-filtering pipeline (Sec. 2.7) are described in Chapter 2, and the transport of the unsegmented trajectory is measured in Chapter 3; the segmentation, the modelling it enables (Chapter 4), and the extension to competition flights define the work ahead.

1.3 Scope of this document

This document reports the current state of the project and is revised as the work proceeds. The statistics it quotes, and the tables and figures that carry them, are generated from the dataset by the analysis code rather than typed, so those numbers cannot drift from the data they describe. Chapter 2 follows the data end to end: its acquisition and the reproducibility of the pipeline (Sec. 2.1); the tracklog format and the flight catalog (Secs. 2.2–2.3); the glider categories and the coverage statistics (Secs. 2.4–2.5); the known data-quality caveats (Sec. 2.6); the pre-processing that turns the raw logs into analysis-ready trajectories (Sec. 2.7); a first characterization of the cleaned dataset (Sec. 2.8); and a one-map summary of the whole pipeline that closes the chapter (Sec. 2.9). Chapter 3 measures the transport of that ensemble on the whole trajectory, without segmenting a flight: the estimator and the exponent it returns (Secs. 3.2 and 3.4), and what a second moment cannot say (Sec. 3.7). Chapter 4 sets out the phase segmentation and the modelling it enables; none of it is implemented yet.

Throughout, the discussion keeps to the scientific and statistical reasoning behind each choice. Implementation-level detail (code modules, configuration values, sample sizes, computation times, reproduction commands) is collected in Implementation and Computational Details, from p. 72. That part is organised chapter by chapter to mirror the body, and the body links into it at each point where an implementation detail was trimmed out of the text, as "implementation details: Sec. I2.7.2". Its numbers carry an `I`, so a cross-reference always says which of the two parts it points into: Sec. I2.7.2 is the implementation side of Sec. 2.7.2.

Chapter 2

The dataset

This chapter follows the data end to end: how the raw archive is acquired, verified and kept reproducible (Sec. 2.1); what the records contain and what the collection looks like (Secs. 2.2–2.6); and the pre-processing that turns a raw tracklog into the analysis-ready trajectory the transport analysis consumes (Sec. 2.7), closed by a first characterization of the cleaned dataset (Sec. 2.8) and a one-map summary of the pipeline (Sec. 2.9).

2.1 Data acquisition

2.1.1 Source: the FFVL Coupe Fédérale de Distance

The data come from the *Coupe Fédérale de Distance* (CFD), the cross-country competition run by the French free-flight federation (FFVL). The federation operates two parallel competitions on separate sites: one for paragliders (`parapente.ffvl.fr`) and one for hang gliders (`delta.ffvl.fr`). Pilots upload their flights; each entry carries metadata—among them pilot, date, take-off and landing sites, scored distance, duration, and glider, with the full set of fields listed in Sec. 2.3—and, in most cases, the raw GPS tracklog in IGC format (the standard flight-recorder format, Sec. 2.2). A "season" is identified by its starting year, e.g. the year 1999 denotes the 1999–2000 season; the paraglider archive starts in 1999–2000 and the hang-glider archive in 2001–2002.

2.1.2 Collected data

For every season the pipeline retrieves two things:

- the **season XML export**, archived verbatim as provenance (`raw_xml/<year>.xml`); it is the authoritative list of flights and links;
- the **IGC tracklog files** referenced by the XML, one file per flight that has a track.

From these, a derived **catalog** is built (`catalog.csv`, one row per flight, its columns detailed in Sec. 2.3) together with a per-season index (`seasons_index.csv`). The raw data (tens of gigabytes) live on an external disk and are never committed to the repository. The catalog is itself rebuildable from the archived XMLs and likewise uncommitted (Sec. 2.3).

2.1.3 Pipeline and tooling

Acquisition is performed by a dedicated, installable Python package, exposed through two command-line tools, one per discipline, sharing the same subcommands for fetching, downloading, cataloguing, and verifying. The design favours robustness for a long-running, large job: downloads are resumable and safe to interrupt, a response is accepted only once validated as real IGC content, and download requests are rate-limited so as not to load the server (implementation details: Sec. I2.1.1).

2.1.4 Reproducibility

The destination directory is not hard-coded: it is set per machine through an environment variable or a configuration file, of which the repository ships only a placeholder, so no machine-specific path is committed. Archiving the season XMLs verbatim keeps the full provenance (implementation details: Sec. I2.1.2).

2.1.5 Integrity

Because the data sit on an external drive, integrity is re-checked independently after download, rather than trusted at download time (implementation details: Sec. I2.1.3). Both FFVL collections have been verified in full: all 186,052 paraglider and 6,716 hang-glider downloaded tracklogs pass the same content check applied at download — a readable file whose first record is an `A` header and which carries at least one `B` fix record (Sec. 2.2). These are the tracklogs actually fetched; they cover a subset of the flights that carry a track at all (Sec. 2.5).

2.1.6 Further data sources

The FFVL CFD is the primary source of the present study, but the dataset is not assumed to be final. The paraglider data are the main analysis target; the hang-glider data are collected in parallel and enable the cross-type comparison of Sec. 2.4 (the counts for both are given in Sec. 2.5). Additional public soaring repositories are being pursued to enlarge and diversify the sample and, above all, to add the third glider category, *sailplanes* (Sec. 2.4). A large sailplane source with a machine-readable interface is *WeGlide*, which exposes flight metadata and IGC tracks through an API. The default access is rate-limited (of order sixty requests per day), which rules out a bulk export of the archive; a request for research access with a higher quota has been sent and is pending. Paraglider and hang-glider coverage may likewise be broadened through further portals such as XContest. A comparable research-access request has been sent to XContest; as of 20 July 2026 neither WeGlide nor XContest has replied. All of these can be acquired with the same approach, each producing a catalog in the same format.

2.2 The IGC tracklog format

A flight track is stored as an *IGC file*, the plain-text format defined by the FAI/IGC Technical Specification for IGC-approved GNSS flight recorders [10], decoded from fixed character positions; every downloaded track is checked at acquisition to be IGC content — an `A` record followed by at least one `B` record (Sec. 2.1).

Each line is a *record* whose type is set by its first character: `A` the flight-recorder identifier, `H` header fields (date, pilot, glider, datum, . . .), `I` the declaration of optional extra fields, and `B` the GPS fixes that form the body of the file. A `B` record decodes to a UTC time, latitude and longitude on the WGS84 datum, a fix-validity flag, and *two* altitudes in metres — barometric (pressure) then GNSS. The flag concerns the *GNSS* solution only: `V` marks a GNSS altitude drop-out and leaves the latitude, longitude and, on a logger with a working pressure sensor, the barometric altitude usable — the channel this analysis adopts (Sec. 2.7.1), a choice that matters because the vertical velocity derived from it feeds the phase segmentation.

Decoding can succeed numerically on a corrupted byte that is not a legal encoding, so the parser also checks format validity at decode time (a valid time, valid coordinate fields, a valid hemisphere letter), dropping the record otherwise, on the same footing as a record of the wrong length (implementation details: Sec. I2.3). This is distinct from the *plausibility* checks of fix-level cleaning (Sec. 2.7.2), which target fixes that decode validly but are physically implausible given the rest of the trajectory.

The sequence of `B` records is a time-stamped three-dimensional trajectory at a sampling period typically of a few seconds, not guaranteed uniform and varying across recording devices — a point the analysis must handle explicitly.

2.3 Flight metadata and the catalog

Every flight in the season XML carries a set of metadata as record attributes, which the acquisition preserves. Beyond the identifiers and links, the available fields are collected in Table 2.1.

Table 2.1. Flight metadata preserved from the season XML (raw attribute names), beyond the identifiers and links.

Field(s)	**Content / caveat**
date	flight date
pilot, club	pilot and club
take-off, landing, department	the two sites and the French department
`aile`	wing model name (raw XML field; French for "wing")
`aile_class`	wing class, one system per discipline (Sec. 2.4)
flight type	the *scoring* category: free distance, out-and-return, flat or FAI triangle, hike-and-fly, . . . (*caveat*: a scoring label, not a glider or manoeuvre type)
distance, points	scored distance and points
duration, speed	duration and average speed, where present

From these, the pipeline builds a single derived table, the **catalog** (`catalog.csv`), one row per flight across all seasons: the metadata above plus a `local_path` linking each flight to its track on disk and a `downloaded` flag, so a flight traces to its file and back to its metadata and public page without an external lookup. A smaller **season index** (`seasons_index.csv`) summarises each season by flights declared, carrying a track, and downloaded, and is the source of the statistics below. Both are regenerated from the archived XMLs on demand (`build-catalog`).

2.4 Glider categories

Soaring kinematics depend strongly on the aircraft, so the glider must be identifiable for any stratified analysis. Three categories exist, in increasing order of performance: **paragliders** (flexible canopy; slowest, glide ratio ∼8–11, tight turns), **hang gliders** (delta wing, flexible or rigid; faster, glide ratio ∼10–15), and **sailplanes** (rigid aircraft; fastest, glide ratio $\gtrsim$25) — different enough in cruise speed, glide ratio and turn radius that the reference study [9] analyses each separately rather than pooling them.

The category is set by the *data source*: the FFVL competition is split by discipline (Sec. 2.1), so the present dataset spans both paragliders and hang gliders, already enough for the cross-type comparison of [9]. Sailplanes are collected by different portals and not yet in hand.

Within each discipline the flights can be further stratified by *performance class*, recorded in `aile_class`; the two disciplines use different class systems, collected in Table 2.2. Paragliders

Table 2.2. Class labels recorded in `aile_class`, one system per discipline. The EN ladder is ordered by increasing demand on the pilot, and that ordering is all the stratification needs from it; the FAI classes are *not* a ladder, being defined by the method of control. Sources: the FAI Sporting Code, Common Section 7 [11] (§§1.2, 1.4.1) for the FAI classes and the sub-class ceilings; the EN class descriptions are condensed from Table 1 of the standard, on the caveat explained in the text.

Class	Notes
Paragliders — EN/LTF ladder	
EN A	maximum passive safety, extremely forgiving; any level of training
EN B	good passive safety, forgiving; ceiling of the FAI *Standard* sub-class
EN C	moderate passive safety, dynamic reactions; recovery needs precise input; ceiling of the FAI *Sport* sub-class
EN D	demanding, potentially violent reactions; for pilots practised in recovery
CCC	CIVL Competition Class racing wings
tandem / non-certified	two-seater and uncertified wings, pooled in this field
Hang gliders — FAI Class O (paragliders are Class 3 of the same scheme)	
Delta Class 1	weight-shift as the sole method of control
Delta Class Sport	sub-class of Class 1: type-certified production wing with a king post
Rigide Class 2	movable aerodynamic surfaces as the primary control
Rigide Class 5	as Class 2, but in the roll axis only; no fairings

follow the EN/LTF certification ladder, which does track performance (higher classes fly faster and flatter) and is therefore the natural within-source stratification variable. The hang-glider scheme is *not* a ladder: the FAI classes are defined by the method of control, so they stratify the fleet by design rather than by performance, and any ordering imposed on them is an assumption to be checked rather than a property of the scheme.

The two ladders are not documented on equal terms: the FAI classes come from a freely published Sporting Code, the EN classes from a paid CEN standard with no public authoritative text, so the EN entries of Table 2.2 are condensed from the closest openly available source rather than the published standard itself (implementation details: Sec. I2.5). Nothing in this thesis depends on the wording — the class field is a stratification label, and what the analysis needs is that the labels are distinct and ordered, not what each one certifies.

Two caveats bear on any use of `aile_class`: it must be mapped onto one canonical label per discipline before it can serve as a stratum, since the field carries the FFVL's own code ("C ou 2", "non homologuée") rather than the EN/FAI label of Table 2.2; the placeholder and a blank entry are left unclassified rather than guessed (implementation details: Sec. I2.5). And the class certifies the wing, not the pilot, which no processing can remove and which bounds what a class-stratified result can be taken to mean.

2.5 Coverage and statistics

The statistics below describe the FFVL CFD dataset currently in hand; further sources may be added later (Sec. 2.1). It comprises 212,602 flights in total, of which 192,971 carry a GPS tracklog and 192,768 have been downloaded and verified, split across two disciplines.

The *paraglider* collection spans 27 seasons, from 1999–2000 to 2025–2026: 203,343 flights, of which 186,225 (91.6 %) carry a tracklog and 186,052 (99.9 %) are downloaded and verified (Table 2.3). The *hang-glider* collection spans 25 seasons, from 2001–2002 to 2025–2026: 9,259 flights, of which 6,746 (72.9 %) carry a tracklog and 6,716 (99.6 %) are downloaded and verified (Table 2.4). The hang-glider archive is roughly twenty times smaller, reflecting the smaller active population of the discipline.

In both disciplines the number of flights carrying a track grows by orders of magnitude over the period, reflecting the adoption of GPS loggers and online scoring: the early seasons contribute a handful of tracks each, while recent seasons contribute thousands of tracks for paragliders and several hundred for hang gliders.

Table 2.3. FFVL CFD *paraglider* flights per season (`parapente.ffvl.fr`): total declared, with a GPS track, and downloaded. (implementation details: Sec. I2.6)

Season	Flights	With GPS	Downloaded	Coverage (%)
1999–2000	10	10	10	100.0
2000–2001	832	1	1	100.0
2001–2002	1,150	3	3	100.0
2002–2003	1,996	123	123	100.0
2003–2004	1,470	337	337	100.0
2004–2005	1,765	574	574	100.0
2005–2006	2,162	894	894	100.0
2006–2007	2,386	1,016	1,016	100.0
2007–2008	3,081	1,667	1,667	100.0
2008–2009	3,974	2,614	2,613	100.0
2009–2010	4,171	3,059	3,059	100.0
2010–2011	4,872	4,185	4,169	99.6
2011–2012	6,744	6,075	6,065	99.8
2012–2013	7,873	7,106	7,075	99.6
2013–2014	9,804	9,157	9,127	99.7
2014–2015	13,793	13,293	13,254	99.7
2015–2016	13,982	13,703	13,696	99.9
2016–2017	15,760	15,534	15,528	100.0
2017–2018	14,917	14,682	14,676	100.0
2018–2019	16,702	16,517	16,508	99.9
2019–2020	10,966	10,868	10,863	100.0
2020–2021	9,500	9,483	9,480	100.0
2021–2022	14,552	14,507	14,502	100.0
2022–2023	11,590	11,561	11,560	100.0
2023–2024	10,372	10,362	10,360	100.0
2024–2025	10,906	10,885	10,884	100.0
2025–2026	8,013	8,009	8,008	100.0
Total	203,343	186,225	186,052	99.9

Table 2.4. FFVL CFD *hang-glider* flights per season (`delta.ffvl.fr`): total declared, with a GPS track, and downloaded. (implementation details: Sec. I2.6)

Season	Flights	With GPS	Downloaded	Coverage (%)
2001–2002	248	0	0	0.0

Season	Flights	With GPS	Downloaded	Coverage (%)
2002–2003	281	6	6	100.0
2003–2004	256	28	25	89.3
2004–2005	244	35	25	71.4
2005–2006	365	80	76	95.0
2006–2007	337	100	96	96.0
2007–2008	388	136	132	97.1
2008–2009	344	155	155	100.0
2009–2010	324	136	136	100.0
2010–2011	425	147	147	100.0
2011–2012	117	103	103	100.0
2012–2013	428	406	406	100.0
2013–2014	423	409	409	100.0
2014–2015	628	595	591	99.3
2015–2016	417	413	413	100.0
2016–2017	468	462	462	100.0
2017–2018	342	331	331	100.0
2018–2019	604	594	594	100.0
2019–2020	378	373	373	100.0
2020–2021	328	328	328	100.0
2021–2022	570	568	568	100.0
2022–2023	376	374	374	100.0
2023–2024	437	437	436	99.8
2024–2025	353	352	352	100.0
2025–2026	178	178	178	100.0
Total	9,259	6,746	6,716	99.6

2.6 Data quality and caveats

A few limitations are already known and will shape the analysis:

- **Flights without a track.** About 8 % of paraglider entries and about 27 % of hang-glider entries have no IGC file (older or manually declared flights); they carry metadata only. These are the flights outside the "With GPS" column of Tables 2.3 and 2.4.
- **A small residue of unrecoverable files.** A few flights advertise a track whose download fails or returns something that is not valid IGC content (broken or placeholder links); these are recorded as failures and excluded.
- **Heterogeneous logging.** Sampling rate, availability of the two altitude channels, and per-fix positioning noise depend on the device; trajectories must be resampled and quality-filtered before any ensemble statistic is computed.
- **Altitude channel.** A sizeable minority of loggers record no pressure altitude – almost a third of paraglider and roughly a sixth of hang-glider flights – which fall back to the GNSS altitude, with the channel actually used recorded per flight (Sec. 2.7.1).
- **Glider metadata.** Flights are paragliders or hang gliders, tagged by source (Section 2.4); the class field `aile_class` must be normalised before it is used to stratify.
- **Incomplete dates.** Some historical entries carry the placeholder date `0000-00-00`, written as a sentinel at acquisition; such flights are discarded, since a date-less flight

cannot be assigned to a season and no downstream step needs it. The cut is negligible and checked per season, not only archive-wide, since a small total share could still empty the smallest early seasons: 8 of 203,343 paraglider entries carry the sentinel (0.0039 %), 0 of 9,259 hang-glider ones, and the worst-affected season loses only 0.17 % of its entries (2001-2002).

These caveats motivate the pre-processing described in the remainder of this chapter.

2.7 Trajectory pre-processing

Each IGC file is parsed into a sequence of fixes $(t_i, \varphi_i, \lambda_i, h_i)$ – UTC time, geodetic latitude and longitude, and altitude. The pipeline then proceeds in a fixed order:

(i) choose the altitude channel (Sec. 2.7.1);

(ii) clean individual fixes (Sec. 2.7.2);

(iii) trim the ground phases (Sec. 2.7.3);

(iv) filter out unfit whole flights (Sec. 2.7.4);

(v) convert to a local Cartesian frame (Sec. 2.7.5);

(vi) enforce a uniform sampling interval within each flight (Sec. 2.7.6);

(vii) smooth and differentiate the trajectory (Sec. 2.7.7).

The output is a tidy, analysis-ready representation written to three tables on the data disk: one carrying a row per fix, one a row per segment, one a row per flight *attempted*. What each file holds and how the disk is organised is set out in Sec. I2.2.

Steps (i)–(iv) act on the *raw geographic* coordinates, since none of them needs a direction. A horizontal distance comes straight from (φ, λ) through the great-circle (haversine) formula of Eq. (2.1); a speed is one such distance over an elapsed time; an altitude is read as logged. The conversion (v) to the metric East–North–Up frame (Sec. 2.7.5) then runs on the cleaned, trimmed track, with the origin at the first fix that survives trimming, so the tangent-plane frame is built from good data. Steps (vi)–(vii) come last because they produce *vector* quantities: 3D position, velocity and acceleration components, which need the Cartesian axes step (v) provides.

Every step below is governed by a handful of numerical thresholds. Each is stated where it is argued, and the whole set is gathered, with the symbols used here, in Table 2.7 at the close of the chapter, beside the map of the pipeline. All of them are *working values*: fixed a priori, on physical scales rather than on the transport observables, and audited a posteriori on the fraction of data each rule removes (Table 2.6).

2.7.1 Choice of altitude channel

The analysis adopts the *barometric* altitude over the GNSS altitude for the vertical dynamics: both are written to 1 m resolution by the IGC format itself, but the barometric channel carries far less high-frequency noise. Its errors are slow — a constant offset from the ICAO reference (tens of metres, Fig. 2.1a) and a drift already reaching tens of metres within the half-hour window of Fig. 2.1b — both traceable to the gap between the day's actual sea-level pressure and the ICAO reference of 1,013.25 hPa (implementation details: Sec. I2.7.1), and annihilated by the differencing every dynamical quantity involves (a vertical velocity, a displacement increment, an autocorrelation). The GNSS altitude is referenced to the WGS84 ellipsoid, so it carries no atmospheric offset, but the vertical coordinate is the worst-determined one geometrically, and multipath biases the range on timescales of minutes [12] (implementation details: Sec. I2.7.1). This is why free-flight variometers derive the climb rate from the pressure sensor rather than from GNSS.

Empirical confirmation. Figure 2.1 compares the two channels on 3,000 flights per discipline (Appendix 2.A) (implementation details: Sec. I2.7.1). At low frequency the two agree, tracking the real climbs and glides; at high frequency the *typical* flight is comparable in both, the median spectra nearly coinciding at the plateau set by the metre rounding common to both channels (panel c). What separates the channels is the *spread* across flights: the barometric band stays tight, while the GNSS band fans out by one to two orders of magnitude towards the Nyquist frequency, from receiver tracking noise amplified by poor vertical dilution of precision on a substantial minority of flights. Since the vertical velocity is a finite difference that amplifies exactly that band, and which flights have poor GNSS reception cannot be known in advance, the uniformly clean barometric channel is the safe choice — a deliberate departure from the reference study [9], whose pipeline derives the vertical coordinate from GNSS.

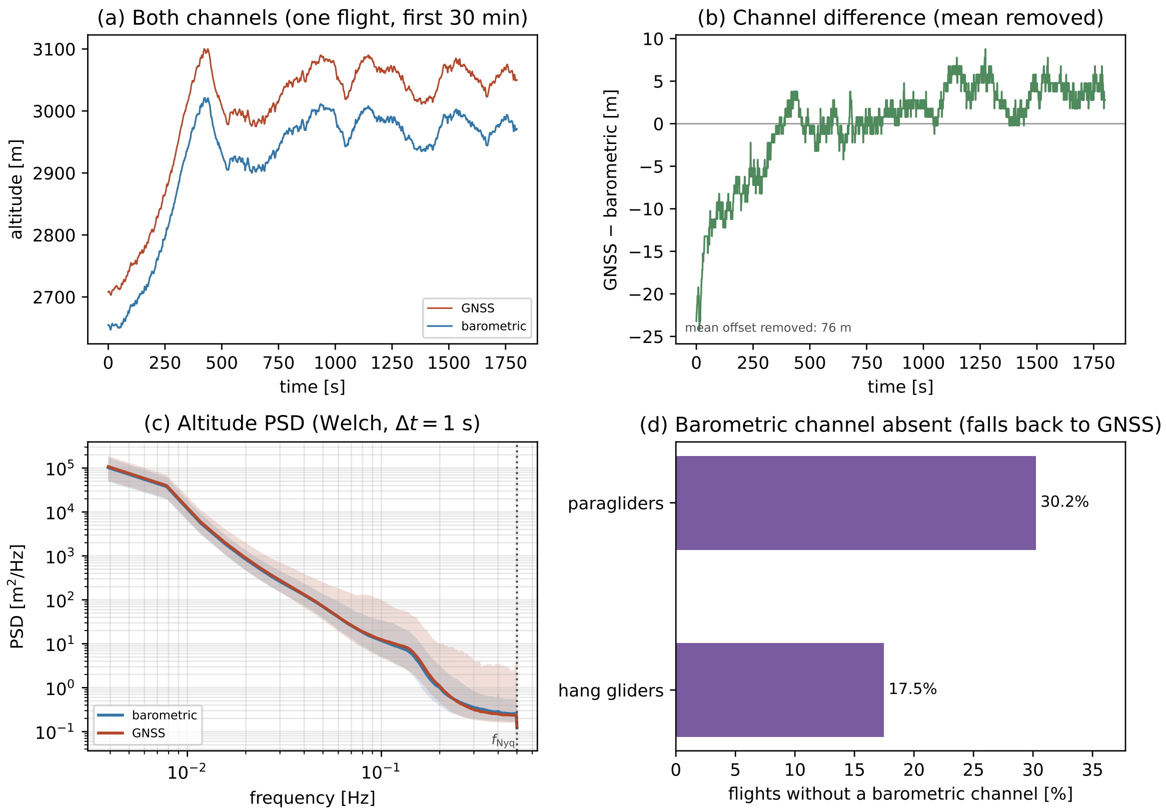

Figure 2.1. Barometric vs GNSS altitude noise. (a) One representative flight over a 30 min window, the two raw channels on a shared axis. The vertical gap between the curves is the offset between their reference surfaces (pressure vs. ellipsoidal). (b) Their difference, mean removed: the offset is not constant but drifts by tens of metres already within this 30 min window. (c) Per-frequency *median* Welch spectrum of each channel (Appendix 2.A), pooled over the two disciplines, with the 10 to 90 percentile band across flights; the medians nearly coincide, while the GNSS band fans out towards the Nyquist frequency $f_{\text{Nyq}} = 1/(2\Delta t)$ (dotted). The median is used because the per-flight spectra are heavy-tailed (Sec. I2.7.1). (d) Fraction of flights whose barometric channel is absent, per discipline. (implementation details: Sec. I2.7.1)

Four practical points follow. A trajectory uses a single channel end-to-end, never spliced within a flight, since a mid-flight switch offset by tens of metres would inject a spurious jump; which channel it uses is recorded as a per-flight attribute, so the two source groups stay separable downstream (implementation details: Sec. I2.7.1). A flight is used with its barometric channel when it has one; the few individual fixes with a missing value are dropped rather than back-filled from GNSS — 4.2 % of barometric paraglider flights (2.1 % hang) miss any value at all, with a median of 3 missing fixes among them against $\sim 10^4$ in a typical track (implementation details: Sec. I2.7.1), the hole handled downstream like any other gap (Sec. 2.7.6). A logger with no pressure sensor writes the whole channel as zero, or a stuck sensor writes a constant non-zero value: a flight counts as barometric only when the channel covers at least 95 % of its fixes *and* its total range exceeds 30 m, else it falls back to GNSS, unfiltered — a non-negligible minority (Figure 2.1d), measured at 30.2 % of paraglider and 17.5 % of hang-glider flights scanned (implementation details: Sec. I2.7.1). That liveness test catches only a sensor dead for the whole recording, not one that freezes partway through, and nothing downstream closes that gap (implementation details: Sec. I2.7.1). Nor does the presence test check what it falls back to: 107 paraglider flights sit just under its cut rather than at zero, and for these the assumption that GNSS is the more complete channel goes untested, since nothing in the pipeline measures GNSS completeness to compare it against (implementation details: Sec. I2.7.1).

Mixed altitude sources. The mix is acceptable. The horizontal, two-dimensional transport analysis — the main subject of this work — does not use the altitude at all, so every flight is usable there regardless of channel. Altitude enters only the vertical velocity, hence the segmentation and the phase-resolved observables, where the two source groups are kept distinct through the per-flight record above and checked for compatibility on the fraction of airborne time each phase accounts for, per discipline (implementation details: Sec. I2.7.1). Forcing GNSS on every flight to avoid the mix would needlessly degrade the vertical signal of the barometric majority to match the worst loggers.

2.7.2 Fix-level cleaning

Geometry on the raw coordinates. The cleaning runs on the raw geodetic coordinates (φ_i, λ_i), before the Cartesian conversion (Sec. 2.7.5), so distances between fixes are great-circle (haversine) distances [13] on a sphere of mean radius $R_\oplus = 6{,}371$ km:

$$\Delta s_i = 2R_\oplus \arcsin\sqrt{\sin^2\frac{\varphi_{i+1}-\varphi_i}{2} + \cos\varphi_i \cos\varphi_{i+1}\, \sin^2\frac{\lambda_{i+1}-\lambda_i}{2}}\,, \tag{2.1}$$

giving the horizontal speed $v_{xy,i} = \Delta s_i/(t_{i+1}-t_i)$ and, from the barometric altitude, the vertical velocity $v_{z,i} = (h_{i+1}-h_i)/(t_{i+1}-t_i)$; the same distance also yields the whole-flight *path length* $L = \sum_i \Delta s_i$ and *extent* $D = \max_i \Delta s(\mathrm{fix}_0, \mathrm{fix}_i)$. The spherical approximation is ample: replacing the WGS84 ellipsoid by a sphere perturbs an inter-fix distance by at most $\lesssim 0.3\,\%$ (the flattening scale) (implementation details: Sec. I2.7.2), about 3 cm on a 10 m step, far below the metre-scale noise of consumer GNSS receivers.

Design principle. Corrupt or physically impossible fixes must be removed without discarding the flight around them; an over-aggressive cleaning biases the result as much as dirty data would. A spurious fix that survives lies at worst tens of metres from the true track, a displacement the smoothing absorbs, whereas a genuine fix removed from a slow, localized phase (a search, a climb) erases a real trapping event that nothing downstream can restore. The cleaning therefore defaults to keeping the data: a fix is removed only on positive evidence of a defect, and no detector deletes a fix that still carries a usable horizontal position — a corrupt altitude is marked missing and restored at resampling (Sec. 2.7.6), a corrupt horizontal position is deleted outright.

Three classes of defect. Past the format-validity check made at decode time (Sec. 2.2), a surviving fix can still be wrong in one of three ways, summarized in Table 2.5. A fix *off the trend* — a position spike, or a jump out and back within one step — is found by a robust local-outlier test, the *Hampel identifier* [14, 15]. Let the window of fix k be the fixes within $\pm w$ of t_k, and $\tilde{\varphi}_k, \tilde{\lambda}_k$ the component-wise window medians. The residual is the great-circle distance from that median point,

$$r_k = \Delta s\big((\varphi_k, \lambda_k), (\tilde{\varphi}_k, \tilde{\lambda}_k)\big), \tag{2.2}$$

with local scale $\mathrm{MAD}_k = \mathrm{med}\{r_j : |t_j - t_k| \le w\}$, and the fix is flagged when

$$r_k > \max(k\,\sigma_k,\ \varepsilon_{\min}), \qquad \sigma_k = 1.4826\,\mathrm{MAD}_k, \tag{2.3}$$

σ_k the normalized MAD [16]: a departure of more than k robust standard deviations *and* an absolute floor $\varepsilon_{\min}$ of a few times the GPS noise. The working values are $k = 5$, a half-window $w = 20\,\mathrm{s}$ and $\varepsilon_{\min} = 15\,\mathrm{m}$, with the test evaluated only where the window holds at least 5 fixes (rationale in Sec. I2.7.2). A flag does not delete: what deletes is the absolute v_{xy} bound, a fix removed when unreachable from the last accepted fix at that bound *and* the removal of its block lets the track rejoin within it. The identifier flags and attributes, and supplies the per-flight anomaly count, but is not required for deletion (implementation details: Sec. I2.7.2) — a genuine tight thermalling circle produces the same local-median departure as a spike, at a fraction of the speed a spike would imply, so the protection lies in the speed bound that does the deleting rather than in the flag.

A fix *on the trend* agrees with its neighbours yet is still wrong as a record – a duplicated or backward timestamp, or a run of frozen positions from a lost GNSS lock; its residual is near zero, so it escapes the outlier test and needs a structural rule instead.

A value *out of range* concerns the vertical channel: an impossible altitude or climb rate is caught by an absolute bound that needs no context.

An unreturned vertical step. The $|v_z|$ rule recognises two shapes: an *out-and-back* spike is censored, a coherent *run* is a sustained manoeuvre and is kept. A single super-threshold step that is never undone — a barometric re-reference, or one corrupt record — is neither, and could carry a written vertical velocity of up to $5{,}117\,\mathrm{m\,s^{-1}}$ before being caught, on 8.3 % of paraglider and 6.2 % of hang-glider flights. The shape is detected and counted per flight, as `n_alt_level_shift`, so it is visible rather than silently present (implementation details: Sec. I2.7.2); what to *do* about it is left open, since the vertical channel feeds the phase segmentation of Chapter 4 rather than the horizontal transport measurement of Sec. 3.1, so the choice belongs with that analysis.

Why the vertical channel is treated differently. The obvious symmetric design — a Hampel test on z too, corroborated by the climb-rate bound — is wrong here, for three reasons (implementation details: Sec. I2.7.2). An absolute bound is sufficient on z and insufficient on xy: the vertical envelope is narrow and known, while the same horizontal spike passes or fails the speed bound depending on the logger's cadence, so only the horizontal case needs a local, scale-free test. A Hampel test on z would flag the physics: the residual from a local median is largest exactly at thermal entries and exits, the transitions the phase segmentation is built on, and the vertical channel lacks the corroborating witness (speed) that lets a horizontal circle be told from a spike. And the costs are asymmetric: a false vertical flag is cheap to fix by interpolation except across a thermal entry, where interpolation erases the very transition it should preserve — so keeping the vertical rule to a bound that only fires on the physically impossible avoids that failure mode altogether.

The absolute bounds are placed on data, where the per-fix distributions of Fig. 2.2 stop decaying; the Hampel and frozen-lock parameters are working values fixed a priori (Table 2.7), checked a posteriori on the fraction of fixes each rule removes (*Completeness, the integrity gate, and validation* below).

Removal policy. A speed bound alone cannot attribute a defect, since the speed belongs to the segment *between* two fixes: attribution comes from the bounded-removal test, the block whose removal restores the trend being the corrupt one, so an isolated spike is removed without touching its neighbours. What happens next depends on *which* value is corrupt, by a single criterion — keep every genuine horizontal sample, never hide a hole in the horizontal record — and in either case the corrupt value is simply discarded, with reconstruction deferred to the single resampling step (Sec. 2.7.6). A corrupt *altitude* leaves the fix kept and only that channel marked missing, since the horizontal position and timestamp are intact and still sample the path; a corrupt *horizontal position or time* deletes the whole fix, because keeping it with the position marked missing would hide the hole from the gap rule that decides whether to bridge or split (Sec. 2.7.6), and an invented straight line could then be laid across a full thermalling circle (implementation details: Sec. I2.7.2). The IGC validity flag needs no rule of its own: a `V` flag certifies a degraded GNSS altitude (Sec. 2.2), so on a barometric flight the fix is untouched, and on a GNSS-fallback flight it reduces to the corrupt-altitude case.

An impossible step is a boundary, never flown path. What must hold whatever the scan resolves is the invariant the rest of the pipeline leans on: *no step between two surviving fixes may break the v_{xy} bound.* Where a block is kept because nothing could rejoin it, the step by which the track returns from it is itself impossible and, left alone, enters a segment as flown motion — a jump of tens of kilometres inside one second, in the archive. Such a step is a transition whose course is unknown, so it is made a segment boundary on the same footing as a long gap (Sec. 2.7.6), counted per flight as `n_boundaried` (implementation details: Sec. I2.7.2).

A corrupt first fix is a blind spot no bounded-removal test can close: the scan carries the last accepted fix as an anchor and never questions its own first one, and that fix becomes the origin of the local frame (Sec. 2.7.5), so the error does not average out with record length — five paraglider flights left the first full-archive run some 4,500 km from their own origin, enough to raise the ensemble MSD by seven orders of magnitude at the shortest lags. Two attribution rules were tried and both failed, in opposite directions (implementation details: Sec. I2.7.2), so the discontinuity is left as a discontinuity and the guarantee is taken one stage later: a record out of reach of its own first fix is refused whole (Sec. 2.7.4). The invariant Chapter 3 rests on, $|\mathbf{r}(t)| \le v_{xy}^{\max}\, t$, holds either way, checked directly on the written tables (Sec. I2.2).

Frozen-position runs (GNSS lock loss). When the receiver loses its GNSS lock, many loggers repeat the last position for tens of seconds while the clock keeps running. Such a run passes every test above: horizontal speed zero, altitude in range, valid timestamps. Kept, it contaminates the tail of the waiting-time density $\psi(\tau)$ with a false long wait; cut too aggressively, it removes real trapping events (tight thermalling climbs, also slow and horizontally localized) from the same tail (implementation details: Sec. I2.7.2). A run is therefore cut only when three independent signatures agree:

(i) *Spatial diameter.* The run's bounding diameter — the largest great-circle distance between any two of its fixes — stays below δ_{xy} (implementation details: Sec. I2.7.2); a circling climb instead traces a disc of diameter $2R \approx 30$ m to 100 m, at least twice δ_{xy}, and clears the test on geometry alone.

(ii) *An independent witness — the barometric altitude.* A GNSS lock-loss freezes only the GNSS position; a horizontally motionless run whose barometric altitude keeps changing at a plausible rate is a tight climb, and is kept. Only a byte-identical coordinate repeat may overrule this witness.

(iii) *Recorder declarations.* A byte-identical repeat of the coordinate fields is conclusive on its own — no moving receiver rewrites the same bits. The `V` flag and a zeroed GNSS altitude are weaker, certifying a degraded GNSS altitude rather than a frozen fix, and decide only on a GNSS-fallback flight, where no barometric witness exists.

A run is classified as a loss of signal only when spatially collapsed, witnessed, and sufficiently long,

$$\underbrace{\mathrm{diam}(\mathrm{run}) < \delta_{xy}}_{\text{(i) collapsed}} \wedge \underbrace{W}_{\text{witness (ii)/(iii)}} \wedge \underbrace{\Delta t_{\mathrm{run}} \geq \tau_{\mathrm{freeze}}}_{\text{duration}},$$
$$W = \begin{cases} |\Delta z_{\mathrm{baro}}| < \delta_z \vee \text{byte-identical} & \text{barometric flight,} \\ \texttt{V}/\text{zero-alt} \vee \text{byte-identical} & \text{GNSS-fallback flight.} \end{cases} \tag{2.4}$$

and is then marked as a gap and the trajectory *split* there, never interpolated across, with segments keeping their parent's origin and clock (Sec. 2.7.6). The working values are $\delta_{xy} = 15\,\mathrm{m}$, $\delta_z = 5\,\mathrm{m}$ and $\tau_{\mathrm{freeze}} = 60\,\mathrm{s}$, the last of them argued below.

A real thermalling climb fails the cut on two independent instruments at once: a circling wing turns at $R = v^2/(g\tan\phi)$ (implementation details: Sec. I2.7.2), giving $2R \approx 30\,\mathrm{m}$ for the tightest paraglider circles and $2R \gtrsim 50\,\mathrm{m}$ for hang gliders, missing the diameter test (i) by a factor of at least 2–3; and a climb of order $1\,\mathrm{m\,s^{-1}}$ sustained over τ_{freeze} moves the barometer well past the flatness tolerance δ_z. The duration threshold is set on the motion's own timescale — about two thermalling periods — so a run long enough to be cut spans at least two full circles of any genuine climb: on a barometric flight the rule is therefore conservative by construction, both margins having to fail at once, well outside their designed range (implementation details: Sec. I2.7.2). Below τ_{freeze} the rule deliberately abstains — a collapsed run and a genuine slow, localized stretch are not distinguishable by these signatures at durations of a few tens of seconds — at a bounded cost, since a kept short freeze lengthens one waiting time by at most τ_{freeze}, swept in the $\psi(\tau)$ sensitivity checks (Sec. 2.7.3).

Residual corner cases. The frozen-lock rule weakens where the barometric channel is absent, since the GNSS altitude cannot witness a suspected GNSS lock loss — it comes from the very receiver whose lock is in doubt. On GNSS-fallback flights the witness slot of Eq. (2.4) is therefore filled by the recorder's declarations alone, the diameter and duration conditions unchanged (implementation details: Sec. I2.7.2). Three error cases follow: a tight climb within δ_{xy} whose fixes carry a `V` flag is *wrongly cut* on GNSS-fallback flights, with no barometer to defend it; an undeclared *jittering* freeze — re-estimated each second rather than repeated byte-for-byte — is *wrongly kept*, also GNSS-fallback only; and a jittered freeze under a climbing barometer is *kept at bounded cost* on barometric flights, since motion during the run goes unrecorded (the path length under-counted, a spurious wait inserted) while the displacement across the run stays correct, the first position after re-acquisition being exact again. The first two are expected to be rare and cannot be counted directly, since a rule that fails does so silently; what measures the error rate instead is the injected-defect calibration below.

Completeness, the integrity gate, and validation. The segmentation (Ch. 4) needs a defined position and altitude at every time within a *segment* — a maximal stretch between two splits — not everywhere; cleaning does not guarantee this fix by fix, and the guarantee

Defect	Detected by	Action
Off the trend — *the horizontal position itself is corrupt*		
Position spike, out-and-back jump	impossibility gate: unreachable from the last accepted fix at the absolute v_{xy} bound ($45/55\,\mathrm{m\,s^{-1}}$, para/hang), and its block's removal lets the track rejoin. The Hampel identifier [14, 15] flags and attributes; it does not gate	delete the fix; the short gap is bridged at resampling
Position discontinuity (re-acquisition offset)	impossible step that no bounded removal rejoins	split at the step, like a long gap
Impossible step still standing	any surviving step past the v_{xy} bound, whatever the scan resolved (postcondition)	make it a segment boundary
Corrupt block closing the record	a block still open when the track ends: nothing follows it to rejoin, and by construction it spans at most w	delete the block, instead of splitting
On the trend — *record structurally wrong, horizontal position agrees with its neighbours*		
Duplicate timestamp	two or more fixes in one UTC second	merge to one (centroid)
Backward timestamp	time decreases (not a midnight roll-over)	delete the fix
Frozen-lock run	diam $< \delta_{xy}$ for $\geq \tau_{\text{freeze}}$, witnessed (Eq. 2.4): baro. flat or byte-identical; `V`/zero-alt on GNSS-fallback	mark as gap, *split* there
Out of range — *horizontal position intact, only the altitude value is impossible*		
Missing / out-of-band altitude, or $\lvert v_z \rvert$ spike	altitude channel absent, outside $[-100, 6000]$ m, or baro. $\lvert v_z \rvert > 13\,\mathrm{m\,s^{-1}}$	drop the altitude only; the fix stays
Invariant: a usable horizontal position is never discarded. A fix is deleted only when its *own* horizontal position is corrupt or invented (frozen lock); when only the altitude is wrong, the fix is kept and that channel is marked missing, to be restored at resampling (Sec. 2.7.6).		

Table 2.5. Fix-level cleaning, grouped by how each defect presents; in the order of the table, the three groups map onto a robust local test, a structural rule, and an absolute bound. The speed and altitude bounds are audited on the per-fix distributions of Fig. 2.2; the robust-test and frozen-lock parameters (k, w, $\varepsilon_{\min}$, δ_{xy}, δ_z, τ_{freeze}) are working values, stated with the rules that use them and argued in Sec. I2.7.2; Table 2.7 gathers them with the rest. The v_{xy} bound is per-discipline because the speed envelopes differ (a `sailplanes` entry follows once that source is in). The altitude lower bound is -100 m, not 0: the barometric altitude is referenced to the ICAO standard atmosphere rather than to the day's sea-level pressure, so on a high-pressure day a sea-level take-off can read a few tens of metres negative (the offset scale is visible in Fig. 2.1a). (implementation details: Sec. I2.7.2)

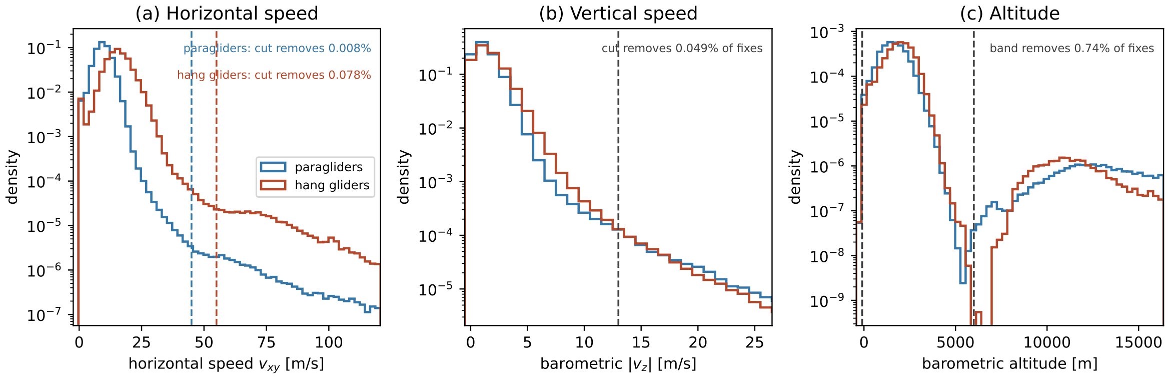

Figure 2.2. Fix-level cleaning: auditing the absolute bounds. Per-fix distributions of the quantities the absolute bounds test—bounds that serve both the out-of-range rule and, through the v_{xy} bound, the impossibility gate that deletes an off-the-trend fix (Sec. 2.7.2) – (a) horizontal speed v_{xy}, (b) barometric $|v_z|$, (c) barometric altitude – pooled over a large paraglider subsample and the complete hang-glider population, with the adopted bounds dashed (Table 2.5); the y-axis is logarithmic so the sparse tail the bounds act on is visible. In panel (a) each discipline shows the flat plateau of sustained speeds with no credible flight explanation that places its own bound: $45\,\mathrm{m\,s^{-1}}$ to $55\,\mathrm{m\,s^{-1}}$ for paragliders, $55\,\mathrm{m\,s^{-1}}$ to $70\,\mathrm{m\,s^{-1}}$ for hang gliders. Binning choices and sample sizes: paragraph *Diagnostic sample* of Sec. I2.7.2 (p. 90). This figure audits the *absolute* bounds only; the Hampel and frozen-lock detectors have no per-fix distribution to read a threshold off, and are validated instead by the removal-fraction sweep described below.

is established one step later, at resampling (Sec. 2.7.6) (implementation details: Sec. I2.7.2). Cleaning also counts how much it had to do over the airborne window (defects in a ground phase that trimming removes anyway must not condemn the flight): a flight whose cleaning removed or reconstructed more than 10 % of its fixes is dropped whole.

All detector parameters are validated by one procedure, run once the cleaning is built: the thresholds are frozen a priori, never tuned on the transport observables; the removal fraction of each rule is audited over the full archive, expected to touch only a small share of fixes; each threshold is swept around its working value and judged well placed when what it removes stays on a plateau rather than growing sharply (implementation details: Sec. I2.7.2); and, since an audit of removals says nothing about what a rule misses, defects of known size are injected into clean tracks and scored on recovery — the only direct measurement of the error rates the firing counts cannot give (implementation details: Sec. I2.7.2).

2.7.3 Trimming of ground phases

Outer ground phases (take-off and landing). The logger starts recording before take-off and stops after landing; these ground phases must be stripped before any kinematic analysis, using the horizontal speed v_{xy} alone. With $v_0 = 5\,\mathrm{m\,s^{-1}}$ and $T_0 = 30\,\mathrm{s}$ (Table 2.7), the retained airborne segment is $[t_{\mathrm{on}}, t_{\mathrm{off}}]$ with

$$t_{\mathrm{on}} = \min\{t:\ v_{xy} > v_0 \text{ on } [t,\, t+T_0]\}, \qquad t_{\mathrm{off}} = \max\{t:\ v_{xy} > v_0 \text{ on } [t-T_0,\, t]\}, \tag{2.5}$$

the same sustained-speed rule read forward and time-reversed, so a wing soaring into wind at near-zero ground speed is not mistaken for having landed. T_0 is set between two scales: from below it must outlast a ground gust or GPS glitch (a few seconds), from above it costs little when wrong, misfiring by at most T_0 at each end against the 40 min minimum duration a retained flight must reach (implementation details: Sec. I2.7.3). A flight with no sustained stretch at all is discarded. The one structural caveat is a slow phase sitting right at the start or end of the flight (a launch straight into ridge lift) being clipped along with the ground phase; the trimmed fraction is recorded per flight so the exposure can be bounded and the tail inspected directly (implementation details: Sec. I2.7.3).

Interior ground stints (mid-flight landings). A mid-flight landing and relaunch recorded in one log would otherwise survive as an interior near-zero-speed stint and read downstream as a false long wait in the tail of $\psi(\tau)$. An interior-ground detector, built like the frozen-lock rule, cuts a stint only when *both* $v_{xy} < v_0$ continuously for at least $T_{\text{ground}} = 10\,\text{min}$ — far beyond any search phase — *and* the adopted altitude channel is flat over the whole stint, its detrended central span staying within $\delta_z^{\text{gr}} = 5\,\text{m}$, the same independent-sensor argument as Eq. (2.4). The span is *central*, not a full range, because a full range grows without bound with the stint length on noise alone and would fail a motionless pilot on a long stint while passing the same pilot on a short one (implementation details: Sec. I2.7.3); and on a GNSS-fallback flight, where the tolerance for the noisier channel has not yet been calibrated, the guard abstains conservatively rather than risk excising a genuine climb, at the cost of never flagging a mid-flight landing on those flights (implementation details: Sec. I2.7.3).

A cut stint is excised and the flight split there, exactly as at a long gap (Sec. 2.7.6); the flight-level cuts of Sec. 2.7.4 are not re-applied to the individual segments, but to the *parent* flight — its airborne duration and path length summed over its segments — since a segment is short because a gap happened to fall there, which says nothing about whether the flight is genuinely cross-country (implementation details: Sec. I2.7.6). Trimming therefore runs before the flight-level cuts in the pipeline order (step (iii) before step (iv)), so a duration or path length is only meaningful once the ground phases are gone. Stints shorter than T_{ground} are not cut but *flagged*: a wait with sustained $v_{xy} < v_0$ and a flat barometric trace is marked suspect whatever its length, and the $\psi(\tau)$ fits are re-run with and without the suspect waits as a sensitivity check, reported per discipline alongside the fits.

2.7.4 Flight-level filtering

After cleaning and trimming, some flights are still unfit for the ensemble analysis and are dropped whole. Too aggressive a cut biases the transport statistics — a high minimum duration, for instance, preferentially keeps successful long flights and inflates the apparent MSD exponent at long lags — so filtering uses a *minimal-threshold strategy*: each threshold is set to the smallest value that still clears non-genuine flights (sled runs, tows, aborted or truncated logs), read off the full-archive distribution and retained-fraction curve of its own quantity (Fig. 2.3), on the stretch where that curve is flat — where moving the cut changes almost nothing, the signature of removing junk rather than signal. Five track criteria gate a flight, plus a metadata stratifier; the two that sum over the flight (duration, path) accumulate *within* a segment and never across a boundary, whether left by an earlier stage or an inter-fix gap wider than the resampling's own $g_{\max}$ (Sec. 2.7.6), so a flight is measured over the intervals it will actually be analysed over (implementation details: Sec. I2.7.4).

(1) *Recorded duration* $T \geq 40\,\text{min}$, with an upper sanity bound $T \leq 16\,\text{h}$ against loggers left running past the flight: 15 paraglider "flights" of 16 h to 166 h (none among hang gliders), which would poison the long-lag analyses.

(2) *Flown path length* $L \geq 20\,\text{km}$, catching 236 of 184,583 paraglider and 7 of 6,638 hang-glider flights above the duration floor — order 10^{-3} of either ensemble, duration already clearing tows and sled runs.

(3) *Altitude activity.* The adopted channel's whole-flight range must reach 600 m: a cross-country flight of at least 40 min lives by climbing, so a record spanning less was not soaring. The cut is cheap: moving it from 75 m to 600 m changes the retained share by

under two points (implementation details: Sec. I2.7.4). The cached census holds only the barometric extremes, so on a GNSS-fallback flight this cut, and its diagnostic in Fig. 2.3c,f, is read on barometric flights alone (implementation details: Sec. I2.7.4).

(4) *Plausibility.* The mean ground speed of the retained track, L/T, must not exceed the discipline's own fix-level bound $v_{xy}^{\max}$ — the weakest statement that catches a log corrupt beyond fix-level repair. On the present archive it removes no flight at all, a result rather than a redundancy: an impossible step is now made a segment boundary before it can enter L, so what such a record fails instead is the reach criterion below (implementation details: Sec. I2.7.4).

(5) *Reach.* No fix may lie further from the first than the bound could have carried it in the elapsed time, $\Delta s(\mathrm{fix}_0, \mathrm{fix}_i) \leq v_{xy}^{\max}\, t_i$ for every i — stated per fix, not as the whole-flight extent against the whole duration, since that weaker form would license a fix 100 km out ten seconds into the record on the grounds that the flight went on for three hours. This is what catches a corrupt first fix once its step is no longer summed into L (Sec. 2.7.2): it removes 228 paraglider and 7 hang-glider flights, order 10^{-3} of either ensemble, and is checked again on the written tables as the invariant $|\mathbf{r}(t)| \leq v_{xy}^{\max} t$ the whole of Chapter 3 assumes (Sec. I2.2).

(6) *Glider class* (metadata). Flights whose `aile_class` cannot be normalised (Sec. 2.4) are dropped from class-resolved sub-analyses but kept in the pooled one; stratification by discipline is already possible.

A further criterion, sampling regularity, is applied later, where a flight's time base is examined (Sec. 2.7.6) (implementation details: Sec. I2.7.4).

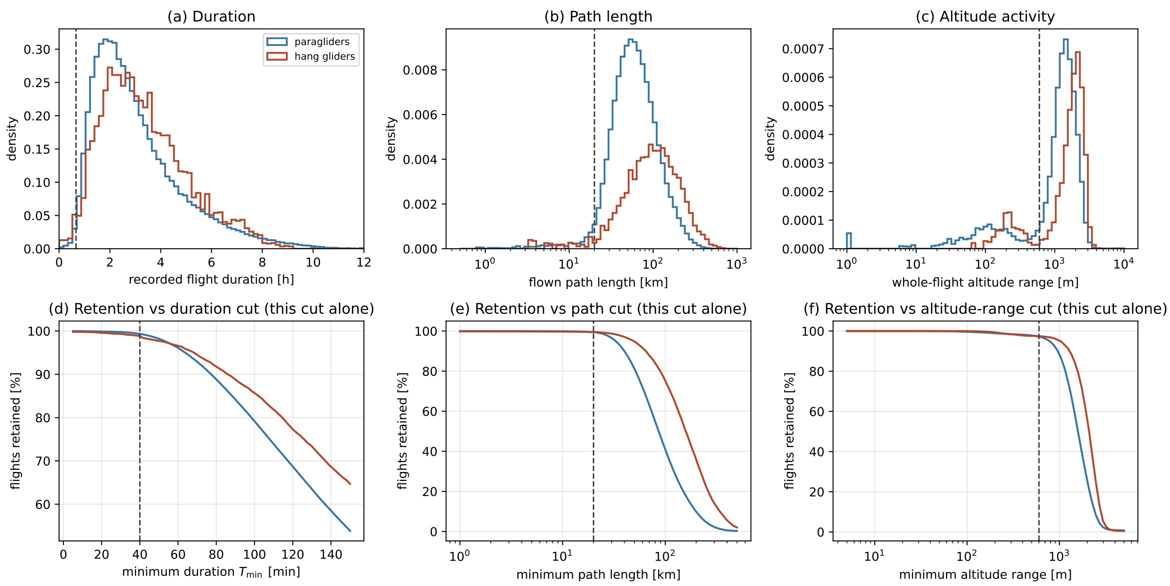

Figure 2.3. Flight-level filtering diagnostics, per discipline. Top: distributions of (a) recorded flight duration, (b) flown path length and (c) whole-flight altitude range. Bottom: the fraction of flights retained versus (d) the duration cut, (e) the path-length cut and (f) the altitude-range cut, each computed for *that cut alone* (marginal, not cascaded). Dashed lines mark the adopted cuts ($T \geq 40$ min, $L \geq 20$ km, range ≥ 600 m). The duration and path cuts sit far out in the tail. Panels (c) and (f) are drawn over the *barometric* flights only: the cached scan stores no GNSS altitude extremes, so a fallback flight would enter them at a range of zero for a reason unrelated to how it flew (Sec. 2.7.4). (implementation details: Sec. I2.7.4)

2.7.5 From geographic to local Cartesian coordinates

The cleaned, trimmed track is next mapped from geographic to local Cartesian coordinates: resampling and smoothing operate on the coordinates *componentwise* (Sec. 2.7), and latitudes and longitudes cannot play that role, being angles on a curved surface whose componentwise differences are not displacements. The mapping proceeds in two steps.

Step 1: geodetic to ECEF. Each fix is converted to Earth-Centred, Earth-Fixed (ECEF) coordinates on the WGS84 ellipsoid (semi-major axis $a = 6{,}378{,}137\,\mathrm{m}$, flattening $f = 1/298.257223563$, $e^2 = f(2-f)$) [17, 18],

$$N(\varphi) = \frac{a}{\sqrt{1 - e^2 \sin^2\varphi}}, \qquad \begin{aligned} X &= (N+h)\cos\varphi\cos\lambda, \\ Y &= (N+h)\cos\varphi\sin\lambda, \\ Z &= (N(1-e^2)+h)\sin\varphi, \end{aligned} \tag{2.6}$$

$N(\varphi)$ the prime-vertical radius of curvature and h the ellipsoidal height along the normal, derived step by step in Appendix 2.B.

Geodetic versus geocentric latitude. The latitude φ is *geodetic*: the angle between the equatorial plane and the ellipsoid *normal* (Figure 2.4), not the geocentric latitude ψ measured to the Earth's centre O, since the normal does not pass through it. The gap $\varphi - \psi \approx (e^2/2)\sin 2\varphi$ (Appendix 2.B) peaks at $\varphi = 45°$ and reaches $\approx 0.19°$, which if confused would misplace a fix by up to $\sim$20 km. No conversion is needed here: WGS84 — hence the GPS receiver and the IGC `B` records (Sec. 2.2) — already reports geodetic latitude, and Eq. (2.6) with $N(\varphi)$ is precisely the transform that consumes it; these are also the (φ_0, λ_0) that orient the local frame of Step 2 (Figure 2.5).

Step 2: ECEF to local ENU. Taking the first fix $(\varphi_0, \lambda_0, h_0)$ of the *trimmed* track as the origin — close to but not the take-off point, since the ground phase has been removed (Sec. 2.7.3) — the ECEF displacement $\mathbf{\Delta X} = \mathbf{X} - \mathbf{X}_0$ is rotated into the local East–North–Up (ENU) frame tangent to the ellipsoid at that point,

$$\begin{pmatrix} E \\ N \\ U \end{pmatrix} = \begin{pmatrix} -\sin\lambda_0 & \cos\lambda_0 & 0 \\ -\sin\varphi_0\cos\lambda_0 & -\sin\varphi_0\sin\lambda_0 & \cos\varphi_0 \\ \cos\varphi_0\cos\lambda_0 & \cos\varphi_0\sin\lambda_0 & \sin\varphi_0 \end{pmatrix} \mathbf{\Delta X}, \tag{2.7}$$

derived in Appendix 2.B: an orthogonal matrix of unit determinant, so lengths and angles are preserved and only the axes change. This ENU frame is the local Cartesian frame the whole analysis is carried out in (E east, N north, U along the local vertical); the ECEF coordinates of Step 1 are only an intermediate representation, no observable is ever expressed in them. Over the spatial extent of a single flight the tangent-plane approximation is accurate to well below the GPS noise: it shortens a distance d along the surface by $\approx d^3/(24R_\oplus^2)$, about 1 cm at $d = 20\,\mathrm{km}$ and 13 cm at 50 km (Appendix 2.B).

The altitude h entering this transform is the adopted channel of Sec. 2.7.1. The horizontal coordinates are insensitive to that choice: h enters the horizontal ECEF components only through the radius factor $(N+h)$, so switching channel, which changes h by at most $\sim 100\,\mathrm{m}$ (Fig. 2.1a), rescales them by a relative $\delta h/(N+h) \lesssim 2\times10^{-5}$ — under a metre across a 50 km flight, below the GPS noise on those coordinates. For the vertical, the analysis does not use the rotation's U component: it keeps the adopted altitude channel itself, $z(t) = h(t)$, at its measured value, the first fix's altitude *not* subtracted. Nothing is gained by re-zeroing: every vertical quantity that matters downstream (the vertical velocity, the climb/sink discrimination of the segmentation) depends on increments, invariant to the reference, while the measured value preserves the absolute height, an observable in its own right (Sec. 2.7.8).

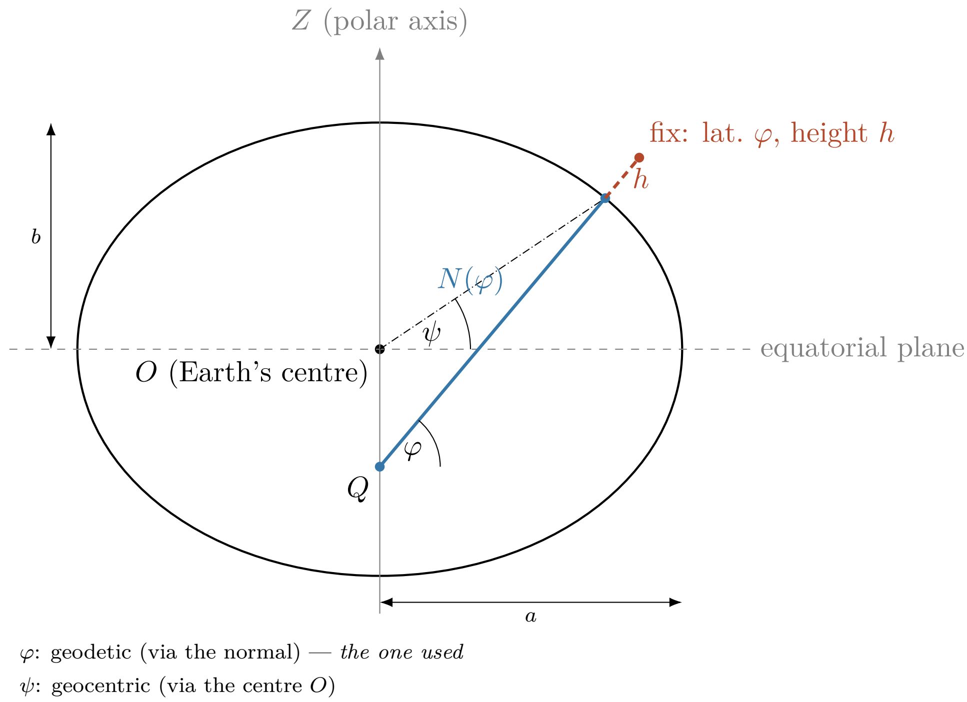

Figure 2.4. Geodetic latitude and height on the reference ellipsoid, in meridian cross-section (flattening exaggerated; the true WGS84 $f \approx 1/298$ would be indistinguishable from a circle): the semi-axes a and b; the geodetic latitude φ, the angle between the ellipsoid normal and the equatorial plane; the height h along that normal; and $N(\varphi)$, the distance along the normal from the surface to the point Q on the polar axis. The normal does not pass through the centre O, so φ differs from the geocentric latitude ψ (dash-dotted). Longitude λ is out of the page here; it is shown in Figure 2.5.

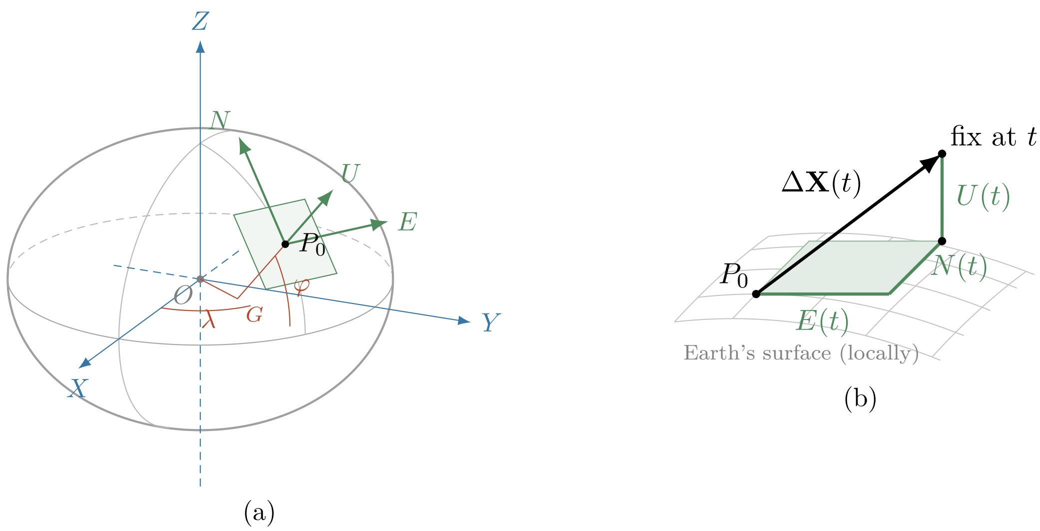

Figure 2.5. The ENU rotation: it turns a vector's global ECEF components into local East, North, Up ones. (a) The global ECEF frame (X, Y, Z) in blue, centred at the Earth's centre O; the two red angles that locate the origin fix P_0 (the first fix of the trimmed track, Sec. 2.7.5): longitude λ, measured in the equatorial plane from the prime meridian, and geodetic latitude φ, the tilt of the ellipsoid normal, measured at the point G where the normal meets the equatorial plane; and the local ENU frame in green, tangent at P_0. Flattening exaggerated as in Figure 2.4: U follows the normal G–P_0, not the radius from O. The green arrows are the rows of the rotation matrix, as unit vectors. (b) Close-up of the tangent plane: a later fix's ECEF displacement $\mathbf{\Delta X}(t)$ is the space diagonal of the box whose edges are its ENU components $E(t)$, $N(t)$, $U(t)$.

2.7.6 Uniform sampling within a flight

The smoothing and differentiation below assume a *uniform* sampling interval, so each surviving flight's time base is examined next. Loggers record at a fixed nominal rate — paragliders mostly at 1 s (69 % of flights), hang gliders spread across several rates from 1 s to 10 s (Fig. 2.7) — but the realised series can carry jitter and drop-outs. Each flight is kept at its *native* rate, since the derivatives are read off the local polynomial the smoothing fits to each flight separately rather than by finite differences across flights (Sec. 2.7.7): what matters is uniformity *within* a flight, not across flights.

Per flight, with Δt the median inter-fix interval, three cases arise. A **uniform** series (all intervals equal to tolerance) is used as is. A **mildly irregular** one — isolated jitter or drop-outs, no gap beyond the split bound $g_{\max}$ and less than 10 % of the grid missing — is resampled onto the uniform grid at its native Δt, interpolating across the small gaps only. One **broken by a long gap** — native, or opened where cleaning removed a frozen-lock run or a position discontinuity, both sides genuine and only the transition unknown — is not bridged, since interpolating across it would fabricate dynamics the smoothing could not tell from real motion: the flight is *split* at the gap into independently analysed segments. The missing-fraction check runs *per segment*, after splitting rather than among the flight-level cuts, since it needs the time base and the splits to exist first (implementation details: Sec. I2.7.6): a segment exceeding 10 % residual missing fraction is dropped alone, and the flight is lost only if no segment survives.

The split bound has two scales. The relative part, $10\,\Delta t$, tolerates isolated dropped fixes in proportion to the cadence but alone would let a slow logger bridge holes approaching the thermalling period, letting an interpolated straight line replace a full circle. The absolute cap of 20 s is pinned by three constraints: at least twice the slowest common cadence, so a single missed fix never splits a flight; at most about two thirds of a thermalling period, so a bridged gap can never hide a full circle; and well below τ_{freeze}, for consistency with the frozen-lock rule. A floor at $2\,\Delta t$ keeps the first constraint true at every cadence, giving $g_{\max} = \min\left(10\,\Delta t,\, \max(20\,\mathrm{s},\, 2\,\Delta t)\right)$: the relative bound sets it at the dominant 1 s rate, the absolute cap from 2 s to 10 s, the floor above that (implementation details: Sec. I2.7.6). On the census this leaves 15.3 % of paraglider and 40.5 % of hang-glider flights with at least one split, against 11.8 % and 14.5 % had the relative part acted alone; since a split censors the phase in progress, preferentially removing long phases from the duration distributions, the cap is treated as a working value and swept in the validation procedure of Sec. 2.7.2 (implementation details: Sec. I2.7.6). The bound is the same for every channel, altitude included: the interpolation error is set by the signal's curvature, not its speed, and a wide altitude gap would erase the climb–glide transitions that v_z, the segmentation's discriminant, exists to resolve (implementation details: Sec. I2.7.6).

Consequences of a split. A split is bookkeeping rather than a new trajectory: the ENU conversion (step v) runs before this step, so every fix already carries coordinates and elapsed time referred to the parent origin, and **a segment keeps its parent's origin and clock**. Ensemble quantities (the MSD, the propagator) need the position at a given elapsed time, not the path in between, so a split costs them nothing; quantities that *accumulate along the path* (path length, phase duration, any time-averaged window) are always summed inside one segment, never across a gap. Three consequences follow: a phase truncated at a boundary is censored and excluded from the duration fits and $\psi(\tau)$ rather than counted as complete (implementation details: Sec. I2.7.6); a segment starting at parent time $t_0 > 0$ is used as an observation window onto a process already t_0 old, not re-zeroed into a fresh flight, since re-zeroing would mix statistics of young and aged trajectories; and the flight-level cuts of

Sec. 2.7.4 are not re-applied to segments, which would bias against gaps concentrated in particular conditions, so a segment need only be long enough to carry the smoothing window, $T_{\mathrm{seg}} = 90\,\mathrm{s}$ (Table 2.7).

Interpolation across short gaps. What is filled is only the short holes, up to $g_{\max}$, inside a segment, per channel. The time base and a channel come apart in one place: $g_{\max}$ bounds the interval between two *fixes*, not an interval over which a fix exists but carries no value — the vertical channel can be absent while the horizontal record is perfect, so no time gap opens and no split is declared, and the linear fill spans the hole whatever its length. Such a hole does not split the flight, since the horizontal trajectory Chapter 3 measures is intact across it; but the reconstruction must be visible, tracked by its own per-fix flag `z_reconstructed` and per-segment share, distinct from `interpolated` (a statement about the time base alone) (implementation details: Sec. I2.7.6).

The horizontal coordinates are interpolated by a monotone piecewise cubic (a shape-preserving Hermite interpolant, chosen so it cannot swing outside the range of the points it passes through, which an unconstrained spline could) and the adopted altitude linearly, an asymmetry that follows from what each channel is used for: the altitude is differentiated once more before it is read, so an interpolation artefact in z propagates into a phase assignment where the same artefact in E or N is absorbed by the smoothing (implementation details: Sec. I2.7.6). The linear error over a hole of width g is bounded by $\max|\ddot{z}|\,g^2/8$ (implementation details: Sec. I2.7.6); holes are bridged inside a phase far more often than across a transition, where the barometric signal has a nearly constant slope, so the error stays at the metre scale even at the 20 s cap.

The result is the completeness invariant promised at cleaning: within a retained segment, every grid time carries a defined (E, N, z). Every filled point carries a flag — `interpolated` where the time base had no fix, `z_reconstructed` where a fix was there but its altitude was not — so downstream statistics can distinguish measured from reconstructed samples, and a phase boundary the segmentation places inside a bridged stretch is censored exactly as at a split. The gap bound $g_{\max}$ is the sampling-regularity criterion of the flight-level filtering (Sec. 2.7.4); Figure 2.6 makes the fraction of flights it leaves whole auditable. One caveat carries downstream: a few observables, the turning angle above all, are intrinsically Δt-dependent by definition, so where such observables are compared across flights, the ensemble is stratified by rate or restricted to the dominant one.

2.7.7 Smoothing and differentiation

Velocity and acceleration drive the kinematics and the segmentation, but differencing raw GPS coordinate series amplifies the high-frequency noise of Sec. 2.7.1. A **Savitzky–Golay** filter [19] solves both problems at once: it fits a low-order polynomial to a sliding window by least squares and reads off that polynomial, or its derivatives, at the window centre, returning a smoothed value, velocity and acceleration in one pass per component.[1] The fit is done in units of samples, so velocity and acceleration in physical units are $a_1/\Delta t$ and $2a_2/\Delta t^2$ with the flight's own Δt — the only place the cadence enters, and why flights of different cadences need no common grid. The window never crosses a segment boundary; near one it is fitted on the nearest full window inside the segment and evaluated off-centre, carrying higher variance, flagged the same way as an interpolated point (implementation details: Sec. I2.7.7).

The two hyperparameters, window length w (odd, in samples) and polynomial order p, are fixed from the measured noise spectrum rather than by eye. From the ensemble Welch

[1] In the window of w samples centred on t_i, the coefficients of $P(t) = \sum_{k=0}^{p} a_k\,(t - t_i)^k$ minimize $\sum_j \big(x_j - P(t_j)\big)^2$; the smoothed value is a_0, the velocity a_1, the acceleration $2a_2$, and since the grid is uniform the whole filter is a convolution with precomputed coefficients [19].

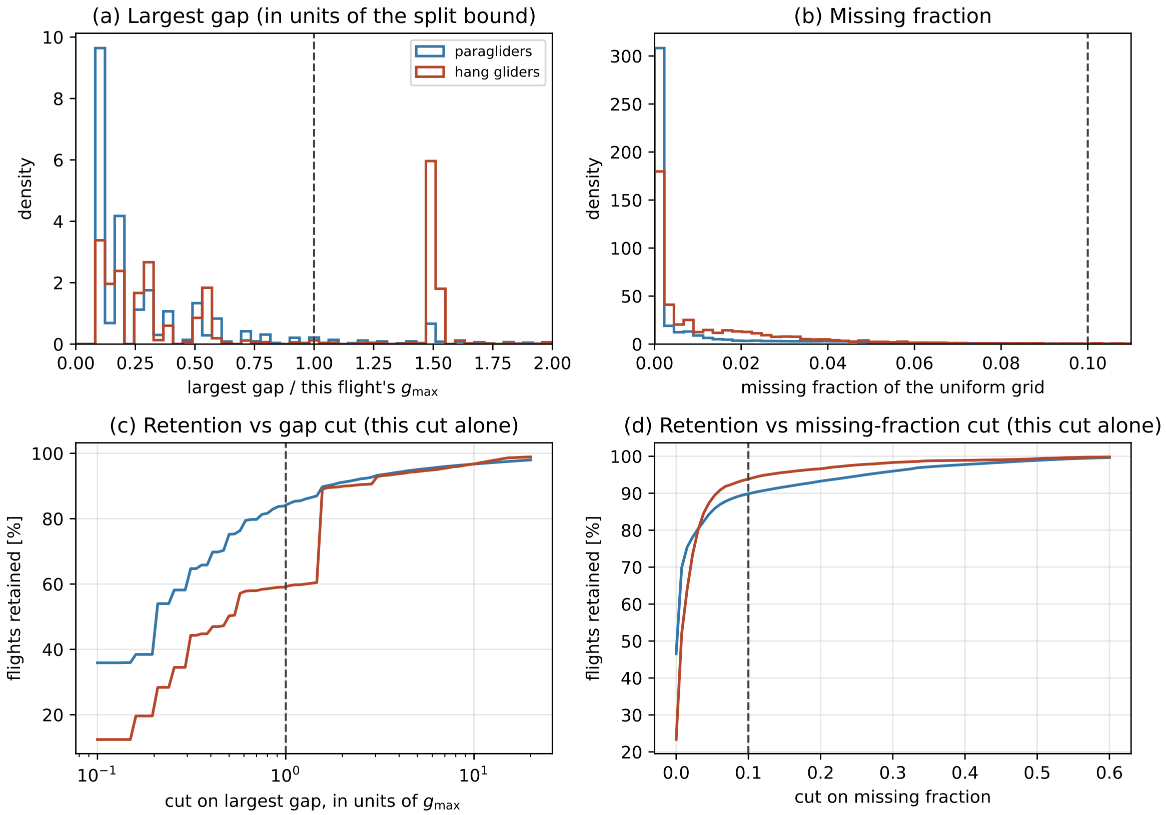

Figure 2.6. The two sampling-regularity criteria and their retention curves, per discipline. Computed on the same full track scan as Figure 2.3—every parsed flight, not only those passing the flight-level cuts, because a cut must be audited on the population it acts on, before its own effect hides part of the picture—what would hide it is the flight-level cleaning applied first, not this cut. (a)/(b) Distributions of the two tested quantities: the largest single gap of the flight and the missing fraction of a uniform grid at its native interval. The gap is plotted in units of *that flight's own* $g_{\max}$, not of Δt, so the criterion actually in force sits at 1 (dashed) for every flight; because $g_{\max}$ is the two-scale bound and not a fixed multiple of Δt, a plot against Δt alone would draw a cut that binds at no cadence but 1 s. The distribution is discrete below the cut, since a gap is usually a run of wholly missed fixes and hence an exact multiple of the native interval. (c)/(d) The fraction of flights each cut alone would retain as the threshold is moved: *marginal* curves, i.e. each cut applied by itself to the full population, not after the other cuts. Both panels are recomputed once the fix-level and flight-level stages run in full; the census behind them predates that pipeline. (implementation details: Sec. I2.7.6)

spectrum of the raw ENU components, the frequency f_c of the knee where the dynamical signal meets the flat GPS noise floor gives a smoothing timescale $\tau_c = 1/f_c$; w is $\tau_c/\Delta t$ rounded up to the nearest odd integer, floored at $w = 5$ (the smallest window a cubic fit meaningfully determines). At the dominant 1 s cadence the two rules coincide; at every slower cadence the floor binds and the window spans more than τ_c, a cost accepted since those cadences carry little noise to remove at the noise scale in the first place and are excluded from Δt-sensitive observables by the rate stratification (Sec. 2.7.6). The order $p = 3$ is the lowest that works: an acceleration needs curvature ($p \geq 2$), and one order more is needed for the velocity, since on a symmetric window an even-order term leaves the odd-order coefficients untouched — the velocity of a $p = 2$ fit is identical to a straight-line fit's, and $p = 3$ is the first order that responds to the acceleration varying across the window (implementation details: Sec. I2.7.7). The vertical is treated on the same footing as the horizontal: the three channels' knees coincide at $f_c \approx 0.2$ Hz ($\tau_c = 5$ s), since the flat high-frequency floor of every channel comes from the IGC format's own rounding rather than from receiver noise, so what distinguishes the

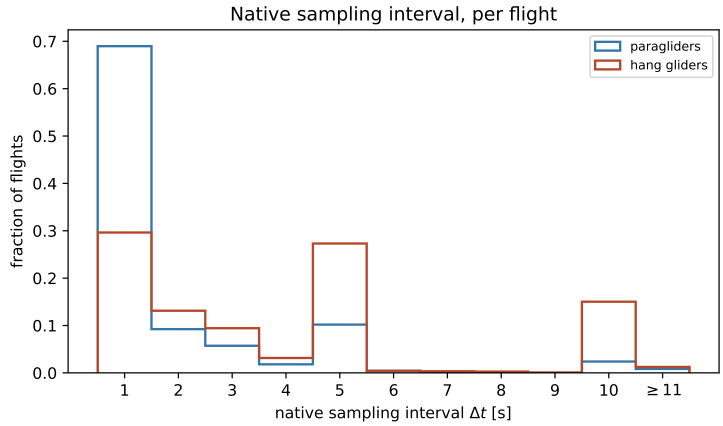

Figure 2.7. Native sampling interval Δt (median inter-fix time) per flight, on the full dataset per discipline. Δt is in practice discrete – it lands on an exact whole second for nearly every flight (Sec. I2.7.6) – so bins are one second wide, centred on each integer; each bar is the fraction of flights at exactly that rate. Paragliders concentrate at 1 s; hang gliders spread across several rates, with a thin tail beyond 11 s (last bin, aggregated). A thin tail is present in both disciplines; what differs is the bulk, with the 10 s rate far commoner among hang gliders than among paragliders. Each flight keeps its own Δt (Sec. 2.7.7). (implementation details: Sec. I2.7.6)

noisy GNSS minority is how high its floor sits, not where the knee is (implementation details: Sec. I2.7.7).

The choice is checked on a held-out sample against three criteria: the residual spectrum (raw minus smoothed) flat above f_c; the smoothed kinematics within the physical bounds of Table 2.7; and stability when w is changed by one step (implementation details: Sec. I2.7.7).

A fourth criterion, on the observable itself. A fourth criterion is a property of what the filter is for: the smoothing must not remove the displacement the transport analysis is about to measure. The window is floored at 5 samples, binding for every logger slower than 1 Hz, so in seconds the window is $\max(\tau_c, 5\,\Delta t)$: 5 s at a 1 s cadence, 50 s at 10 s, reaching the lower end of the range Chapter 3 fits its exponent over. Measured on synthetic trajectories of known exponent ($\alpha = 1.8$, the regime Chapter 3 finds) and no measurement noise, the filtered MSD sits within 2.3 % of the true one at a lag of two samples and within 1 % from three samples upward: a cubic filter reproduces any locally cubic trajectory exactly, and a strongly correlated trajectory is locally close to cubic. What does move the short-lag MSD is measurement noise, in the opposite direction: independent noise of standard deviation σ adds $\approx 2\sigma^2$ at every lag, and with $\sigma = 3$ m the measured-to-true ratio is 8.1 at one sample, falling to 1.05 at twelve and 1.003 at sixty-four — smoothing de-biases rather than erases it, and the floor it leaves, $2\sigma^2 \approx 18\,\mathrm{m}^2$, sits orders of magnitude below the measured MSD everywhere in the fitted range (implementation details: Sec. I2.7.7). The consequence is that the first decade of Fig. 3.1 is instrument, not transport, and the fit does not begin there; a second, related effect is that a flight answers a lag with its own nearest fix, so a logger slower than 1 Hz shows a short-lag staircase — both effects die out well below the fit range.

2.7.8 Notation

Throughout:

- **Horizontal position.** $\mathbf{r}(t) = (x(t), y(t)) = (E(t), N(t))$, in this local frame. By construction each trajectory starts at the horizontal origin, $\mathbf{r}(0) = \mathbf{0}$.

- **Vertical position.** $z(t)$, the adopted altitude channel (Sec. 2.7.1). Unlike $\mathbf{r}$, it is *not* re-zeroed: it is kept at its measured value, so the height at the start of free flight is retained (Sec. 2.7.5).
- **Time.** t, the elapsed flight time, with the clock reset to $t = 0$ at the start of free flight (the first fix of the trimmed track, Sec. 2.7.3). A segment created by a split at a long gap keeps its parent flight's origin and clock (Sec. 2.7.6).
- **Velocity.** $\mathbf{v}(t) = \dot{\mathbf{r}}(t)$, the horizontal velocity, with magnitude the horizontal speed $v_{xy} = |\mathbf{v}|$; $v_z = \dot{z}$, the vertical velocity.
- **Displacement.** $\Delta\mathbf{r}(t) = \mathbf{r}(t_0 + t) - \mathbf{r}(t_0)$, the horizontal displacement over a lag t.

A terminological rule holds throughout the document: a bare "position" means $\mathbf{r}$, the horizontal one; a statement about the full three-dimensional point says *3D position* explicitly.

2.8 Preliminary characterization

The point of this section is to establish the basic statistics of the retained ensemble. None of the checks below requires the phase segmentation, so all run as soon as the pipeline does.

Trajectory statistics. For each retained flight: airborne duration T, total horizontal path length, number of valid fixes and native sampling interval Δt. Their distributions (Fig. 2.8) describe the dataset the results rest on, expose any outlier that survived the filtering, and, through Δt, fix the per-flight smoothing window (Sec. 2.7.7). Over the 155,788 retained paraglider flights the median airborne duration is 2.63 h (interquartile range 1.78 h to 3.93 h), the median flown path 87 km (57 km to 134 km) and the median record 7,129 fixes; 69.8 % of them log at 1 Hz. The 6,132 hang-glider flights are longer and faster—3.07 h and 151 km in the median—and slower-logging, with a median Δt of 3 s against 1 s and only 27.6 % at 1 Hz.

Nothing in the retained ensemble is an outlier in the sense the filtering was meant to catch: the on-disk tables of the retained flights carry 0 non-finite coordinates and 0 non-monotone clocks, and the fastest step between two *measured* samples is $49\,\mathrm{m\,s^{-1}}$ against the fix-level bound of $45\,\mathrm{m\,s^{-1}}$—an excess of under a tenth, on 32 steps out of 1.4×10^9, which Sec. I3.1 accounts for.

One feature of panels (a) and (b) is worth reading correctly. Both are measured over what a flight keeps after resampling, not over the trimmed flight the cuts of Sec. 2.7.4 were applied to. The two stop agreeing where the sparsity rule of Sec. 2.7.6 takes most of a record: the flight is retained on its surviving segments, its duration and path become theirs alone, and it can land below cuts it did pass. That is the thin left tail below 40 min and 20 km, 535 paraglider flights (0.34 %) and 9 hang-glider ones. On the quantities the cuts did test, no retained flight is under either.

What the cascade costs, and whether it costs it evenly. Table 2.6 reads the discards as a cascade: end to end it leaves 83.7 % of attempted paraglider flights standing against 91.3 % of hang-glider ones.

For paragliders one criterion carries the cost, the resampling sparsity rule: 21,089 flights, 11.9 % of the remaining flights yield no valid points post-resampling, which is 4.4 times the impact of the next filter, namely altitude activity below the floor (2.7 %). Unlike the cuts of Sec. 2.7.4, set by the minimal-threshold strategy against the retained-fraction curve, the missing-fraction and minimum-duration gates of Sec. 2.7.6 are config defaults, not swept or derived, and the rule falls unevenly by native cadence: from 0.1 % to 22.6 % across the six most common ones. Not by season: Fig. 2.9b plots the cascade's total retention rate, every

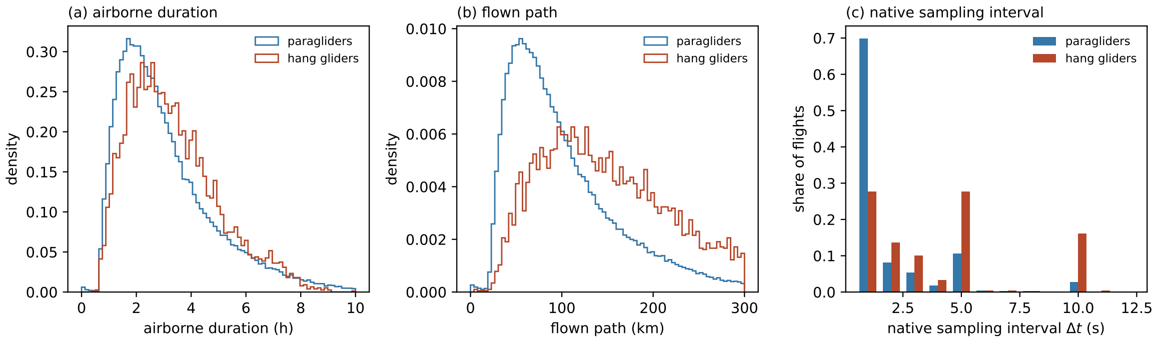

Figure 2.8. The records the pipeline retains. (a) Airborne duration, (b) flown path and (c) native sampling interval Δt, per discipline, over the retained flights rather than over every parsed flight—which is what distinguishes these from Figs. 2.6 and 2.7. Duration and path are measured over the segments a flight keeps after resampling (Sec. 2.7.6), not over the trimmed flight the cuts of Sec. 2.7.4 tested, which is why a thin tail falls below those cuts.

criterion combined, but since the sparsity rule accounts for most of the paraglider loss its unevenness would show up there; it does not, staying between 70.3 % and 91.6 % over the 20 seasons carrying at least a thousand flights.

For hang gliders the same criterion is again the largest, 3.9 % of those standing, but only 1.7 times the next cut, altitude activity again (2.3 %), against 4.4 for paragliders: no single criterion dominates the hang-glider cascade the way the sparsity rule dominates the paraglider one.

Where the flights are. The distribution of take-off sites is presented across three separate panels to account for the geographic coverage of the CFD: Figure 2.10 displays these sites against coastline and national borders.

- **Metropolitan France.** 94.7 % of retained paraglider flights launch here, against 96.7 % of hang-glider ones — the hang gliders are slightly more domestic.
- **La Réunion.** 3.2 % (4,939 flights), the one overseas department with a substantial share.
- **Elsewhere.** 2.2 % (3,359 flights): the Canaries, Andalusia, the Italian Alps, and further out Colombia, Nepal, India, South Africa and Brazil.

Within metropolitan France the ensemble does not sample the country evenly. It occupies 5,332 distinct take-off sites, where distinct sites are launches rounded to a hundredth of a degree (∼1 km); by share of paraglider launches:

- **Alps.** 58.8 %.
- **Pyrenees.** 7.8 %.
- **Massif Central.** 8.2 %.
- **Elsewhere in France.** 19.8 %.

The orographic groups are longitude and latitude boxes, drawn on Fig. 2.10a and listed with their edges in the mirror of this section (implementation details: Sec. I2.8).

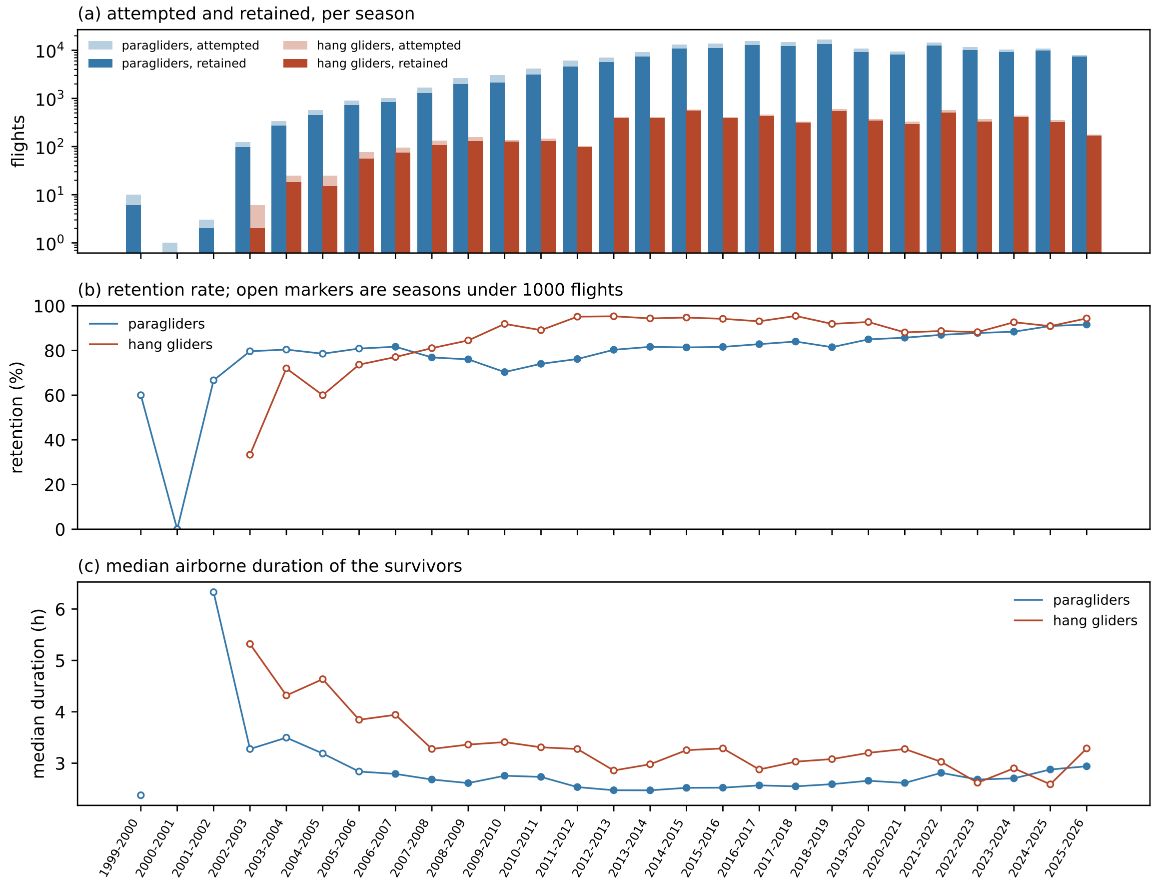

Figure 2.9. The archive season by season. (a) Flights attempted and retained, on a logarithmic scale: the ladder grows by three orders of magnitude over its life. (b) Retention rate and (c) median airborne duration of the survivors; open markers are seasons of fewer than 1,000 flights. Retention is approximately flat across the substantial seasons, which is what says the cascade does not discard one era of the archive preferentially.

Isotropy The transport analysis would like to study the one-dimensional marginal of the propagator in place of the full two-dimensional density (Sec. 3.1), a substantial simplification. The reduction is legitimate only if the process is isotropic, and isotropy cannot be assumed a priori here. It's important to highlight that a single flight could be strongly directed: a chosen route, the orography, the valley winds. Isotropy can therefore hold at best at the level of the *ensemble*, and the check is run there.

The primary test is the per-component variance ratio: for an isotropic process $\langle E(t)^2\rangle = \langle N(t)^2\rangle = \mathrm{MSD}(t)/2$ at every t, so the ratio $\langle E(t)^2\rangle/\langle N(t)^2\rangle$ should be identically 1. The check is necessary, not sufficient: isotropy proper asks whether the conditional distributions $P(E(t) = x \mid \mathbf{r}_0, t_0)$ and $P(N(t) = x \mid \mathbf{r}_0, t_0)$ (here $\mathbf{r}_0 = \mathbf{0}$ and $t_0 = 0$, take-off) coincide, of which equal variances are only one necessary consequence. Operationally the ratio screens whether that stronger comparison is worth pursuing at all: if the 10–90% band intersects unity over a zone of lags, the two distributions could still coincide there; where it does not, the weaker condition already fails and isotropy is ruled out without needing the fuller comparison. Figure 2.11 shows it is not. The curve plotted is the median of a cluster bootstrap resampling (200 times) whole take-off-site-and-day groups, with a 10–90% band. Across all evaluated lags, the transport analysis shows that the paraglider band remains bounded between 0.70 and 0.86, maintaining an approximately constant offset from unity without ever intersecting the unity line. The hang gliders oscillate mildly instead: unlike the paragliders the ratio is not steady,

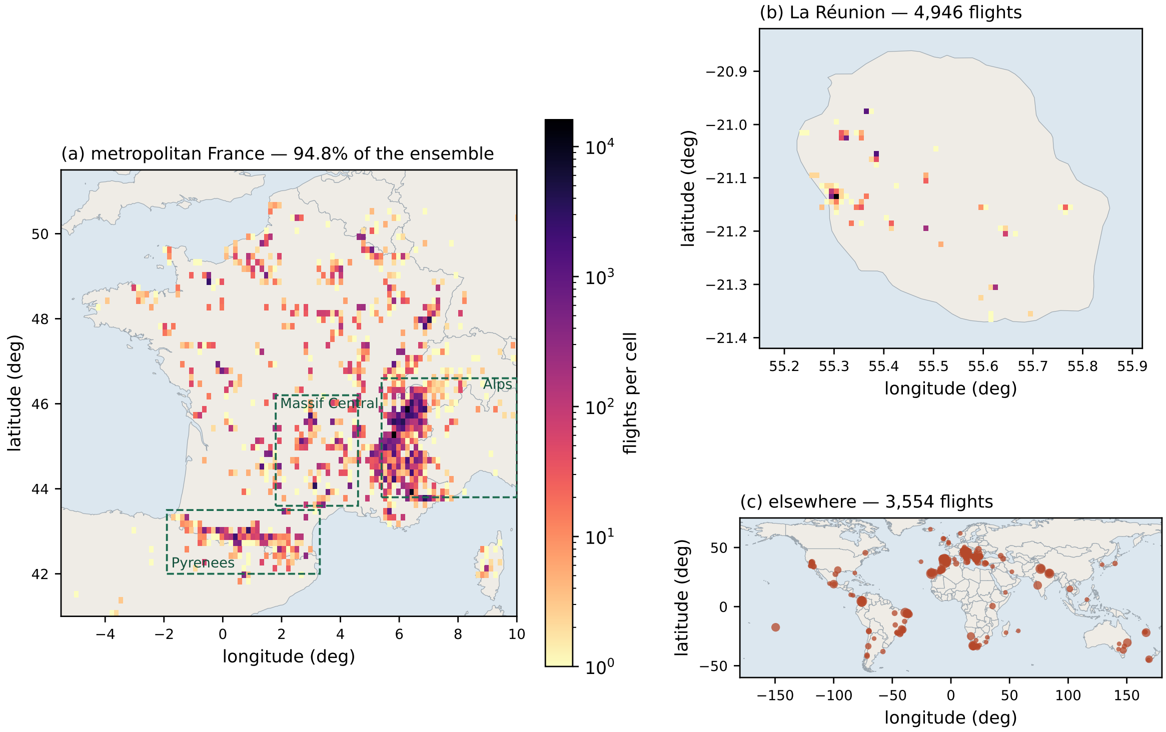

Figure 2.10. Take-off sites of the retained ensemble, on three frames, pooling both disciplines throughout. (a) Metropolitan France, density on a 0.15° cell; the dashed boxes are the orographic groups used as strata in Fig. 3.3 (Sec. 3.3). (b) La Réunion, on a 0.01° cell, the one overseas department with a substantial share. (c) Everything else, one marker per 0.5° cell with the area set by its count.

swinging between values below unity and above it, from 0.55 to 1.43.

The consequence is operative. The process cannot be treated as isotropic, so the two marginals $P(E(t) = x \mid \mathbf{r}_0, t_0)$ and $P(N(t) = x \mid \mathbf{r}_0, t_0)$ are analysed *separately* rather than merged into a single one-dimensional sample and read as a stand-in for the 2-D propagator. Both directions are studied on their own, alongside the modulus $|\mathbf{r}(t)| = \sqrt{E(t)^2 + N(t)^2}$.

2.9 Pipeline summary

Figure 2.12 closes the chapter with the whole processing chain in one map: every step the data undergo between the raw IGC tracks of Sec. 2.2 and the ensemble handed to the transport analysis, in the order the pipeline applies them. Each stage carries the level it acts at and its effect, and points back to the section where that choice is argued.

Two decisions organise the chain.

- **The order.** Every check on the scalar record as logged (times, geodetic coordinates, altitude) comes first (stages 1–4), and the vector geometry is built only from what survives (stages 5–7). Converting coordinates does not clean a record: it recomputes each fix in a new frame, corrupt or not. Running it before the cleaning stages would therefore buy nothing, and would cost rework whenever trimming moved the fix the frame is anchored to, or filtering dropped the record outright.

- **The level.** Each stage acts per fix, per flight or per segment, and every parameter it adapts is read off the record it is acting on rather than off the archive around it. No flight's cleaning therefore depends on which other flights happen to be in the dataset. What is shared across flights is only the thresholds of Table 2.7, fixed a priori.

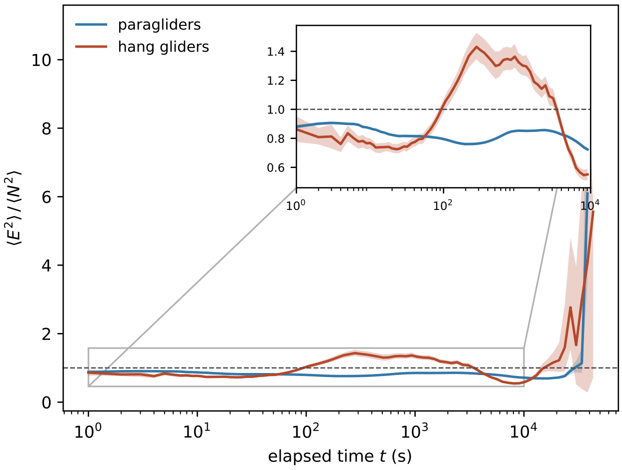

Figure 2.11. The per-component variance ratio $\langle E^2 \rangle / \langle N^2 \rangle$ against elapsed time, median and 10–90% band over a cluster bootstrap resampling take-off site and day (200 resamples): unity is isotropy, and across the decades of lag explored neither band sustains a zone of overlap with unity. Where the two meet at all, it is at isolated points, never over a range of lags.

What the chain retains. Run over the whole archive, the cascade retains 155,788 of 186,052 paraglider flights (83.7 %) and 6,132 of 6,716 hang-glider flights (91.3 %). Table 2.6 says where the rest went. The numbers are a *cascade*, not a set of marginal effects: a flight counted against one criterion was still standing when that criterion tested it, so the table reads in pipeline order and the counts do not add up to what each cut would remove on its own (those are the `\StatScan*` numbers quoted in Secs. 2.7.4 and 2.7.6, measured on the raw population).

Inside the retained flights the cleaning is light, which is what the design asked for: it deletes 1.84 % of paraglider fixes and 0.75 % of hang-glider ones, and invalidates an altitude on a further 0.016 % and 0.009 %. The local outlier test flags far more than is deleted, and by design: it never gates, so 0.68 % and 0.97 % of fixes are flagged and *kept*. The flag is a per-fix record and a per-flight anomaly count, and what condemns a fix is the impossibility gate (Sec. 2.7.2). Resampling reconstructs 0.58 % and 0.32 % of grid points. A split is common but not the rule: 6.9 % of retained paraglider flights and 16.3 % of hang-glider ones carry at least one, the difference following the cadences of Fig. 2.7 as Sec. 2.7.6 anticipated.

Table 2.7 gathers the thresholds themselves, by stage and with the symbols the chapter uses.

Figure 2.12. The processing chain, from the raw archive to the analysis ensemble. Each stage carries its effect and the section that argues the choice; the badge to the right is the level it acts at (per fix, per flight, per segment). Every check on the scalar record (stages 1–4) precedes the construction of vector quantities (stages 5–7), so no coordinate conversion is computed for a record about to be repaired or dropped. The thresholds each stage applies are listed, with their symbols, in Table 2.7.

Table 2.6. The pre-processing cascade over the full archive, in the order the pipeline applies it. *Standing* is how many flights had survived every earlier criterion when this one ran and *share* is the removal as a fraction of those, not of the archive: a criterion late in the cascade tests a population the earlier ones have already thinned, so a percentage of the archive would understate it. Read this way one cut dominates, and it is not one the text argues a threshold for—21,089 paraglider flights, 11.9 % of those still standing (249 hang-glider flights, 3.9 %), leave nothing behind after resampling. The marginal effect of each cut on the raw population is a different quantity and is what the `\StatScan*` numbers of Secs. 2.7.4 and 2.7.6 report. Generated by `generate_dataset_stats.py` from the `flights_meta` table the pipeline itself writes.

	paragliders			
Why the flight was dropped	**standing**	**removed**	**share (%)**	**remaining**
unreadable track	186,052	27	0.01	186,025
never airborne	186,025	225	0.12	185,800
integrity gate	185,800	1,442	0.78	184,358
shorter than the duration cut	184,358	2,107	1.14	182,251
longer than the duration cut	182,251	3	0.00	182,248
path below the floor	182,248	182	0.10	182,066
altitude activity below the floor	182,066	4,960	2.72	177,106
out of reach of its own first fix	177,106	228	0.13	176,878
no segment survived resampling	176,878	21,089	11.92	155,789
no segment carries the smoothing window	155,789	1	0.00	155,788
retained				**155,788**

	hang gliders			
Why the flight was dropped	**standing**	**removed**	**share (%)**	**remaining**
never airborne	6,716	15	0.22	6,701
integrity gate	6,701	73	1.09	6,628
shorter than the duration cut	6,628	86	1.30	6,542
path below the floor	6,542	1	0.02	6,541
altitude activity below the floor	6,541	153	2.34	6,388
out of reach of its own first fix	6,388	7	0.11	6,381
no segment survived resampling	6,381	249	3.90	6,132
retained				**6,132**

Table 2.7. Working values of the pre-processing pipeline, by stage. Symbols are those used in the text; every value is generated from `configs/preprocessing.yaml`, and the per-parameter rationale is given in the section listed. Config keys and implementation notes: Sec. I2.7.2 onward.

Symbol	Value	What it controls
Fix-level cleaning (Sec. 2.7.2) — removes corrupt fixes, keeps every genuine one		
k	5	robust-σ multiplier of the Hampel test, Eq. (2.3)
w	20 s	half-width of the Hampel window
$\varepsilon_{\min}$	15 m	absolute floor of the Hampel test: no flag below this residual, whatever the local scale
—	5 fixes	smallest window population for which the test is evaluated
$v_{xy}^{\max}$	$45\,\mathrm{m\,s^{-1}}$ (para), $55\,\mathrm{m\,s^{-1}}$ (hang)	absolute inter-fix speed bound; the impossibility gate, on its own (Sec. 2.7.2)
$\lvert v_z\rvert^{\max}$	$13\,\mathrm{m\,s^{-1}}$	absolute climb/sink bound
$[h_{\min}, h_{\max}]$	$[-100, 6000]$ m	sensor-plausibility window on the altitude
δ_{xy}	15 m	diameter within which a frozen-lock run must stay
δ_z	5 m	altitude flatness tolerance of the same rule
τ_{freeze}	60 s	minimum duration of a run before it counts as frozen
—	95 %	smallest share of a flight's fixes carrying a non-zero pressure altitude for the barometric channel to count as present (Sec. 2.7.1); below it the flight falls back to GNSS
—	30 m	smallest whole-flight barometric range below which the channel is treated as absent (Sec. 2.7.1)
—	10 %	share of fixes above which the cleaning is declared to have failed and the flight is dropped
Ground trimming (Sec. 2.7.3) — keeps only the airborne stretch		
v_0	$5\,\mathrm{m\,s^{-1}}$	ground-speed threshold separating ground from flight
T_0	30 s	how long $v_{xy} > v_0$ must persist for take-off (and, time-reversed, landing)
T_{ground}	10 min	minimum duration of an interior ground stint before it is excised
—	5 m	altitude flatness tolerance over such a stint
Flight-level filtering (Sec. 2.7.4) — discards whole flights that cannot support the statistics		
—	40 min	minimum airborne duration
—	20 km	minimum flown path length
—	16 h	maximum duration, above which the record is not a single flight
—	600 m	minimum whole-flight altitude range
—	$v_{xy}^{\max}$ (above)	plausibility bound on the mean ground speed of the cleaned track; reuses the fix-level bound, read over a whole flight (Sec. 2.7.4)
—	$v_{xy}^{\max}$ (above)	reach bound: the extent of the flight may not exceed what that same speed would cover in its duration (Sec. 2.7.4)
Resampling (Sec. 2.7.6) — one uniform grid per flight, holes bridged or split		

continued on the next page

Table 2.7 (continued)

Symbol	**Value**	**What it controls**
$g_{\max}$	$\min\big(10\,\Delta t,\ \max(20\,\mathrm{s},\ 2\Delta t)\big)$	longest gap that is interpolated instead of splitting the flight
—	10 %	largest share of grid points a retained *segment* may have had to interpolate
—	90 s	shortest segment kept: below this a segment cannot carry the smoothing window
Smoothing (Sec. 2.7.7) — Savitzky–Golay filter and its derivatives		
p	3	polynomial order of the filter
τ_c	5 s	correlation time setting the window, on all three channels. Per-channel keys exist—E/N, and the altitude split by `alt_source`—for the noisy GNSS minority, and all currently hold this same value, because the spectral knees of the three coincide (Sec. 2.7.7)
—	5 samples	floor on the window, one sample more than a cubic determines. It is this floor and not τ_c that fixes the smoothing scale for every logger slower than 1 Hz: in seconds the window is $\max(\tau_c,\ 5\,\Delta t)$

The analysis ensemble, defined. Everything downstream reads one object, and this is it. The *analysis ensemble* is the set of trajectories written to `fixes.parquet`: one row per grid point of every *retained* segment of every *retained* flight, carrying the elapsed time t in the parent clock and the local Cartesian coordinates (E, N, z) with their first two derivatives. A flight is retained when it passes stages 1–4 of Figure 2.12 and at least one of its segments survives stage 6; a segment is retained when it is long enough and dense enough on its own uniform grid. That leaves 155,788 paraglider flights of 186,052 (83.7 %) in 179,395 segments, and 6,132 hang-glider flights of 6,716 (91.3 %) in 6,951. No consumer re-applies a retention rule: a flight or segment that should not be analysed has no rows to read. Every segment keeps its parent's origin and clock, so a position is always measured from that flight's take-off and a time is always elapsed since it, whether or not the record is continuous up to that point. Section 2.8 characterises this ensemble, Chapter 3 analyses it, and `docs/guide/data-on-disk.md` states which columns each estimator reads and what a segment boundary means to it.

Chapter 3

Global transport

This chapter measures the transport of the retained ensemble without segmenting a single flight. What can be established that way is a *class* of transport: how far a wing gets in a given time, how that growth is distributed, how long the motion remembers its direction, and how many distinct regimes the answer needs. What cannot be established is which flight behaviour produces it, and that is Chapter 4.

The measurements are made on the analysis ensemble assembled in Chapter 2: 155,788 paraglider and 6,132 hang-glider flights, holding 179,395 and 6,951 retained segments. Every statistic below is computed within a retained segment and never across the gap that ends one (Sec. 2.7.6) (implementation details: Sec. I3).

One constraint from Sec. 2.8 holds throughout: the horizontal components are anisotropic — the ratio of their mean squares sits at 0.80 for paragliders and 1.19 for hang gliders, never touching unity — so the two marginals are measured separately and a pooled one-dimensional marginal never substitutes for the two-dimensional propagator. The discipline is also heterogeneous (Sec. 3.3): read against the pooled ensemble MSD, the typical wing class departs by 51 % and the typical orographic group by 62 %, against 12 % across seasons. Wing class and terrain are relevant strata and season is not, so every exponent below is reported by stratum as well as pooled, and the two disciplines are never pooled into each other.

Over 60 s to 2,000 s, on an estimator that annihilates any polynomial trend in the trajectory, cross-country soaring gives $\alpha_2 = 2.02 \pm 0.10$ for paragliders and 1.94 ± 0.13 for hang gliders. It is not the Lévy walk this thesis was framed around: the spectrum of moments is straight, $\nu(q)$ moving by 0.019 and 0.009 across the range of q read. It is not Gaussian either: the non-Gaussian parameter is +0.042 for paragliders and +0.181 for hang gliders in the median, rising to an interior maximum near 266 s, so one exponent governing every moment states how the increment distribution scales and nothing about its shape.

3.1 Displacement from take-off measures the launch geometry

Every flight starts at the origin, $\mathbf{r}(0) = \mathbf{0}$, so the position after an elapsed time t and the displacement from take-off are the same vector. Its mean square admits two averages that answer different questions. The ensemble average

$$\mathrm{MSD}(t) = \langle\, |\mathbf{r}(t)|^2 \rangle, \tag{3.1}$$

taken over flights at a fixed elapsed time, fixes the transport regime through $\mathrm{MSD}(t) \sim t^{\alpha}$ — $\alpha = 1$ Brownian, $\alpha > 1$ super-diffusive, $\alpha < 1$ sub-diffusive, $H = \alpha/2$ — and is the observable the anomalous-transport question is normally posed with. The time average

$$\overline{\delta^2(\tau)} = \langle\, |\mathbf{r}(t_0+\tau) - \mathbf{r}(t_0)|^2 \rangle_{t_0}, \tag{3.2}$$

taken over starting times t_0 inside one record, answers how far a wing travels in a window of duration τ placed anywhere in its flight. The ensemble average is synchronised at take-off, so a feature it carries at a fixed elapsed time is a property of that common origin; the time average slides its window with no privileged origin, so a feature the first shows and the second does not belongs to the origin rather than the motion.

Both are computed on the analysis ensemble of Sec. 2.8 (155,788 paraglider, 6,132 hang-glider flights, 179,395 and 6,951 retained segments), within a retained segment and never across the gap that ends one (Sec. 2.7.6), with a flight answering a lag only when it holds a fix within half its own native interval of it (implementation details: Sec. I3). Figure 3.1 shows both estimators.

The ensemble curve is not a power law. Over the range each estimator is fitted on, the ensemble curve departs from its own fitted line by 15.2 % rms for paragliders and 31.2 % for hang gliders, against 3.8 % and 3.6 % for the time average. That residual exceeds the standard error on each point by 51 and 22 on the two ensemble curves, against 28 and 5 on the time averages: a departure exceeding the error on each point many times over is a shape, not scatter about a power law (implementation details: Sec. I3).

The shape is an arch above the ballistic value. The ensemble local slope falls to a minimum of 1.31 near 137 s, rises to 2.21 near 1,504 s, and settles back towards 2; hang gliders do the same between 1.10 and 2.52. The maximum lies *above* the ballistic value, which no sum of power laws with positive coefficients and exponents at or below 2 can produce: for $f = \sum_i a_i t^{\alpha_i}$ with $a_i > 0$,

$$\frac{\mathrm{d}\log f}{\mathrm{d}\log t} = \frac{\sum_i a_i \alpha_i t^{\alpha_i}}{\sum_i a_i t^{\alpha_i}} \tag{3.3}$$

is a weighted mean of the α_i and cannot exceed the largest of them, so a slope above 2 requires the ballistic component to switch on at a finite time rather than to be present from $t = 0$. The arch is larger than the estimator's own uncertainty (implementation details: Sec. I3).

Three controls rule out an artefact. A changing population, survivorship, and a shared heading could each produce the same arch without it belonging to the motion. Splitting by native cadence removes any drift from a growing population: across the 3 paraglider classes the local slope spreads by only 0.06 against a swing of 0.91. Fixed-duration cohorts remove survivorship: the local slope spreads across cohorts by 0.11 and 0.21, where a synthetic ensemble with only the *speed* correlated to duration reproduces neither feature (implementation details: Sec. I3). And a shared heading is bounded by the part of the mean square the mean vector carries, which never exceeds 1.8 % for paragliders and 10.0 % for hang gliders. The flights do not share a heading; they share a starting point.

The origin of the crossover. A wing does not leave its launch site immediately: it climbs out, often circling close to it, and only then commits to a course. Until it does, its distance from take-off is bounded by the local soaring area however fast it is flying, which depresses the local slope at intermediate lags; once the population has committed, the distance grows with the flown speed, which returns the slope towards 2. Because commitment happens at a spread of times rather than at $t = 0$, the transition overshoots. Taking the first crossing of 5 km from take-off as the departure onto the cross-country leg — well inside the median flown path of 89 km — 95.1 % of paraglider flights depart, with a median departure time of 1,725 s, against a slope peak at 1,504 s; for hang gliders 96.4 % depart at a median of 1,530 s against a peak at 1,334 s. The radius is chosen rather than fitted, so the agreement corroborates the account rather than measuring it. The ensemble MSD about take-off is therefore a statement about launch geometry over the decade where the overshoot sits, and an exponent fitted across that crossover — $\alpha_{\mathrm{EA}} = 1.918$ and 1.896 over 121 s to 13,021 s — is a chord across a bend rather than a growth law the motion obeys.

The time average shows no arch. Panel (b) is Eq. (3.2) on the same ensemble. Over 121 s to 6,341 s and 121 s to 7,149 s it gives $\alpha_{\mathrm{TA}} = 1.732$ ($H = 0.866$) for paragliders and 1.688 ($H = 0.844$) for hang gliders, both above Brownian and below ballistic, with a smaller residual, and a local slope that reaches its minimum at 1,504 s and 1,334 s and rises from there, with no interior maximum at all. Since the two averages differ only in what they average over, the crossover is a property of the common origin, confirmed by its absence from panel (b); both exponents are reported as *effective* exponents over a stated range.

The gap between the two averages is not evidence of ageing. Fitted over their respective ranges the ensemble slope exceeds the time-averaged exponent by 0.187 for paragliders and 0.208 for hang gliders, a gap of the kind that signals a non-ergodic, ageing process [20]. Reading it that way here presupposes both curves are exponents, and the ensemble figure is a chord across a curve whose local slope covers 0.91, so changing the fitted range moves it by more than the gap. What the measurement supports is weaker: displacement from take-off grows faster, over the upper part of its range, than displacement over a window of the same duration placed anywhere in the flight. Separating ageing from launch geometry needs the phase decomposition of Sec. 4.1 and the ergodicity-breaking parameter of Appendix 3.A, neither of which needs a curve to be a power law.

What the two estimators return. The time average gives an effective exponent on a curve close to a single power law. The ensemble average gives a timescale, not an exponent: displacement from take-off crosses over near 1,504 s for paragliders and 1,334 s for hang gliders, where the population leaves its launch area, and no single power law spans that crossover. Neither curve carries a confidence interval here: a bootstrap over flights would treat the archive's records as independent, which they are not, and a curve that is a systematic arch has no single exponent for a sampling error to be an error *of*. The resampling unit and the range dependence are measured in Sec. 3.4, on the estimator that replaces this one.

Below the fitted range the curves carry two pipeline artefacts rather than flying ones: a coverage rule meeting the integer lag grid, and the smoothing evaluated off-centre at a segment's first samples, which moves the curve by up to 49.4 % below 10 s on the 0.14 % of fixes it touches (implementation details: Sec. I3). Above it, the ensemble thins to a survivorship-selected minority, holding only 2,319 flights at 30,146 s. Nothing outside the fitted range in Figure 3.1 is read as motion.

A displacement statistic on this archive is contaminated by the common origin just diagnosed; by each flight's own course, which does not simply inflate it, since 56 % of retained paraglider flights fly a closed triangle that saturates displacement at a scoring-rule timescale rather than a flying one; and by the heterogeneity of Sec. 3.3, which lets a superposition of ordinary processes with different persistence times imitate a single anomalous one. None of the three is repaired by subtracting an estimate of it; the next section meets the specification by filtering instead.

3.2 The filtered variation

The displacement statistic leaves a specification: a quantity measured from no common origin, requiring no estimate of the flown course, and not turned into a single anomalous law by a

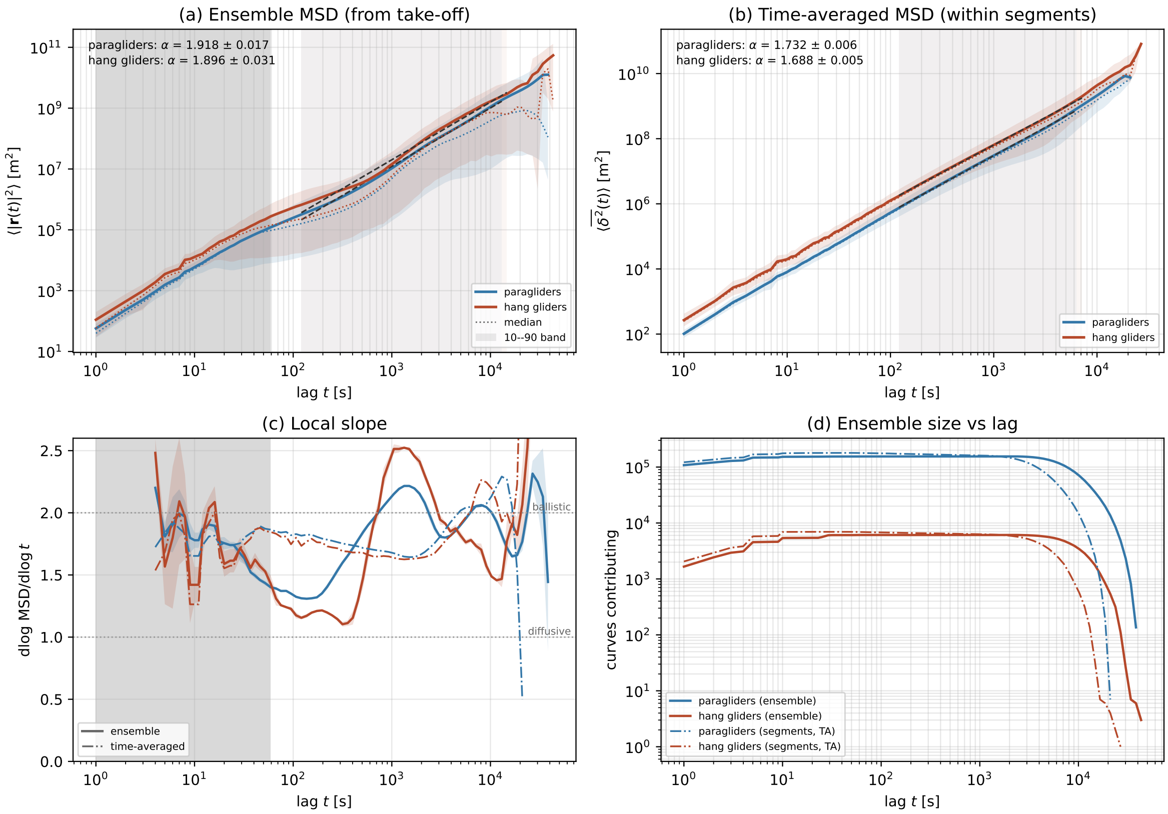

Figure 3.1. Mean-squared displacement of the retained ensemble, by two averages. (a) Ensemble MSD against elapsed flight time, log–log, fitted line and range shaded, 10 % to 90 % band across flights. (b) Time-averaged MSD, cut at half a segment's length. (c) Local logarithmic slope of both, ballistic and diffusive values marked. (d) Flights contributing at each lag, which sets the upper end of each fit range.

mixture of populations. The first two are met by construction, the third by measurement below.

Filtering rather than subtracting. The course could be subtracted rather than filtered, but subtraction fails here: the natural estimate $[\mathbf{r}(T) - \mathbf{r}(0)]/T$ is built from the record's own endpoints, and subtracting it turns the trajectory into a bridge whose increments must sum to zero by construction. On synthetic fractional Brownian motion of known exponent the operation returns a value well below the truth at every lag, worst towards the record's own length (implementation details: Sec. I3).

The estimator. A finite difference removes a trend without estimating it, because it annihilates the trend identically. Writing A_p for the difference of order p taken at spacing Δ,

$$A_p\,\mathbf{r}(t;\Delta) = \sum_{k=0}^{p} (-1)^k \binom{p}{k}\, \mathbf{r}(t + (p-k)\Delta), \tag{3.4}$$

the operator vanishes identically on every polynomial trajectory of degree $p-1$, and the *filtered variation* is its mean square,

$$V_p(\Delta) = \left\langle \left|A_p\,\mathbf{r}(t;\Delta)\right|^2 \right\rangle \;\sim\; \Delta^{\alpha_p}, \tag{3.5}$$

averaged over every admissible t and flight and summed over the two horizontal components, which is what makes it invariant under a rotation of the frame: an anisotropic ensemble must not return a different exponent for having been measured with north somewhere else. The propagator of Sec. 3.5 does the opposite and keeps the components apart, because the ensemble is anisotropic (Sec. 2.8) and a one-dimensional marginal is exactly what it reads; the two are different statistics of the same trajectories rather than one of them pooled. For a self-similar process with stationary increments $\alpha_p = 2H$ at every order, with only the prefactor depending on p: $p = 1$ is the plain increment and is blind to the origin, $p = 2$ to a constant velocity, $p = 3$ to a constant acceleration, so no drift is estimated at any order and none can be estimated badly. The three orders are not successive refinements of one number, and are not expected to agree: α_1 is the exponent of displacement with the flown course still inside it, α_2 is what that displacement does once a mean velocity — the course — is set aside, and a course that is no part of the wandering still adds to how far a wing gets, so the two answer different questions even when both are read off the same trajectories. A difference of order p spans $p\Delta$ of record, so each order costs coverage at the long end of the grid and the scan stops at $p = 3$.

The scan over orders is itself a measurement. The difference $\alpha_1 - \alpha_2$ is the part of the apparent exponent that was the flown course, read off directly rather than modelled, and the agreement between α_2 and α_3 is the statement that nothing polynomial is left for a higher difference to find. On synthetic data with a constant course imposed on a process of known exponent, order 1 returns a displaced value and orders 2 and 3 both recover the truth (implementation details: Sec. I3). Everything below is computed inside a retained segment, never across the gap that ends one (Sec. 2.7.6), and never on displacement from take-off.

The fitted range. 60 s to 2,000 s, 1.5 decades, declared in advance for both disciplines and all three orders, with both ends set by the system rather than the curve. Below, the Savitzky–Golay window has a floor of 5 samples (Sec. 2.7.7), and 60 s sits above that window for the slowest logger retained. Above, the declared task takes over: the CFD scoring rule rewards a closed course, and a closed course comes home at a time set by the task rather than the air. The fit is ordinary least squares on $\log V_p$ against $\log \Delta$, applied to the mean of the per-flight curves rather than to per-flight exponents averaged afterwards.

Table 3.1. The order scan over 60 s to 2,000 s. Order 1 is the plain increment and still carries each flight's course; orders 2 and 3 annihilate a constant velocity and a constant acceleration, and agree with each other, which is what says that nothing polynomial is left. The quoted interval combines a day-and-site bootstrap with the movement of the exponent when the fitted range is halved; the two are given separately because they are not of the same size, and the section that follows takes them apart.

	order p	$\boldsymbol{\alpha_p}$	**sampling**	**range**
	1	1.75 ± 0.06	0.001	0.062
paragliders	2	2.02 ± 0.10	0.001	0.104
	3	2.04 ± 0.14	0.001	0.140
	1	1.70 ± 0.06	0.002	0.063
hang gliders	2	1.94 ± 0.13	0.003	0.128
	3	1.97 ± 0.20	0.003	0.198

The order scan. Table 3.1 gives the scan and Figure 3.2 the curves behind it. At second order the archive returns $\alpha_2 = 2.02 \pm 0.10$ for paragliders and 1.94 ± 0.13 for hang gliders, over 60 s to 2,000 s in both — agreement that was not guaranteed, since the two disciplines differ in speed, glide ratio and where they fly. The arch that dominates the ensemble MSD near 1,504 s is three times smaller in V_2 and of the opposite sign: a dip rather than an arch, which is what removing the common origin and the flown course was supposed to achieve (Figure 3.2).

Removing the course raises the exponent. α_1 and α_2 differ by -0.27 for paragliders and -0.24 for hang gliders, and the sign is the opposite of what a residual drift would leave: a drift normally inflates a displacement statistic, but on this archive the declared course *suppresses* it, since 56 % of retained paraglider flights and 63 % of hang-glider flights fly a triangle that returns towards its start, saturating the plain increment at a task-set lag which the second-order variation does not see at all. Fitting closed and open tasks separately confirms it directly: the gap between them (open minus closed) is $+0.11$ at order 1 for paragliders and $+0.03$ for hang gliders — closed tasks reading lower — and it reverses to -0.10 and -0.07 at order 2, with the order-1 gap growing substantially past the fitted range and the order-2 gap not moving. An exponent read with no trend removed is therefore expected to sit below one measured with the course filtered out, by the difference between the two orders.

Orders two and three agree, and order one checks against the time average. α_2 and α_3 agree to 0.02 for paragliders and 0.02 for hang gliders, well inside either interval: nothing polynomial survives the second difference for the third to remove. And α_1, which *is* $\overline{\delta^2(\Delta)}$ under another name, computed by an independent implementation over a different range, agrees with α_{TA} of Sec. 3.1 more closely than either moves when its own range is halved — 1.75 against 1.732 for paragliders and 1.70 against 1.688 for hang gliders.

What an exponent near two does not mean. $\alpha_2 = 2.02$ is $2H$ with $H = 1.009$, at the upper boundary of the self-similar family and marginally past it, and α_1 and α_2 do not share a ceiling: reading the second against the first's bound is the only way this looks like a violation. A plain increment cannot report above the ballistic value on any process with stationary increments: the displacement over n steps is a sum of n increments, Minkowski's inequality bounds its norm by the sum of theirs, and $V_1(n\Delta) \le n^2 V_1(\Delta)$ follows, so $\alpha_1 \le 2$ whatever the motion does — a bound the archive respects with room to spare, at 1.75 and 1.70. The second-order variation answers to a different bound because it is a different quantity: on a differentiable trajectory $A_2\,\mathbf{r} \simeq \ddot{\mathbf{r}}\,\Delta^2$, so $V_2 \sim \Delta^4$, and its ceiling is therefore 4, twice the ballistic value, reached only by a trajectory smooth enough that its curvature does not wash out as $\Delta \to 0$. A straight line sits at the opposite extreme, where the second difference vanishes identically, since it has no curvature to detect; a persistent random walk, calibrated the same way, returns a value between the two, closer to 4 the longer it stays correlated. 2.02 and 1.94 sit just past $2H = 2$, far below that ceiling of 4: read against the calibration that actually bounds this estimator, not against the ballistic value, which bounds a different one, they say the trajectory stays correlated and rough — persistent, not straight — at every lag of the range, not that it moves faster than a bound it was never subject to. A Hurst exponent above unity has no meaning for an exactly self-similar process, so the figure is quoted as an effective exponent over a declared window; how far the archive is from exact self-similarity is measured in Sec. 3.5, on the distribution of the increments rather than its second moment.

3.3 Compatibility across strata

The dataset spans more than four decades, two glider types and a range of performance classes (Sec. 2.4) whose kinematics differ substantially. Before treating the flights of one discipline as draws from a single statistical process, the subpopulations are checked. The test is the horizontal MSD computed within each stratum and read against the pooled curve, the same statistic calculated across all flights of the discipline without any stratum split, over the same lags; the two disciplines are never pooled, and a stratum below 2,000 flights is not drawn: below that size, sampling noise risks dominating the curve. The class normalisation of Sec. 2.4

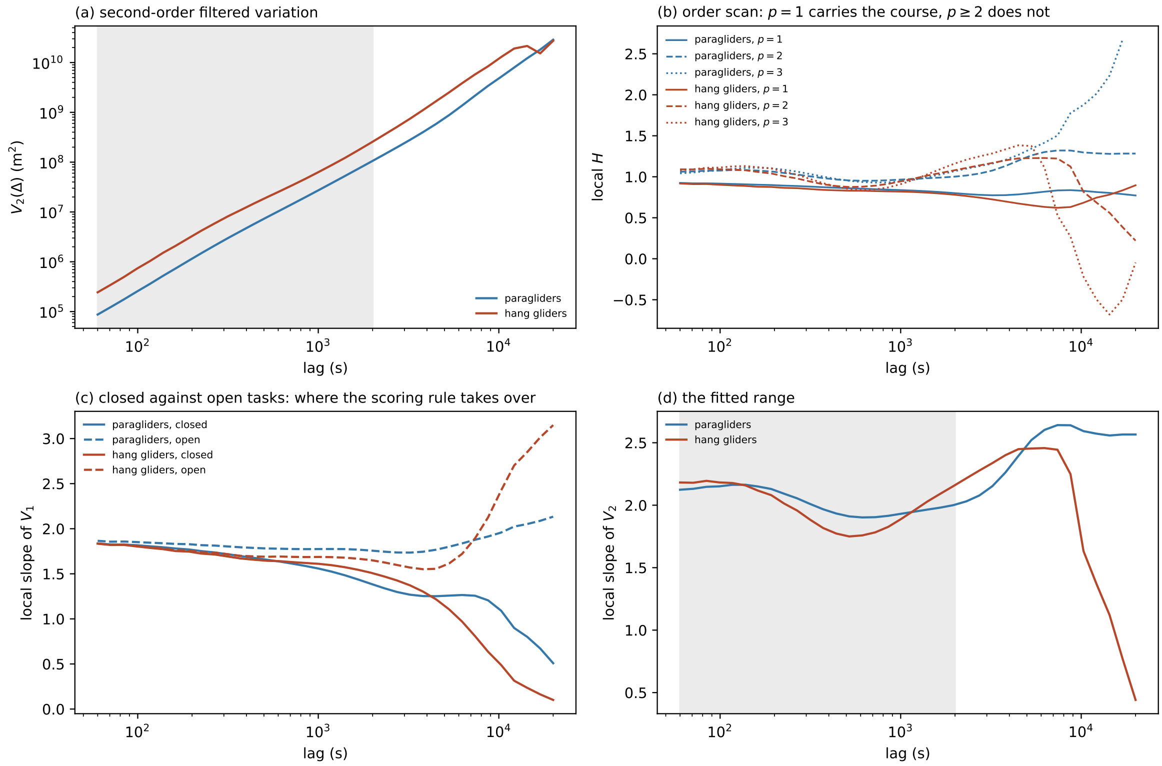

Figure 3.2. The filtered variation, on which the exponent is read. (a) $V_2(\Delta)$ against lag, log–log, fitted range shaded. (b) Local $H = \frac{1}{2}\,\mathrm{d}\log V_p/\mathrm{d}\log\Delta$ at the three filter orders: $p = 1$ runs below $p = 2, 3$, the flown course read as a lower exponent. (c) Local slope at $p = 1$ split by declared task; closed and open courses part company past 3,000 s. (d) Local slope of V_2, the quantity the fit averages.

runs first, since as long as one wing class appears under several spellings a stratum is not a well-defined set of flights.

Figure 3.3 gives three axes and two answers. Across the 6 wing classes the curves separate: the typical stratum departs from the pooled curve by 51 % and the extreme by 90 %, with the CIVL competition wings above the pooled curve throughout and the EN A wings well below it. Across the 5 orographic groups they separate further, by 62 % typically and 130 % at the extreme, with the Massif Central at half the pooled curve over the intermediate lags and the Alps above it. Across the 17 seasons they do not: the typical departure is 12 % and the largest 21 %, and the latter belongs to the smallest season in the archive.

So the ensemble may be pooled over time and may not be pooled over wing class or over terrain. Season is not a relevant stratum and the archive's two-decade span is not a confound. Class and geography are relevant strata, and the difference is itself a result rather than a nuisance: a competition wing covers ground faster than a recreational one, and thermals over a mountain range are anchored to slopes and organised along ridges whereas over flat terrain they drift with the wind, so the persistence of the glides and the spacing of the climbs need not be the same. The exponent below is therefore reported by wing class as well as pooled, alongside cuts by native cadence, season and declared task (Table 3.3); the orographic cut is made here, on the ensemble MSD, which is the statistic it separates. A pooled exponent is read as an average over strata that are known to differ, not as a property of a homogeneous population.

3.4 The uncertainty on the exponent

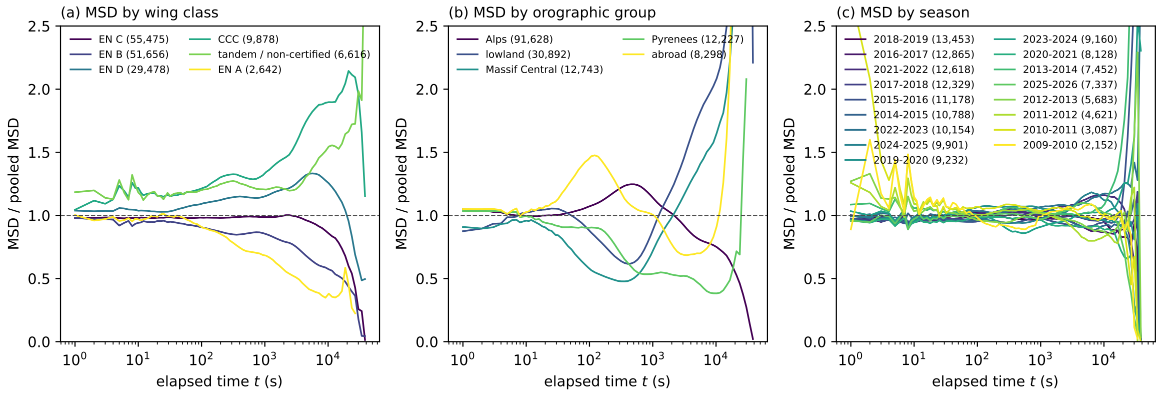

Figure 3.3. Whether the retained ensemble may be pooled: the ensemble MSD computed within each stratum and divided by the pooled curve, for paragliders, (a) by wing class, (b) by orographic group and (c) by season; strata below 2,000 flights are not drawn. Class and terrain separate the curves, season does not.

The archive holds 155,788 paraglider and 6,132 hang-glider flights, and 84,132 and 4,331 occasions on which one site was flown on one day. An interval built on the first pair of counts is not an interval built on the second, so which of them carries the independent measurements has to be settled before either is quoted.

The resampling unit. The quantity clustered is the logarithm of each flight's own first-order variation at the lag nearest the geometric centre of the fitted window (372 s), the per-flight input every fit in this chapter averages over. Its intraclass correlation at a level ℓ is

$$\mathrm{ICC}_\ell = \frac{\sigma^2_{\text{between},\,\ell}}{\sigma^2_{\text{between},\,\ell} + \sigma^2_{\text{within},\,\ell}}, \tag{3.6}$$

the share of the total variance sitting *between* clusters at that level rather than within them; near zero the clustering is irrelevant, near one a cluster is effectively a single measurement. The ladder (Table 3.2) is zero at the flight by construction and largest at the day and site in both disciplines — the air over one site on one day being what two flights actually have in common — so every bootstrap in this chapter resamples days and sites, of which the archive holds 84,132 and 4,331. The correction this buys is small and in the expected direction: resampling flights returns 0.000 on the paraglider α_1 against 0.001 from the clustered resample, both sitting below the last digit of the exponent they qualify.

Table 3.2. Intraclass correlation of $\log V_1$ by clustering level. Zero at the flight is the assumption a per-flight bootstrap makes, not a measurement; largest at the day and site, which sets the resampling unit for every bootstrap in this chapter.

	flight	**day**	**pilot**	**day & site**
paragliders	0.00	0.14	0.29	0.57
hang gliders	0.00	0.25	0.50	0.63

The sampling term and the range term. The two are reported separately. On α_2 the day-and-site bootstrap returns 0.001 for paragliders and 0.003 for hang gliders; refitting each half of the fitted window and halving the difference between the two exponents returns 0.104 and 0.128. The sampling term is negligible against the range term in both disciplines, and the quoted ±0.10 and ±0.13 are the two combined in quadrature — the range term with the sampling term invisible inside it. The same construction gives 0.062 at order one and 0.140 at order three, growing with the order of the filter as a higher difference of a noisier curve should: the second decimal of the exponent is not determined, and a number carrying the bootstrap alone would claim four significant figures on a curve that does not have two.

A third figure, the standard error least squares returns on the fitted slope, measures something else: how well a *straight line* describes the fitted points, treating each residual as an independent draw from a common noise. The residuals of these curves are neither independent (every lag averages the same flights) nor noise (the local slope drifts across the fitted decades), so that figure is dominated by the curve's departure from a power law rather than by sampling, and is not an interval on the exponent (implementation details: Sec. I3).

What sets the two ends of the window. Below, the smoothing window is $\max(\tau_c, 5\,\Delta t)$ (Sec. 2.7.7): its floor is in *samples*, so in seconds it widens with the logger's step. 60 s therefore sits above it for 99.3 % of retained paraglider flights and 99.8 % of hang-glider ones; the slow remainder is smoothed over more than the shortest fitted lag, and enters the average with the few windows so sparse a record supplies. Above, the declared task governs: closed and open courses separate in local slope by 0.03 at 60 s and by 1.12 at 12,156 s for paragliders, from 0.00 to 2.39 for hang gliders (Figure 3.2), growing smoothly with no lag at which the measurement becomes clean; the ceiling 2,000 s is set where that separation is still a small fraction of what it reaches, leaving a window 1.5 decades wide.

Four cuts of the ensemble. Table 3.3 asks whether α_2 belongs to the motion or to who is in the ensemble, on the same row-selection pass used throughout this chapter (implementation details: Sec. I3).

Table 3.3. What α_2 does when the paraglider ensemble is cut four ways, over the same range. The comparison that matters is with Sec. 3.3, where the same wing classes departed from the pooled ensemble MSD by 51 % and the orographic groups by 62 %. The declared task is entered as the open-minus-closed difference rather than as a spread, and it is the largest of the four.

cut	strata	α_2 range	spread
native cadence	6 classes	2.01–2.03	0.02
wing class	6 classes	1.97–2.03	0.06
season	17 seasons	2.00–2.03	0.03
declared task	closed, open	—	−0.10

The cadence cut is a check on the pipeline: across 6 classes spanning the whole range of smoothing scales in the archive, the exponent spreads by only 0.02, so whatever the estimator is reading, it is not the filter. Wing class and season are the heterogeneity confound measured on the ensemble MSD in Sec. 3.3 (departures of 51 % and 62 %): here the exponent spreads by only 0.06 across 6 wing classes and 0.03 across 17 seasons, since a mixture of populations with different speeds and courses is a mixture of trends, and a trend is what the second difference removes. The heterogeneity that makes the ensemble estimator unusable does not reach this one. The declared task is the largest of the four, at −0.10 for paragliders and −0.07 for hang gliders — the one cut that divides flights by something other than who was flying — and no entry in the table exceeds the interval the exponent already carries.

The task stratum and the range term are one effect, not two. The task gap at order two, −0.10, and the range dependence, 0.104, are the same size and plausibly the same effect: both measure how much of the fitted exponent is fixed by the long end of the window. The quoted ± 0.10 already carries this systematic through its range half, so entering the task stratum as a further term would count the scoring rule twice. For hang gliders the two figures, -0.07 and 0.128, are of the same order but not the same size, so the identification is weaker there and the interval is left as the range term alone.

How many exponents the range supports. A piecewise-linear fit to $\log V_2$ against $\log \Delta$, with k breaks scored by a Bayesian information criterion of the standard form,

$$\mathrm{BIC}(k) = n_{\mathrm{eff}} \log(\mathrm{RSS}_k/n_{\mathrm{eff}}) + p_k \log n_{\mathrm{eff}}, \tag{3.7}$$

RSS_k the fit's residual sum of squares and p_k the number of free parameters at k breaks, but with $n_{\mathrm{eff}} = 2.0$ the same effective degrees of freedom used throughout this chapter substituted for the 22 nominal lags, so a curve whose residuals are correlated cannot buy a break with points it does not independently have. This selects 1 break for paragliders at 267 s — separating 2.16 from 1.93, bracketing the single-exponent value by about the size of the range dependence — and 2 for hang gliders. Run on surrogates that are a single power law plus the measured sampling noise, the same procedure invents a break 59 % and 58 % of the time (implementation details: Sec. I3), too often for either count to be trusted: the correlation across lags puts 22 lags at no more than 2.0 effective points, a budget for one exponent and not for a count of regimes.

3.5 The exponent from the increment quantiles

Every exponent quoted so far is a second moment of a displacement, and this archive has already shown what a second moment can measure instead of the motion: averaged about a common origin, it returned the geometry of the launch site. What is wanted is a route to the exponent that uses neither a moment nor a common origin.

First-passage times, tried and withdrawn. The first-passage time $T(L)$, the first time a flight is a distance L from take-off, gives $\langle T(L)\rangle \sim L^{1/H}$ for a self-similar process [21], so its slope in the space domain was tried as a cross-check on α_2. Read naively it exceeds the ballistic value because of right censoring (only the slow minority of flights survive to the largest radii); a Kaplan–Meier median removes the bias and lands inside the paraglider interval it was meant to test, but moves across sub-ranges of radius by several times that interval (implementation details: Sec. I3). $T(L)$ is measured from take-off, so it carries the same launch-geometry origin into the space domain — the crossover of Figure 3.1 read along the space axis instead of the time axis — and the check is withdrawn as not independent of it.

The propagator of the increments. Write $\Delta\mathbf{r}(t_0, \Delta) = \mathbf{r}(t_0 + \Delta) - \mathbf{r}(t_0)$ for the horizontal increment over a lag Δ, inside a retained segment and never across the gap that ends one (Sec. 2.7.6). Pooled over starting times and flights, the distribution of its modulus $r = |\Delta\mathbf{r}|$ is the propagator of the increments, and for a self-similar process with stationary increments it takes the scaling form

$$P(r, \Delta) = \Delta^{-H}\, F\!\left(\frac{r}{\Delta^{H}}\right), \tag{3.8}$$

with F a fixed density independent of the lag. Every quantile of r then grows as Δ^{H}: writing $q_p(\Delta)$ for the pth quantile, $\log q_p$ against $\log \Delta$ is a straight line of slope H at every p, fixed by the middle of the distribution where every flight contributes and a heavy tail moves it not at all. Neither of the two defects of the earlier estimates enters here: no moment for a few large increments to set, and no absolute position for the launch site to bias. The four levels read are the lower quartile, the median, the upper quartile and the ninetieth percentile, from one streaming pass keeping a histogram per lag and per native cadence (implementation details: Sec. I3).

Why the modulus, and what the components are still for. The exponent has to be measured on the same quantity as the estimators it is being checked against. The ensemble MSD averages $E^2 + N^2$ and the filtered variation sums the squared difference over both components to stay invariant under a rotation of the frame (Sec. 3.2); both are statements about $|\Delta\mathbf{r}|$, so only a modulus exponent can be set beside them in Table 3.4. Reading H off one component and comparing it with those would be comparing two different quantities, and on this ensemble they are numerically different as well.

That is not the same as pooling the two components into one histogram, which would be the error Sec. 2.8 forbids: pooling presupposes that E and N share a distribution, and they do not. The modulus presupposes nothing — it is a scalar built from both, and it is the scalar the other two estimators already use. The components are therefore measured separately as well, and kept apart, because their disagreement is itself a result: it is reported below and carried into Chapter 4 as a constraint. So the pass histograms all three of $|\Delta x|$, $|\Delta y|$ and $|\Delta\mathbf{r}|$; the modulus answers what the transport exponent is, and the components answer whether one exponent can stand for both directions.

What it returns. $H = 0.884$ for paragliders and 0.873 for hang gliders, over 60 s to 2,000 s, from 245,444,138 and 5,830,745 increments in 6 and 2 native cadence classes — doubled into the units the filtered variation is quoted in, $\alpha = 1.77$ and 1.75. Across the paraglider cadence classes, spanning a factor of ten in smoothing scale, the estimate moves between 0.878 and 0.891, so the smoothing scale does not set it.

Table 3.4. Three routes to H, on the same ensemble and the same window, 60 s to 2,000 s. The first and the third are second moments and differ in the frame they are taken in: the plain increment retains each flight's course, the second-order variation annihilates it. The second is not a moment. The first is the quantity a published exponent for human soaring flight measures; the third is the route the chapter's α_2 comes from, at $\alpha_2 = 2H$.

route	what it uses	H paragliders	H hang gliders
plain increment V_1	second moment, course retained	0.876	0.852
increment quantiles	four order statistics	0.884	0.873
second-order variation V_2	second moment, course filtered	1.009	0.972

Three routes, and what they bracket. Table 3.4 sets the quantile exponent beside the two second moments. They bracket it: H lies between 0.876 and 1.009 for paragliders and between 0.852 and 0.972 for hang gliders, nearer the plain increment. The gap between the two moments is the flown course, measured by the order scan and read here from the other side, and it fixes which row a published value is compared against: Vilpellet et al. [9] report a global H for human soaring flight from a displacement statistic with no trend removed — the plain increment of the first row, not the second-order variation. The paraglider value in that row rounds to the reported one, on an independent archive and at the precision it is quoted to; the second-order variation sits above it, with the sign of the gap the one the order scan predicts, since the archive's closed tasks suppress displacement rather than inflate it. The bracket is the result rather than any of its decimals: two second moments taken in different frames and one statistic that is no moment at all, none of them measured from take-off.

What the quantiles cost the scaling form. Equation (3.8) claims the four quantiles share a slope. They do not: fitted separately, H falls from 0.937 at the lower quartile and the median to 0.868 at the upper quartile and the ninetieth percentile for paragliders, and from 0.948 to 0.834 for hang gliders, in the same direction in both disciplines, on the modulus and on both components alike, and at every native cadence measured (Figure 3.4b draws the three variables at 1 s). Table 3.5 gives the four fits one at a time, on the modulus at the fastest cadence, each with an error that resamples whole day-and-site clusters rather than trusting the lags of the fit to be independent points. That error is small — 0.0008 at worst for paragliders and 0.0067 for hang gliders — and it is small for a reason that is not a courtesy of the estimator: 60 blocks of day-and-site clusters over an archive this size leave almost nothing for the sampling to move. The comparison that matters is therefore between that error and the spread down the column, which is 145 and 24 times it. The quantiles disagreeing is not sampling noise, and no error bar of this kind will make it one.

It is worth saying which way that cuts, because the naive alternative points the other way. The least-squares error of the same fit is 0.0055 and 0.0063, *larger* than the resampled one by up to a factor of twenty. That is not the resampling being optimistic: an ordinary least-squares error on a log–log fit measures how far the points sit off a straight line, and here they sit off it because the relation is genuinely a little curved — which is the finding of this section, not an uncertainty about it. It also treats lags built from the same flights as independent observations, which they are not. So neither number is a reason to write the exponent to three figures: what limits it is not how well the archive is sampled — the archive is large enough that the sampling error is negligible — but the systematic spread, across the quantiles here, across native cadences (0.878 to 0.891 for paragliders), and with the fitted range in Sec. 3.4. The spread across the four quantiles is 0.116 and 0.163, against 0.018 on an exactly self-similar process fitted the same way with less data (implementation details: Sec. I3). It is wider on the modulus than on either component —0.070 east and 0.093 north for paragliders, 0.083 and 0.125 for hang gliders (Figure 3.4b)— and that ordering is the expected one rather than a surprise: a small $|\Delta x|$ is not a small displacement, since the wing may be going due north, so a component's lower quantiles mix barely moving with moving across the axis being read, and flatten the drift by averaging the two. The modulus separates them, and reports more of it.

Rescaling the histograms by Δ^H leaves 0.059 and 0.087 dex of scatter through the bulk — a factor of 1.15 and 1.22 in the rescaled density, close enough that the curves read as one by eye — with visible separation only at the flanks (Figure 3.4c). The scaling form is therefore only approximate: the centre of the increment distribution spreads faster than its flanks, a distribution changing shape with scale rather than a single self-similar one. That qualifies the exponent rather than refuting it — the distance from bulk to flank is small beside the distance of either from the Brownian value, and no reading of these curves puts the transport near it — but it is a second reason, independent of the range dependence of α_2, why a single exponent quoted to three figures would claim more than this archive supports.

The anisotropy reaches the exponent. The two horizontal components differ in exponent as well as amplitude: 0.890 east against 0.908 north for paragliders, and 0.860 against 0.911 for hang gliders, the wider separation of the two. Pooling the components would average that away, which is why they are kept apart here and in Sec. 2.8.

3.6 The exponent, split by component

Sec. 3.5 already split H into its two horizontal components for one route, the increment quantiles, because Sec. 2.8 measures the archive as anisotropic and a pooled one-dimensional marginal would average that away. The question here is whether the split is a property of that

Table 3.5. H fitted separately at each percentile of $|\Delta\mathbf{r}|$, at the fastest native cadence (1 s) over 60 s to 2,000 s. No error column: resampling 60 blocks of whole day-and-site clusters moves each of these figures by at most 0.0008 and 0.0067, which would print as a column of zeros at the precision shown. The differences down the column are one to two orders of magnitude larger than that, and they are the point of the table.

percentile of $|\Delta\mathbf{r}|$	H paragliders	H hang gliders
25th	0.979	0.986
50th (median)	0.897	0.904
75th	0.862	0.835
90th	0.872	0.843

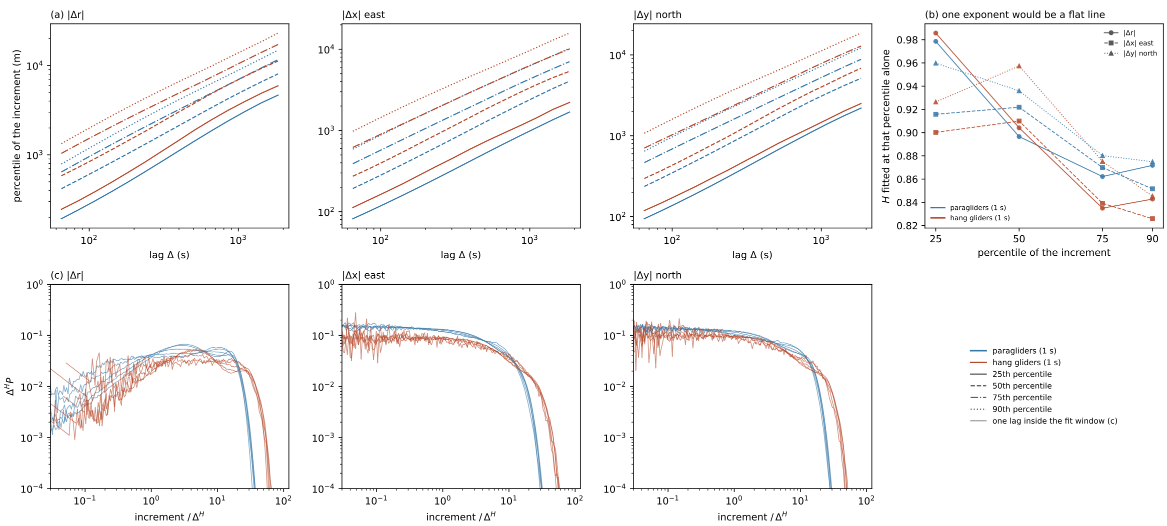

Figure 3.4. The exponent read from the distribution of the increments, not from a moment of it. All seven panels use each discipline's fastest native cadence, 1 s, where the histograms hold the most increments. **(a)** The 25th, 50th, 75th and 90th percentiles against lag, log–log, over the fit window 60 s to 2,000 s, one column for the modulus $|\Delta\mathbf{r}|$ and one for each horizontal component. Four parallel lines would mean one exponent describes every part of that column's distribution. **(b)** The slope of each of those four lines, fitted separately, with the modulus and the two components overlaid in one panel rather than split into columns — it is already the comparison across variables that (a) and (c) split apart. A flat line would mean the same thing as in (a). All three fall from the 25th percentile to the 90th, so the disagreement is not a property of one direction. **(c)** The histograms behind each column of (a), rescaled by Δ^H, one curve per lag inside the fit window and every third lag drawn. H is that column's own value: 0.884 (modulus), 0.893 (east), 0.908 (north) for paragliders, and correspondingly 0.873, 0.870, 0.901 for hang gliders — not a value shared across columns, since H differs by component. If the scaling form held exactly the curves of one column would lie on top of one another; they do through the bulk and separate at the flanks, in every column.

one estimator or of the motion: put the same east/north decomposition through every other route this chapter computes, and see whether they agree.

Which routes, and which is not a fifth exponent. Three routes accept the same treatment for the same reason the quantile route does: $\overline{\delta^2(\Delta)}$ (Sec. 3.1) and the filtered variation at orders 1 and 2 (Sec. 3.2) all reduce, before any sum over components, to a mean square of a within-segment coordinate difference, so reading one coordinate at a time costs no new estimator — only feeding it its own component and a column of zeros for the other. Order 3, the task-systematic check, the cadence/class/season stratification and the regime fit are left pooled, as before.

The ensemble MSD is the fourth, and it is not a fifth measurement of H. Sec. 3.11 withdrew it because displacement from take-off crosses over where the population leaves its launch area, and hand-over item (i) states that the contamination is not confined to the observable that exposed it. Read here per component, it is a diagnostic on that contamination and not a route to the exponent: Table 3.6 keeps its row apart from the other three, and no number in it is to be averaged into the bracket of Sec. 3.5.

Table 3.6. H by component, on every route besides the quantiles that reads one axis at a time. The time average and the two filtered-variation orders are fitted over 60 s to 2,000 s, the window Table 3.4 uses; the ensemble row is fitted on its own, coverage-limited range instead — 121 s to 13,021 s for paragliders, 121 s to 14,680 s for hang gliders (Sec. 3.1) — and is not on the same footing as the three below it.

	route	H east	H north
paragliders	ensemble MSD (diagnostic, *not* an exponent)	0.958 ± 0.001	0.960 ± 0.001
	time-averaged MSD	0.850 ± 0.001	0.877 ± 0.000
	plain increment V_1	0.867 ± 0.034	0.882 ± 0.029
	second-order variation V_2	0.996 ± 0.065	1.021 ± 0.042
hang gliders	ensemble MSD (diagnostic, *not* an exponent)	0.904 ± 0.005	0.988 ± 0.005
	time-averaged MSD	0.822 ± 0.003	0.860 ± 0.002
	plain increment V_1	0.835 ± 0.040	0.864 ± 0.026
	second-order variation V_2	0.958 ± 0.074	0.985 ± 0.056

The three trustworthy routes, against the quantile split. Table 3.4 already reads $H = 0.890$ east against 0.908 north for paragliders, and 0.860 against 0.911 for hang gliders (Sec. 3.5, quoted there without a resampled error, only the quantile self-consistency spread). The time average and both filtered-variation orders order the two components the same way. Three routes built from three different statistics of the same trajectories — a sliding window, a polynomial-annihilating filter at two orders, and an order statistic of the increment distribution — agreeing on which axis carries more of the exponent is a stronger statement than any one of their magnitudes, which the routes do not agree on to the same precision.

The withdrawn route, read per component. The ensemble row of Table 3.6 answers a narrower question: is the launch-geometry contamination itself isotropic? Read against the trustworthy time average beside it, the gap is 0.216 east and 0.165 north for paragliders (0.165 and 0.257 for hang gliders), against 0.187 pooled (Sec. 3.1). Neither gap is confined to one component, so the contamination hand-over item (i) describes is not a one-axis effect in either discipline; which of the two carries the larger gap is not even the same axis in both, east for paragliders and north for hang gliders, so this is a constraint on the launch geometry at each site and not a fifth reading of the transport exponent.

3.7 The archive is not a Lévy walk

An exponent fixes one moment of one distribution, and two processes can agree on it and be different physics. A correlated Gaussian motion and a Lévy walk both give $\langle|\Delta\mathbf{r}|^2\rangle \sim \Delta^{2H}$; what separates them is the rest of the moment spectrum. It is measured on the same unsegmented trajectories, with no estimator in common with the exponent itself, and it refuses the Lévy walk.

The moment spectrum. Write

$$\langle|\Delta\mathbf{r}(\Delta)|^q\rangle \sim \Delta^{\,q\nu(q)} \tag{3.9}$$

and read $q\nu(q)$ as a function of q. A self-similar monofractal process gives a straight line through the origin with $\nu(q)$ constant; a Lévy walk — flight durations drawn from a power-law tail, each traversed at finite speed — gives a bilinear one, with a knee at its tail index and a ballistic right-hand branch of slope one. Over 60 s to 8,000 s, on 24 lags and q from 0.50 to 4, the spectrum is straight (Figure 3.5a): $\nu(q)$ varies by 0.019 for paragliders and 0.009 for hang gliders about a single fitted $\nu = 0.860$ and 0.830, and the rms departure of $q\nu(q)$ from a line through the origin, 0.0109 and 0.0057, sits an order of magnitude below what a Lévy walk's knee leaves on this grid and range of q (implementation details: Sec. I3). The moments are estimable rather than merely computed: the share of each moment carried by the largest hundredth of its increments reaches 28 % at worst for paragliders and 17 % for hang gliders (Figure 3.5b), well short of the half that would make a moment a statement about a few flights rather than an estimate. On the whole trajectory, unsegmented, the archive does not support the Lévy walk this thesis was framed around, and what the spectrum leaves standing is a monofractal process: a single exponent governing every moment over the range read.

The previous section's quantile fit found that H falls from bulk to flank rather than holding a single slope, which looks in tension with a straight spectrum. The two are compatible: a moment of order q integrates the whole increment distribution, so a local exponent drifting between centre and flanks enters $\nu(q)$ weighted by where that moment takes its mass, and the drift partly cancels, while a quantile fit reads the collapse at a fixed probability level and averages nothing. The quantile spread (0.116, 0.163) is accordingly the smaller of the two figures against the spread of $\nu(q)$ (0.019, 0.009), though only the ordering is established, not a derived bound; the refusal of the Lévy walk is untouched either way, since the departure a knee would leave exceeds both.

What survives is a monofractal process, one exponent for every moment over the range read, which fixes how the increment distribution scales and says nothing about its shape.

3.8 The propagator is not Gaussian

A straight moment spectrum says that one exponent governs every moment of the increment distribution. It fixes how that distribution *scales* and says nothing about its *shape*, so calling the process a correlated Gaussian motion does not follow from it. The fourth and second moments already accumulated settle the shape directly, through the non-Gaussian parameter

$$\alpha_2^{\mathrm{NG}}(\Delta) = \frac{\langle|\Delta\mathbf{r}(\Delta)|^4\rangle}{2\,\langle|\Delta\mathbf{r}(\Delta)|^2\rangle^2} - 1, \tag{3.10}$$

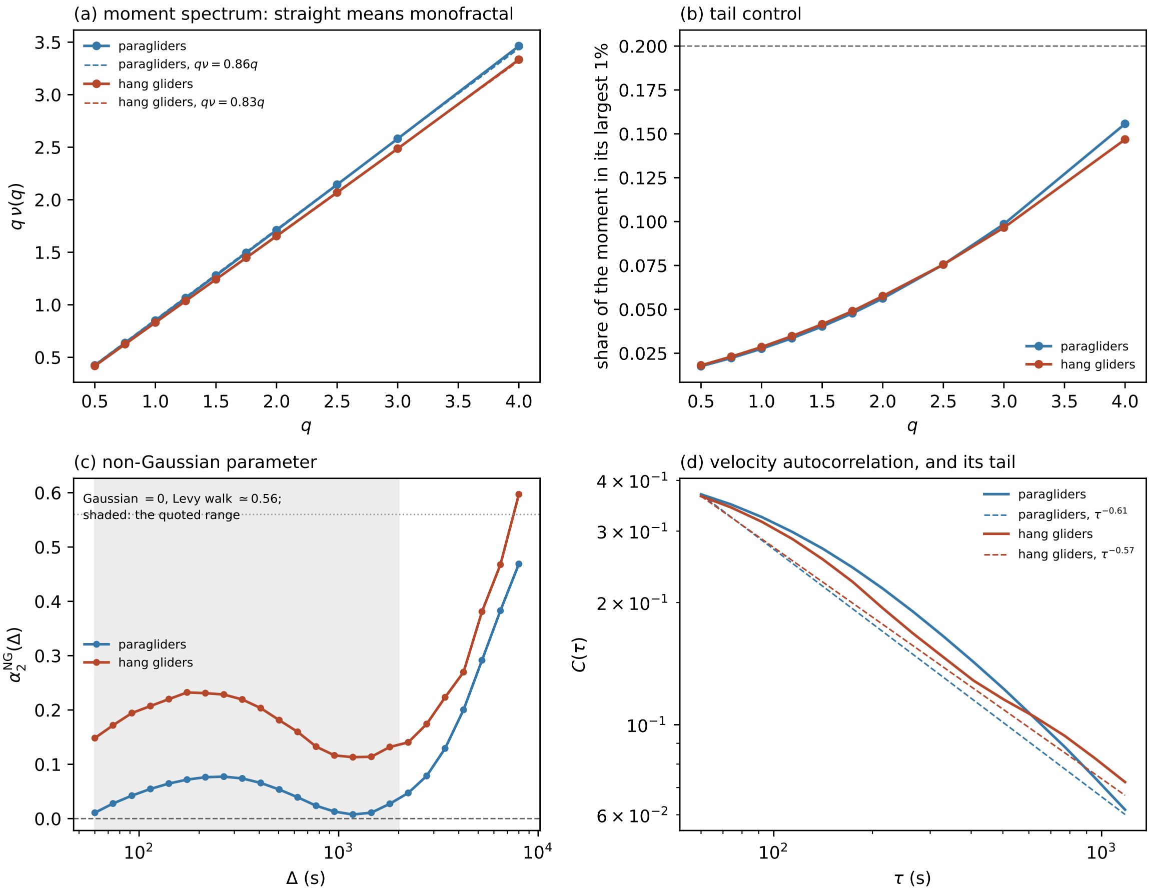

Figure 3.5. What the second moment cannot say. (a) Moment spectrum $q\nu(q)$, straight line dashed: straight means one exponent for every moment. (b) Share of each moment carried by the largest 1 % of increments. (c) Non-Gaussian parameter against lag, Gaussian and Lévy-walk values marked. (d) Velocity autocorrelation, log–log, fitted tail dashed.

evaluated on the increment over a lag Δ rather than the displacement from take-off, normalised so it vanishes for a Gaussian propagator and is positive for heavier-than-Gaussian tails. The superscript distinguishes it from the filtered-variation exponent α_2, an unrelated quantity that happens to share the symbol at filter order $p = 2$.

Over 60 s to 1,805 s the median value is +0.042 for paragliders and +0.181 for hang gliders. Neither is flat: both rise to an interior maximum and fall back — +0.077 at 266 s for paragliders and +0.232 at 174 s for hang gliders, monotone over seven or more consecutive lags on each side, a shape and not scatter. A maximum at two to four hundred seconds is a scale the chapter has otherwise found none of: it is the lag where the *shape* of the increment distribution departs furthest from Gaussian, smaller below it (increments are averages over a smoothing window and a few heading changes) and above it (a central-limit Gaussianisation begins to act). Reported against processes of known shape, both archives are non-Gaussian and neither is close to a Lévy walk: on an exact fractional Brownian motion the same estimator returns a small positive value at a matched sample size, collapsing towards zero as the sample grows, while on a Lévy walk it returns a figure far above either archive's and does not move with the sample (implementation details: Sec. I3). The two archives do not overlap at any lag in the window, a separation larger than the movement either curve shows across it, acting more strongly on the smaller, faster discipline by the ratio of the two median ground speeds (Sec. 3.10).

The pooled parameter is a width mixture, not a tail. The value is pooled by construction: the moments are accumulated over every flight before any ratio is taken, so it is a property of the whole population of increments and not of any flight's own propagator. Writing S_f for a flight's own second moment and a_f for its own non-Gaussian parameter,

$$1 + \alpha^{\mathrm{NG}}_{\mathrm{pooled}} = \frac{\mathbb{E}_n\Big[(1 + a_f)\, S_f^2\Big]}{\mathbb{E}_n[S_f]^2}. \tag{3.11}$$

Setting every a_f to zero — every flight exactly Gaussian — leaves $\mathrm{CV}^2(S_f)$, the excess a population of Gaussian flights of differing amplitude produces on its own; that null is *above* the measurement at every one of 3 lags for paragliders and 3 for hang gliders. Subtracting it, the within-flight excess is *negative* — -0.118 and -0.084 in the median — so a flight's own increment distribution is flatter than Gaussian, not heavy-tailed, which is what a wing holding a characteristic airspeed should give. This changes what the number means rather than what it is: the pooled propagator genuinely is not Gaussian, because a mixture of Gaussians of differing width is not one, but the excess is the width mixture and not a tail, and what has to be reproduced is the between-flight amplitude spread, $\mathrm{CV}(S_f) = 0.35$ against 0.48 at 60 s. The refusal of the Lévy walk is untouched, resting on the moment spectrum, measured per lag and not pooled across flights of different amplitude.

Beyond the fitted range α_2^{NG} climbs, to +0.47 and +0.60 at the longest lags of panel (c): reported and not quoted, since there the declared task governs the trajectory and the moment is carried by the fewest flights. The process is monofractal and its propagator is not Gaussian: one exponent for every moment fixes how the increment distribution scales, and leaves the shape it scales as heavy as it is measured to be.

3.9 The memory of the heading

Writing $\delta\mathbf{v}$ for the horizontal velocity fluctuation about a record's own mean, the velocity autocorrelation is

$$C(\tau) = \frac{\langle\, \delta\mathbf{v}(t_0) \cdot \delta\mathbf{v}(t_0 + \tau)\, \rangle_{t_0}}{\langle\, |\delta\mathbf{v}(t_0)|^2\, \rangle_{t_0}}, \tag{3.12}$$

normalised so that $C(0) = 1$, with the flown course, which would otherwise hold every lag up, removed before the average is taken. It is computed within a retained segment and never across the gap that ends one (Sec. 2.7.6), by the Fourier identity rather than by a double loop, and averaged across records at each lag of the grid the moments use (implementation details: Sec. I3).

Positive at every lag, and small at the first. C is positive throughout Figure 3.5 and changes sign nowhere on the grid: $C = +0.37$ for paragliders at the floor and +0.37 for hang gliders, decaying from there without a zero crossing. A single circling wing would produce a sign change, its velocity pointing the other way half a turn later, but an ensemble of them does not: radius and phase disperse across flights, so the average of many circles cannot go negative even when every record in it turns. What survives an average is the shape, not the sign — a minimum at half a period and a maximum at the period — and both are there, below the first lag Figure 3.5 plots. Measured at native cadence on the 1 Hz segments, C falls to +0.353 at 12 s and recovers to +0.456 at 22 s for paragliders, and to +0.316 at 11 s and +0.449 at 21 s for hang gliders (implementation details: Sec. I3), the minimum sitting at half the maximum's lag in both disciplines — the circle, not a coincidence of the estimator, and nowhere near zero. So the unsegmented ensemble already carries the circling, with a period of 21 s to 22 s, which a phase decomposition in Chapter 4 can be held to rather than asked to invent. And C is already small at the bottom of the fitted grid, falling below $1/e$ by 74 s for paragliders and 60 s for hang gliders, within its first lags — a property of the pipeline as much as of the flying, since the grid starts at the smoothing scale of the slowest logger retained (Sec. 2.7.7).

Integrability, not size, sustains the exponent. A memory spent by 60 s looks incompatible with increments still correlated to 2,000 s, but it is not the size of C that sets the growth law. For a stationary velocity the Green–Kubo relation

$$\langle|\Delta\mathbf{r}(\Delta)|^2\rangle = 2\int_0^{\Delta}(\Delta-\tau)\,C(\tau)\,\mathrm{d}\tau \tag{3.13}$$

makes the growth law a statement about the convergence of an integral: a tail $C(\tau)\sim\tau^{-\gamma}$ with $\gamma<1$ has no finite integral and gives $\langle|\Delta\mathbf{r}|^2\rangle\sim\Delta^{2-\gamma}$ across the window however small C has already become, while $\gamma>1$ gives diffusive motion beyond the correlation time whatever C is worth at the floor; multiplying C by a constant moves every value on it and leaves γ where it was. A single power law fitted to 60 s to 1,179 s gives $\gamma=0.61$ for paragliders and 0.57 for hang gliders, but that fit is itself an arch, the local slope running from 0.30 to 0.84 across the same lags. What the argument needs is not the fitted value: *every* local slope is below unity, in both disciplines and at all 15 lags, so non-integrability holds pointwise rather than on a fitted average — the stronger statement, and one that does not depend on the tail being a power law at all (implementation details: Sec. I3). Two features of the estimate push γ up rather than down (a mean velocity estimated from the same record pulls the tail towards zero, and the first decade of the integral in Eq. (3.13) lies below the grid), so the exponent $2-\gamma$ implied for the displacement is a floor rather than an estimate — 1.39 and 1.43 — and the plain increment, $\alpha_1=1.75$ and 1.70, clears both floors: a consistency between two channels, not a second determination, that excludes a velocity memory integrable over this window.

The turning angles. The persistence has a microscopic form: the angle between successive horizontal displacement vectors, taken at a stride of 5 samples so the smoothing window shares no sample between the two vectors (implementation details: Sec. I3). Its median is 12° for paragliders, over 272,503,728 angles, and 20° for hang gliders, with 63 % and 56 % falling within $\pi/6$ of the previous heading; restricted to matched 1 Hz segments the medians are 11.3° and 17.1°, so the gap between disciplines survives matching, losing about a quarter of its size. Memoryless reorientation, a right-angle median, is nowhere near. A wing that changes heading by 12° at a time is a correlated walk, and a positive $C(\tau)$, a non-integrable tail, and increments still correlated at 2,000 s are the ensemble consequences of the same forward peak — the warrant for the persistent random walk of Sec. 4.4. The angles outside the forward cone are a minority of the same order as the climbing share of Sec. 2.8 (45 % for paragliders, 47 % for hang gliders): pooled, climbing and gliding give one forward-peaked distribution and a $C(\tau)$ with no sign change, which is why the circling is invisible to both statistics here and not absent from the flying. The segmentation of Sec. 4.1 separates them, and the turning angles conditioned on phase are where this result has to reappear.

3.10 Speed and vertical velocity

The horizontal speed and the vertical velocity are marginals of the smoothed kinematics of Sec. 2.7.7, taken over every retained in-segment fix, accumulated in the same streaming pass that builds the increment histograms of Sec. 3.5 (implementation details: Sec. I3). Table 3.7 gives both, each read against a reference fixed outside the archive rather than against itself.

The disciplines separate at every speed percentile, the hang-glider median lying above the paraglider ninetieth percentile, and both ninety-ninth percentiles sit well inside their fix-level

Table 3.7. Marginal kinematics of the retained ensemble, against the fix-level bounds of Sec. 2.7.2: $45\,\mathrm{m\,s^{-1}}$ (paragliders) and $55\,\mathrm{m\,s^{-1}}$ (hang gliders) on the horizontal speed, $\pm 13\,\mathrm{m\,s^{-1}}$ on the vertical velocity. A retained population pressed against them would mean the bound had cut into the fast tail of the flying rather than into instrument artefacts.

	paragliders	**hang gliders**
horizontal speed, median	$9.7\,\mathrm{m\,s^{-1}}$	$15.7\,\mathrm{m\,s^{-1}}$
horizontal speed, P90	$13.6\,\mathrm{m\,s^{-1}}$	$22.1\,\mathrm{m\,s^{-1}}$
horizontal speed, P99	$17.3\,\mathrm{m\,s^{-1}}$	$28.7\,\mathrm{m\,s^{-1}}$
vertical velocity, median	$-0.10\,\mathrm{m\,s^{-1}}$	$-0.12\,\mathrm{m\,s^{-1}}$
vertical velocity, 10–90% band	$-2.02\,\mathrm{m\,s^{-1}}$ to $2.10\,\mathrm{m\,s^{-1}}$	$-2.45\,\mathrm{m\,s^{-1}}$ to $2.56\,\mathrm{m\,s^{-1}}$
climbing fraction	45 %	47 %

bound. The vertical velocity is centred just below zero and stays inside its bound at both ends. Writing $f_\uparrow$ for the climbing fraction of a flight and $\langle \dot z\rangle_\uparrow$, $\langle \dot z\rangle_\downarrow$ for the mean climb and sink rates, a flight that ends near the altitude it began at has $\int \dot z\,\mathrm{d}t \approx 0$, i.e.

$$f_\uparrow \langle \dot z\rangle_\uparrow + (1-f_\uparrow)\langle \dot z\rangle_\downarrow \approx 0 \quad\Longrightarrow\quad \frac{f_\uparrow}{1-f_\uparrow} \approx \frac{|\langle \dot z\rangle_\downarrow|}{\langle \dot z\rangle_\uparrow}, \tag{3.14}$$

the climbing share required near a half whenever the two rates are comparable. Over the median flight of 2.63 h and 3.07 h (Sec. 2.8) both measured shares fall just short of a half — the same fact as the negative median, since the wing spends more of its time sinking and its mean rate of climb exceeds its mean rate of sink. The sign of the vertical velocity is the coarsest climb–glide rule available, so these two shares are what a segmentation in Chapter 4 must return as its threshold goes to zero.

The return probability. The return probability, the increment propagator at its centre $P_0(\Delta) \equiv P(\mathbf{0}, \Delta)$, decays as Δ^{-H} under the scaling form of Sec. 3.5 and so is another reading of H, at a fifth probability level costing a fit and no further traversal of the fix table; it is defined here and not measured. First-passage times are not on this list either: they were measured and withdrawn in Sec. 3.5, since a time to reach a radius from take-off carries the launch geometry into the space domain.

3.11 What this establishes, and what the segmentation receives

Over 60 s to 2,000 s, measured within a retained segment by an estimator that annihilates a polynomial trend rather than estimating one, cross-country soaring gives $\alpha_2 = 2.02 \pm 0.10$ for paragliders and 1.94 ± 0.13 for hang gliders, on 155,788 and 6,132 flights. The wing classes that depart from the pooled ensemble MSD by 51 % (Sec. 3.3) move this exponent by 0.06; native cadence, across a factor of ten in the scale below which the trajectory has been smoothed, moves it by 0.02, season by 0.03, and the declared task by −0.10, the largest of the four. The two disciplines agree within their uncertainties, which is not guaranteed: they differ in speed and in glide ratio.

The class of process. Table 3.8 collects every global number this chapter measures on the unsegmented trajectory. Read together they fix a class rather than a list: over this window the archive is a monofractal (one exponent for every moment, no Lévy knee), non-Gaussian propagator (a between-flight width mixture, not a heavy tail within one record), directionally persistent motion (a non-integrable velocity memory, forward-peaked turning) that is never isotropic enough to summarise with one component instead of two — neither the Lévy walk this thesis was framed around nor a Gaussian one.

Table 3.8. The process, in eight numbers: every global quantity this chapter measures on the unsegmented trajectory, over 60 s to 2,000 s. Together they fix what the transport class is and rule out what it is not; each is measured and qualified in the section of the same name earlier in this chapter.

Property	**paragliders**	**hang gliders**	**reads as**
transport exponent $\alpha_2 = 2H$	2.02 ± 0.10	1.94 ± 0.13	super-diffusive
H bracket, 3 routes	0.876–1.009	0.852–0.972	scaling approximate
moment spectrum, $\nu(q)$ spread	0.019	0.009	monofractal
non-Gaussian, pooled median	+0.042	+0.181	non-Gaussian
non-Gaussian, within one flight	-0.118	-0.084	sub-Gaussian per flight
velocity memory, local γ	0.30–0.84	0.34–0.66	non-integrable
anisotropy, amplitude	0.80	1.19	never isotropic
anisotropy, exponent	0.890 / 0.908	0.860 / 0.911	direction-dependent

The limits. The window is 1.5 decades wide, and neither end is set by the size of the archive. Below it the trajectory has been smoothed over a window whose floor is 5 samples and which therefore scales with the logger's cadence (Sec. 2.7.7); nothing below that scale is read as motion. Above it the declared task governs the displacement, the separation between closed and open courses in the local slope growing from 0.03 at 60 s to 1.12 at 12,156 s. Inside the window the scaling form is only approximate: H falls from 0.937 at the bulk of the increment distribution to 0.868 at its flank, a spread across the quantiles of 0.116 and 0.163 against 0.018 on an exactly self-similar process measured the same way. The exponent itself moves by 0.104 when the fitted range is halved, against 0.001 from resampling at the day and site, and the 22 lags of the fit carry no more than 2.0 effective degrees of freedom, a budget for one exponent and not for a count of regimes. That the transport is characterised by $\alpha = 2.02$ is therefore a different, stronger statement than the effective exponent quoted above, and the range dependence and the quantile spread are each enough to withhold it. The window will not widen with more data: a soaring day ends, so the upper cutoff follows from the system rather than the method.

What was withdrawn. Four measurements are withdrawn. The ensemble MSD about take-off yields a timescale instead of an exponent: displacement from the launch site crosses over where the population leaves its launch area, at a median of 1,725 s for paragliders and 1,530 s for hang gliders on the first crossing of 5 km. The first-passage time was brought in as a route to H in the space domain and withdrawn when its exponent moved across sub-ranges of radius by more than the uncertainty it was meant to check; what it bought is the sharpest evidence that the ensemble curve's failure belongs to the common origin rather than the time axis, since exchanging the time axis for a radius reproduces it (implementation details: Sec. I3). The sign-run count on the MSD residuals is kept as a description of how those residuals are arranged and withdrawn as a test, every lag of that curve being an average over the same flights; what carries the claim instead is the size of the residual against the standard error on each lag. And the reading of a straight moment spectrum as a Gaussian propagator is withdrawn, one exponent for every moment being a statement about how the distribution scales and none about its shape; what it bought is the non-Gaussian parameter, computed from moments the spectrum had already accumulated.

The hand-over. Chapter 4 inherits a class of transport and five constraints on how it may be decomposed.

(i) Any statistic measured from take-off inherits the launch geometry, out to the departure times above, and the contamination is not confined to the observable that exposed it: any phase statistic binned by elapsed time since take-off carries it below those times. The remedy is the one used throughout this chapter, to measure within a segment and never about a common origin (Sec. 2.7.6).

(ii) The estimator is additive over any partition of the sample: $V_2(\Delta)$ is a mean over the placements of a second difference, and labelling those placements by phase and recombining the per-phase means with the weights actually observed returns the pooled curve identically. That is a property of the estimator rather than of the transport, asserting no decomposition of an exponent into phase exponents, only that the arithmetic closes; a recombination that misses the pooled value is evidence about the segmentation and is to be read that way before it is read as physics.

(iii) The two horizontal components are analysed separately and never pooled. They are incompatible in amplitude (Sec. 2.8) and differ in exponent: $H = 0.890$ east against 0.908 north for paragliders, and 0.860 against 0.911 for hang gliders. Pooling presupposes an isotropy this ensemble does not have. Sec. 3.6 puts the same split through the time average and both filtered-variation orders and finds the same ordering; the pooled quantile route is not the only estimator that sees it.

(iv) The pooled non-Gaussian parameter is $+0.042$ for paragliders and $+0.181$ for hang gliders, and against a matched Gaussian null both sit *below* it, by -0.118 and -0.084. What is handed on is therefore a between-flight amplitude spread and not an increment tail: $\mathrm{CV}(S_f) = 0.35$ against 0.48 at 60 s. A phase decomposition has to produce that spread across records, and must not produce a heavy tail within one.

(v) The ergodicity question is handed on in two forms that need no power law: the drift of the exponent with the time already flown, obtained by running the same within-segment estimator on successive stretches of a record (Sec. 4.1); and the ergodicity-breaking parameter of Appendix 3.A, which compares the distribution of the per-flight time average across flights rather than two fitted slopes. The usual comparison of α_{EA} with α_{TA} requires both curves to be power laws, and this archive supplies one of the two.

The class of transport is fixed over one window; which flight behaviour produces it is not. Every number above mixes climbing, gliding and searching into one, and the persistent random walk of Sec. 4.4 takes its warrant from the memory measured here rather than from any single one of them. Chapter 4 separates the phases, and is answerable to the five constraints.

Chapter 4

Flight phases

Chapter 3 establishes which class of transport the archive exhibits, but it cannot say which flight behaviour produces it, since every observable there averages over climbs, glides and searches together. Disentangling them requires segmenting each flight into its phases, and that is the work this chapter plans.

The distinction is worth stating plainly, because it sets what a segmentation is allowed to claim. Pre-segmentation determines *which* class of transport is present; segmentation explains *which mechanism* produces it. A segmentation explains an exponent, it does not measure one — and if the two disagree, the disagreement is evidence about the segmentation rather than about the transport.

None of this chapter is implemented. Each section states what is to be built and what it would settle.

4.1 Segmentation into flight phases

Building on the previous work of Vilpellet et al. [9], each flight is segmented into the three phases of the soaring cycle:

Transition (glide). The inter-thermal cruise: a near-straight, energy-efficient glide toward the next source of lift (sinking, large directed horizontal displacement). This is the directed "step".

Search. The exploratory phase entered when no thermal is found at a comfortable altitude: gentle turns and probing manoeuvres to sample the air mass and locate the next thermal while limiting altitude loss. It mediates the passage from one step to the next trap.

Climb (thermalling). Circling within rising air to regain altitude: persistent turning with small net horizontal displacement. This is the localized "trap".

That work performs the segmentation with a hidden Markov model [22] whose observations are a small set of *binary* trajectory-derived features — the sign of the vertical velocity, the persistence of the curvature angle, and indicators of trajectory straightness.

A first task of this thesis is to scrutinize this segmentation rather than take it for granted: to assess whether those binary variables are adequate and robust on the larger, more heterogeneous dataset assembled here, and whether the feature set or the model should be refined, extended, or replaced. The segmentation method is therefore itself an object of study, and the specific variables are deliberately left open at this stage. This three-state description refines the two-state caricature underlying continuous-time random walks and Lévy walks.

4.2 Phase-resolved observables

The segmentation gives access to a second group of observables, defined per phase occurrence and then as ensemble distributions. These are the building blocks of an intermittent-transport description and, unlike those of Section 3.1, cannot be obtained without first segmenting the flight.

The waiting phase: climb and search. Between two consecutive glides the flier advances little horizontally: it searches for a thermal and climbs in it. For a transport model these two phases together form the *waiting time* between directed steps, so we treat them on an equal footing and build three distributions – the climb duration τ_c, the search duration τ_s, and their sum $\tau_w = \tau_s + \tau_c$, the total time spent essentially stationary between glides. The distinction matters: Vilpellet et al. [9] report the search times to be broadly *power-law* distributed while the climb times are closer to *exponential*, so that the heavy tail of the combined waiting time $\psi(\tau_w)$, the quantity that determines whether trapping drives sub-diffusion, would be inherited from the search phase. Whether this still holds on the larger dataset is itself one of the things to re-check once the segmentation is in place. We therefore report $\psi(\tau_w)$ together with the separate $\psi(\tau_c)$ and $\psi(\tau_s)$. Alongside the durations we record the climb kinematics (altitude gained Δz, mean climb rate, thermalling radius and period) and the search geometry (path length, tortuosity, net displacement) – the search being the phase in which prior work locates the signature of pilot skill.

The directed step: glide (transition). Each glide contributes a horizontal step of length ℓ and duration τ_g, with ground speed, glide ratio (horizontal distance per unit altitude lost) and sink rate. Because a glide is close to ballistic, $\ell \approx v_g \tau_g$, step length and duration are expected to be nearly proportional; we test this directly through the joint law of (ℓ, τ_g) – the ℓ–τ_g scatter, the conditional mean $\langle \ell \mid \tau_g \rangle$, and the spread of the ratio ℓ/τ_g (the glide speed). Both marginals matter: the tail of the step-length distribution $\lambda(\ell)$ controls super-diffusive contributions, so we compute the distributions of ℓ and τ_g separately as well as their joint law.

Anatomy of a flight. Number of thermals (climbs) per flight, inter-thermal distances, and the fraction of both time and along-track distance spent in each phase: the coarse composition that sets the weight of each phase in the global transport.

Angular diffusion between glides. The turning angle between the heading of one transition segment and the next – the segment-level counterpart of the turning angle of Section 3.1. How fast successive glide directions decorrelate is the key ingredient of the minimal model below.

Two couplings then decide which transport model applies, and are stated here as explicit objects of study. The first is the *step–waiting coupling*: whether a long glide tends to be followed by a long wait, i.e. whether ℓ and τ_w are correlated. A genuine coupling is the defining feature of a Lévy walk and is what separates it from a decoupled CTRW, in which jumps and waiting times are drawn independently.

The second is the *memory in the phase sequence*. A flight is a discrete sequence over the three states $\{\text{transition}, \text{search}, \text{climb}\}$, and memory can hide in two places. (i) The *order* of the phases, summarised by the transition matrix $P_{ij} = \Pr(\text{next} = j \mid \text{current} = i)$: a perfectly cyclic flight $(\mathrm{T} \to \mathrm{S} \to \mathrm{C} \to \mathrm{T})$ gives a near-deterministic matrix, whereas aborted climbs, repeated searches or skipped phases appear as off-cycle probabilities and reveal how stereotyped the soaring cycle actually is. (ii) The *durations*, through correlations between consecutive phase durations – for instance whether a long climb tends to precede a long glide.

Both probe the *renewal* assumption underlying the textbook CTRW and Lévy-walk results, in which each step and wait is independent of the history: if the transition matrix is structured or the durations are correlated, the process is non-renewal and those formulas must be amended. These measurements determine whether a memoryless model is admissible.

4.3 Reproducing the reference results

A first objective of the analysis, undertaken once the segmentation is in place, is to check whether the enlarged—and eventually multi-source—dataset reproduces the quantitative and graphical results of Vilpellet et al. [9]. That study covers paragliders, hang gliders and sailplanes; the present data cover the first two (Chapter 2), so both the paraglider and the hang-glider comparisons are within reach, while the sailplane figures become reachable once that source is added. Two figures are the main targets.

The first is the per-phase conditional MSD (their Fig. 3): the transition segments should stay ballistic ($\alpha = 2$, i.e. $H = 1$) across time lags; the search phase should be strongly sub-diffusive ($H \approx 0.3$ for paragliders and hang gliders, ≈ 0.2 for sailplanes); and the climb phase should be quasi-ballistic at long lags but with an effective velocity about two orders of magnitude smaller, including the transient anti-correlation near the $\sim 30\,\mathrm{s}$ thermalling period.

The second is the set of per-phase descriptive statistics (their Fig. 4): the glide ratios ($\approx$ 8.5, 10, and 25 for paragliders, hang gliders, and sailplanes), the horizontal-velocity statistics, the heavy-tailed distribution of transition times, the search-time fraction and the effective search radius, the vertical climb velocities, the thermalling radii, and the thin, near-exponential distribution of climb times—together with the global time budget ($\approx$60% transition, $\approx$35% climb, the remainder searching). The global exponent is not on this list: it needs no segmentation, and Sec. 3.11 has already put this archive's value beside the reported $H \approx 0.88$.

Recovering these results would validate the acquisition pipeline and the segmentation and would test the universality of the reported behaviour on a substantially larger sample; systematic departures would be equally informative, pointing to dataset- or method-dependent effects to investigate. This comparison is therefore the bridge between the present data-acquisition stage and the original analysis that follows.

4.4 Modelling and simulation

The aim is a *minimal* stochastic model that reproduces the asymptotic anomalous scaling reported for soaring (a Hurst exponent $H \approx 0.88$) with as few ingredients as possible. A preliminary working hypothesis—to be made precise, and possibly revised, once the empirical phase statistics are available—is that the transport is carried mainly by the transition (glide) phases. The model would therefore resolve the transition segments explicitly while merging search and climb into a single effective non-transition state, so that the soaring cycle reduces to a sequence of directed glides interrupted by reorientations. Anomalous transport would then emerge from the directed transition segments combined with an *angular diffusion* of the heading from one transition to the next: successive glide directions decorrelate gradually rather than abruptly, producing a correlated random walk whose persistence controls the long-time exponent. Simulated ensembles would be compared with the data to test whether these ingredients suffice. These are baseline ideas, to be refined—or changed—as the analysis progresses.

4.5 Tooling

Mirror the acquisition package with an analysis package, and let the figures and tables of this document be generated automatically from the analysis outputs – as the dataset statistics already are – so that this document stays in step with the work and remains a reliable basis for the thesis and for presentations.

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Appendices

Each appendix is numbered after the body chapter that first cites it: Appendices 2.A and 2.B belong to Chapter 2, Appendix 3.A to Chapter 3. They follow one another in the order in which the text first cites them.

Appendix 2.A

Power spectral density and Welch's method

This appendix explains the tool used in Chapter 2 to compare the two altitude channels of an IGC file, namely the *power spectral density* (PSD) estimated with *Welch's method* [23]. No prior familiarity with spectral estimation is assumed.

The power spectral density

Consider a signal sampled at a fixed rate, here an altitude series $z_n = z(n\,\Delta t)$, $n = 0, \dots, N-1$, with sampling interval Δt (so the sampling frequency is $f_s = 1/\Delta t$). Any such finite signal can be written as a sum of sinusoids of different frequencies (its discrete Fourier transform). The **power spectral density** describes how the power (the variance) of the signal is distributed across frequencies. A slow drift puts its power at low frequency; rapid jitter puts its power at high frequency. Formally the PSD $S(f)$ is the Fourier transform of the autocovariance of the signal, and it is normalised so that the total area under it equals the variance of the signal,

$$\int_0^{f_s/2} S(f)\,\mathrm{d}f \;=\; \mathrm{Var}(z), \tag{2.A.1}$$

with units of (metres)^2 per hertz. The upper limit $f_s/2$ is the *Nyquist frequency*: a series sampled every Δt cannot resolve anything faster than one oscillation per two samples. This separates a slowly drifting but locally clean channel (power concentrated at low f) from one with a high-frequency noise floor (power that stays large up to the Nyquist frequency).

The periodogram

The naive estimator of $S(f)$ is the **periodogram**: take the discrete Fourier transform $\hat{z}(f) = \sum_n z_n\, e^{-2\pi i f n \Delta t}$ and form $|\hat{z}(f)|^2$ (suitably normalised). It is asymptotically unbiased but *not consistent*: its variance does not shrink as the record gets longer, because a longer signal gives more frequency points, each as noisy as before. Plotted directly it is a spiky curve, too noisy for a reliable comparison of two channels.

Welch's method

Welch's method turns the noisy periodogram into a usable estimate by *averaging*. It proceeds in three steps:

(i) **Segment.** Split the record into several shorter, overlapping segments.

(ii) **Window.** Multiply each segment by a smooth taper (a Hann window) that falls to zero at its ends. This suppresses the spurious high-frequency power (*spectral leakage*) that an abrupt segment boundary would otherwise inject.

(iii) **Average.** Compute the periodogram of each windowed segment and average them.

Averaging K segments cuts the variance of the estimate by roughly a factor K, at the cost of coarser frequency resolution (set by the segment length rather than the whole record). The result is a smooth PSD estimate whose *level* can be compared meaningfully between two signals; it is the standard, robust spectral estimator. Each segment is typically detrended before transforming (e.g. by removing a linear fit), so that the estimate reflects the fluctuations about the trend, where the sensor noise lives, rather than the trend itself (implementation details: Sec. I2.7.1).

The spectra of the two altitude channels, and the choice of channel they motivate, are discussed in Sec. 2.7.1; this appendix covers only the method.

Appendix 2.B

Geodetic transforms: from (φ, λ, h) to ECEF to ENU

This appendix derives the two formulas that Sec. 2.7.5 uses as given: the geodetic-to-ECEF transform, Eq. (2.6), and the ECEF-to-ENU rotation matrix. It closes with the size of the tangent-plane error quoted there. Notation as in the body: WGS84 semi-major axis a, flattening f, $b = a(1-f)$ the semi-minor axis, $e^2 = f(2-f) = 1 - b^2/a^2$ the first eccentricity squared [17].

2.B.1 Geodetic coordinates to ECEF

The ECEF frame is Cartesian, centred at the Earth's centre O: the Z axis along the polar axis, the X axis through the intersection of the equator and the prime meridian, Y completing the right-handed triad. By symmetry, a point at longitude λ lies in the meridian plane obtained by rotating the X–Z plane by λ; writing p for the distance from the polar axis, its ECEF coordinates are

$$X = p\cos\lambda, \qquad Y = p\sin\lambda, \qquad Z = Z. \tag{2.B.1}$$

The problem therefore reduces to the meridian ellipse: find (p, Z) of a point at geodetic latitude φ and height h.

The surface point. The meridian section of the ellipsoid is the ellipse

$$\frac{p^2}{a^2} + \frac{Z^2}{b^2} = 1. \tag{2.B.2}$$

The geodetic latitude is defined through the surface *normal* (Figure 2.4). The gradient of the left-hand side of (2.B.2) is proportional to $(p/a^2,\ Z/b^2)$, so the outward normal at a surface point (p_s, Z_s) makes an angle φ with the equatorial plane given by

$$\tan\varphi = \frac{Z_s/b^2}{p_s/a^2} \qquad \Longrightarrow \qquad Z_s = \frac{b^2}{a^2}\, p_s \tan\varphi = (1-e^2)\, p_s \tan\varphi. \tag{2.B.3}$$

Substituting (2.B.3) into (2.B.2), with $b^2 = a^2(1-e^2)$,

$$\frac{p_s^2}{a^2}\Big(1 + (1-e^2)\tan^2\varphi\Big) = 1 \quad \Longrightarrow \quad p_s^2 = \frac{a^2\cos^2\varphi}{\cos^2\varphi + (1-e^2)\sin^2\varphi} = \frac{a^2\cos^2\varphi}{1 - e^2\sin^2\varphi}, \tag{2.B.4}$$

that is,

$$p_s = N(\varphi)\cos\varphi, \qquad N(\varphi) \equiv \frac{a}{\sqrt{1 - e^2\sin^2\varphi}}, \tag{2.B.5}$$

and, back through (2.B.3),

$$Z_s = (1 - e^2)\, N(\varphi) \sin\varphi. \tag{2.B.6}$$

The quantity $N(\varphi)$ so defined is the *prime-vertical radius of curvature*; a short computation shows it is precisely the distance, along the normal, from the surface point to the polar axis (the segment Q–P of Figure 2.4): the normal direction in the meridian plane is $(\cos\varphi, \sin\varphi)$, and following it inward from (p_s, Z_s) a length N lands on the axis, $p_s - N\cos\varphi = 0$ by (2.B.5).

Adding the height. The height h of a fix is measured along the same normal, whose unit vector in ECEF components is

$$\hat{\mathbf{u}} = (\cos\varphi\cos\lambda,\ \cos\varphi\sin\lambda,\ \sin\varphi). \tag{2.B.7}$$

Adding $h\,\hat{\mathbf{u}}$ to the surface point (2.B.5)–(2.B.6) in the meridian plane and restoring the longitude via (2.B.1):

$$X = (N + h)\cos\varphi\cos\lambda, \qquad Y = (N + h)\cos\varphi\sin\lambda, \qquad Z = \big(N(1 - e^2) + h\big)\sin\varphi, \tag{2.B.8}$$

which is Eq. (2.6). The asymmetry between the horizontal factor $(N + h)$ and the vertical one $(N(1 - e^2) + h)$ is the ellipsoid's flattening at work: on a sphere $(e = 0)$ both reduce to $R + h$ and the geodetic and geocentric latitudes coincide.

How far the two latitudes differ. The geocentric latitude ψ of a point on the surface is the angle its position vector makes with the equatorial plane, so from (2.B.5)–(2.B.6), at $h = 0$,

$$\tan\psi = \frac{Z_s}{p_s} = \frac{N(1 - e^2)\sin\varphi}{N\cos\varphi} = (1 - e^2)\tan\varphi, \tag{2.B.9}$$

the classic relation: the two agree only where $e = 0$, or at the equator and the poles, where tan is 0 or infinite. To see the size of the gap, write $\psi = \varphi - \Delta$ and expand for small e^2. Using $\tan(\varphi - \Delta) \simeq \tan\varphi - \Delta\,(1 + \tan^2\varphi)$ in (2.B.9),

$$\tan\varphi - \Delta\,(1 + \tan^2\varphi) \simeq (1 - e^2)\tan\varphi \quad\Longrightarrow\quad \Delta \simeq \frac{e^2\tan\varphi}{1 + \tan^2\varphi} = e^2\sin\varphi\cos\varphi, \tag{2.B.10}$$

that is

$$\varphi - \psi \simeq \frac{e^2}{2}\,\sin 2\varphi, \tag{2.B.11}$$

which is the expression quoted in Sec. 2.7.5. It is largest where $\sin 2\varphi = 1$, i.e. exactly at $\varphi = 45^\circ$, and there equals $e^2/2 = 3.347 \times 10^{-3}$ radian, or $0.192^\circ \approx 11.5'$. Mistaking one latitude for the other therefore displaces a fix along the meridian by $R_\oplus\,(e^2/2) \approx 21$ km at mid-latitudes—the $\sim$20 km quoted in the body. The error vanishes at the equator and at the poles and is symmetric about 45°.

2.B.2 ECEF to local ENU

The local frame at the origin fix $(\varphi_0, \lambda_0, h_0)$ is spanned by three unit vectors: East, tangent to the parallel; North, tangent to the meridian, pointing to the pole; Up, along the ellipsoid normal (2.B.7). East is the direction in which the position (2.B.1) changes under λ alone,

$$\hat{\mathbf{e}}_E = \frac{\partial\mathbf{X}/\partial\lambda}{\|\partial\mathbf{X}/\partial\lambda\|} = (-\sin\lambda_0,\ \cos\lambda_0,\ 0), \tag{2.B.12}$$

Up is $\hat{\mathbf{u}}$ of (2.B.7) evaluated at (φ_0, λ_0), and North completes the right-handed triad,

$$\hat{\mathbf{e}}_N = \hat{\mathbf{e}}_U \times \hat{\mathbf{e}}_E = (-\sin\varphi_0\cos\lambda_0,\ -\sin\varphi_0\sin\lambda_0,\ \cos\varphi_0). \tag{2.B.13}$$

The ENU components of a displacement $\Delta\mathbf{X}$ are its projections on the triad, $E = \hat{\mathbf{e}}_E\cdot\Delta\mathbf{X}$ and likewise for N and U; stacking the three row vectors gives exactly the matrix of Sec. 2.7.5,

$$\begin{pmatrix} E \\ N \\ U \end{pmatrix} = \underbrace{\begin{pmatrix} -\sin\lambda_0 & \cos\lambda_0 & 0 \\ -\sin\varphi_0\cos\lambda_0 & -\sin\varphi_0\sin\lambda_0 & \cos\varphi_0 \\ \cos\varphi_0\cos\lambda_0 & \cos\varphi_0\sin\lambda_0 & \sin\varphi_0 \end{pmatrix}}_{R} \Delta\mathbf{X}, \tag{2.B.14}$$

whose rows are orthonormal by construction, so R is a rotation ($R^{\mathsf{T}}R = \mathbb{1}$) [18]. Two remarks used in the body follow directly. First, the rows depend on the *geodetic* latitude because Up is the ellipsoid normal — the same φ that enters Eq. (2.6), which is why no latitude conversion appears anywhere in the pipeline. Second, the matrix is evaluated once per flight, at the origin fix: the frame is a fixed tangent frame, not one that travels with the wing.

2.B.3 Size of the tangent-plane error

Working in a fixed tangent frame replaces distances measured *along* the curved surface by straight-line (chord) distances in the plane. For two surface points separated by an arc d on a sphere of radius $R_\oplus$, the chord is

$$c = 2R_\oplus \sin\frac{d}{2R_\oplus} = d - \frac{d^3}{24\,R_\oplus^2} + O(d^5), \tag{2.B.15}$$

so the horizontal distance error of the tangent-plane picture is $d^3/(24R_\oplus^2)$: about 1 cm at $d = 20$ km and 13 cm at $d = 50$ km — far below the metre-scale GPS noise on the same displacement, which is the comparison made in Sec. 2.7.5. The *vertical* coordinate of a distant surface point is a different matter: the surface falls below the tangent plane by the sagitta $\approx d^2/(2R_\oplus)$, which reaches tens of metres already at $d = 20$ km. This is why the analysis never takes its working altitude from the plane geometry: the vertical coordinate is the barometric altitude itself (Sec. 2.7.5), and the U component serves only the geometric construction.

Appendix 3.A

CTRW with power-law waiting times: EA-MSD vs TA-MSD

This appendix derives the ensemble-averaged MSD (EA-MSD) and the time-averaged MSD (TA-MSD) for the continuous-time random walk (CTRW) with heavy-tailed waiting times, and shows explicitly that the two differ and that the TA-MSD depends on the observation time T.

Model

A walker takes steps of length l_i (iid, symmetric, $\langle l^2 \rangle = \sigma^2 < \infty$) separated by waiting times τ_i (iid) drawn from

$$\psi(\tau) \sim \frac{\alpha\, \tau_0^\alpha}{\tau^{1+\alpha}}, \qquad \tau \to \infty, \quad 0 < \alpha < 1. \tag{3.A.1}$$

The mean waiting time diverges ($\langle \tau \rangle = \infty$), so the walker is anomalously slow. Let $n(t)$ be the number of steps completed by time t, $T_n = \sum_{i=1}^{n} \tau_i$ the time of the n-th step, and

$$x(t) = \sum_{i=1}^{n(t)} l_i \tag{3.A.2}$$

the position at time t.

Ensemble-averaged MSD

Since steps and waiting times are independent,

$$\langle x^2(t) \rangle = \sigma^2 \, \langle n(t) \rangle. \tag{3.A.3}$$

For large t the mean number of renewals in $[0, t]$ follows from the renewal equation and the Tauberian theorem applied to the Laplace transform of ψ. Because $\hat{\psi}(s) \equiv \int_0^\infty e^{-s\tau} \psi(\tau)\, \mathrm{d}\tau \simeq 1 - \Gamma(1-\alpha)\,(\tau_0 s)^\alpha + \cdots$ for $s \to 0$ (a consequence of ψ having a power-law tail with exponent $1+\alpha$ and amplitude $\alpha \tau_0^\alpha$), the renewal theorem gives

$$\langle n(t) \rangle \sim \frac{t^\alpha}{\Gamma(1-\alpha)\,\Gamma(1+\alpha)\,\tau_0^\alpha} = \frac{\sin \pi\alpha}{\pi\alpha} \left(\frac{t}{\tau_0}\right)^\alpha, \qquad t \to \infty. \tag{3.A.4}$$

Hence the **EA-MSD** grows sub-linearly,

$$\mathrm{MSD}(t) = \langle x^2(t) \rangle \sim \frac{\sigma^2}{\Gamma(1-\alpha)\,\Gamma(1+\alpha)\,\tau_0^\alpha}\, t^\alpha \equiv K_\alpha\, t^\alpha, \qquad 0 < \alpha < 1. \tag{3.A.5}$$

The transport is sub-diffusive with Hurst exponent $H = \alpha/2 < 1/2$.

Time-averaged MSD and its mean

For a trajectory of duration T, the time-averaged MSD at lag $t \ll T$ is

$$\overline{\delta^2}(t;T) = \frac{1}{T-t} \int_0^{T-t} \left[x(t_0+t) - x(t_0)\right]^2 \mathrm{d}t_0. \tag{3.A.6}$$

Since $\langle [x(t_0+t) - x(t_0)]^2 \rangle = \sigma^2 \langle n(t_0+t) - n(t_0) \rangle$, the mean of the integrand equals σ^2 times the mean number of steps in any window of width t. The integral $\int_0^{T-t} [n(t_0+t) - n(t_0)]\, \mathrm{d}t_0$ counts, for each step k that lands at time $T_k \in [0,T]$, the length of the t-wide sliding window that contains T_k. For T_k deep in the interior ($t \le T_k \le T-t$) this length is exactly t; the contributions from the two boundary layers of width t are negligible for $t \ll T$. Therefore

$$\int_0^{T-t} [n(t_0+t) - n(t_0)]\, \mathrm{d}t_0 \approx t\, n(T), \qquad t \ll T, \tag{3.A.7}$$

and

$$\overline{\delta^2}(t;T) \approx \frac{\sigma^2\, t\, n(T)}{T}. \tag{3.A.8}$$

Taking the ensemble average and using $\langle n(T) \rangle \sim K_\alpha T^\alpha / \sigma^2$,

$$\mathrm{TAMSD}(t,T) \equiv \langle \overline{\delta^2}(t;T) \rangle \approx \sigma^2 \frac{\langle n(T) \rangle}{T}\, t \sim K_\alpha\, T^{\alpha-1}\, t. \tag{3.A.9}$$

Comparing (3.A.5) and (3.A.9):

- The EA-MSD scales as t^α (sub-diffusive), whereas the TA-MSD scales as t^1 (linear in the lag) – a qualitatively different dependence on t.
- The TA-MSD carries an explicit $T^{\alpha-1}$ prefactor: since $\alpha < 1$ this is a decreasing function of T, so longer trajectories yield a smaller TA-MSD at the same lag. This T-dependence is the signature of *aging*: the increment statistics depend on how long the process has already been observed.
- The ratio $\mathrm{TAMSD}(t,T)/\mathrm{MSD}(t) \sim (t/T)^{1-\alpha} \to 0$ as $T \to \infty$ at fixed t.

Distribution of the TA-MSD: non-self-averaging

Equation (3.A.8) shows that $\overline{\delta^2}(t;T) \approx \sigma^2 t\, n(T)/T$, so the randomness in the TA-MSD comes entirely from $n(T)$, the total number of steps in $[0,T]$. For a power-law renewal process the rescaled count

$$\xi \equiv \frac{n(T)}{\langle n(T) \rangle} \tag{3.A.10}$$

converges in distribution as $T \to \infty$ to a non-degenerate limit law: the *Mittag-Leffler distribution M_α*, whose moments are $\langle \xi^n \rangle = n!\, \Gamma(1+\alpha)^n / \Gamma(1+n\alpha)$ [20]. This is a broad, non-Gaussian distribution on $(0,\infty)$ that does *not* concentrate around $\xi = 1$ for $\alpha < 1$: different trajectories give systematically different values of $\overline{\delta^2}(t;T)$ even in the limit $T \to \infty$. Hence the TA-MSD is *non-self-averaging*: the value from a single trajectory is not a reliable estimate of any ensemble quantity, however long the trajectory.

Quantitatively, the ergodicity-breaking parameter

$$\mathrm{EB}(t,T) \equiv \frac{\langle [\overline{\delta^2}(t;T)]^2 \rangle - \langle \overline{\delta^2}(t;T) \rangle^2}{\langle \overline{\delta^2}(t;T) \rangle^2} = \mathrm{Var}(\xi) \xrightarrow{T\to\infty} \frac{2\,\Gamma(1+\alpha)^2}{\Gamma(1+2\alpha)} - 1 \;>\; 0 \tag{3.A.11}$$

tends to a finite positive constant rather than to zero [20]: the scatter of $\overline{\delta^2}$ across trajectories survives even in the limit of infinite observation time (for $\alpha = 1$ the constant is 0 and ergodicity is recovered). Monitoring the full distribution of $\overline{\delta^2}^{(k)}(t)$ across trajectories, and not only its mean, therefore diagnoses non-ergodicity directly: the distribution collapses with growing T for a self-averaging ergodic process and converges to a non-trivial limit for a non-ergodic one.

For comparison, in an ordinary random walk ($\alpha = 1$, finite mean waiting time) the ratio $n(T)/\langle n(T)\rangle$ tends to 1 in probability as $T \to \infty$, so the distribution of the TA-MSD collapses to a delta function, $\overline{\delta^2}(t;T) \to \mathrm{MSD}(t)$ almost surely, and the process is ergodic.

Implementation and Computational Details

Chapter I1

Overview

This part collects the implementation- and computation-level detail behind the choices made in the main text: the exact code module and script that produced each figure or table, the configuration values in force, sample sizes and computation times, and the status of each piece (implemented and tested, designed but not yet built, or planned). It is organised as a document within the document: its own chapters, numbered independently of the appendices and prefixed with an `I`, mirror the body chapter by chapter: Chapter I2 here is the implementation side of Chapter 2 there, and the prefix keeps the two apart in every cross-reference. It is not meant to be read start to end – the body links to the relevant place at the point where a detail was trimmed out of the text, via a cross-reference such as "(implementation details: Sec. I3.5.2)".

Contents of this part

Table I1.1 is a quick index of which script reproduces which auto-generated figure or table (paths relative to `scripts/reporting/`, in the chapter-named subdirectory shown); the chapters that follow go into the reasoning and numbers behind each one.

Figure/Table	Script	Notes
Stats macros; Tables 2.3, 2.4	ch2_dataset/generate_stats.py	Reads committed data/*/seasons_index.csv; instant; part of the pre-commit hook.
The three processed tables (Sec. I2.2)	../preprocess.py	Runs stages (i)–(vii) over an archive; ~80 min for both disciplines on 8 workers. ../verify_dataset.py checks the result against the invariants of Ch. 2.
Figure 3.1; \StatMsd*	ch3_global_transport/measure_msd.py, ch3_global_transport/generate_msd_figure.py	The former streams fixes.parquet once, through soaring.analysis.derived.stream_flights, and accumulates in that single pass both estimators, their east-only and north-only twins (Sec. 3.6) and the three fixed-duration cohorts (Sec. 3.1); ~20 min on the paraglider archive before the twins, more now, since time_averaged_msd's per-segment FFT runs once per component for them rather than once pooled – the cohorts cost nothing beyond three more accumulators, because the expensive step is the read. The latter fits every curve, runs the bootstraps and draws the figure from what was written; seconds. --redraw re-renders the figure from the written msd_curve.csv in a second more, without even the reduction: nothing there recomputes an estimator, so a change to the drawing need not cost a re-fit.
\StatPipe*, the per-stage shares quoted around Table 2.6	ch2_dataset/generate_pipeline_census.py	Queries flights_meta.parquet; instant.
Table 2.6 itself; \StatData*, \DataCascade*	ch2_dataset/generate_dataset_stats.py	Reads flights_meta.parquet and the catalogue; seconds.
Census macros (\StatScan...)	ch2_dataset/generate_census_stats.py	Reads the cached track scans on the SSD (never rescans); instant; best-effort in the pre-commit hook, so the committed macros refresh whenever the SSD is mounted. Also emits the flight-level filtering census (LongEnough/Overlong/ShortPath families, Sec. 2.7.4) and the missing-barometric-fix statistics (BaroMiss, Sec. 2.7.1), against the thresholds in configs/preprocessing.yaml, and re-exports those thresholds as the \Preproc* macros quoted by the pipeline map (Figure 2.12).
— (all of the above, in order)	../regenerate.sh	Runs every row of this table that depends on the processed dataset, in the one order that is correct, and refuses to start while ../preprocess.py is still writing. The ordering is a real constraint rather than a convenience: the driver rewrites <data_root>/derived/ in place and completes flights_meta only after its

Chapter I2

The dataset

Mirrors Chapter 2 section by section, opening with the data acquisition (Sec. 2.1) and including the pre-processing and preliminary- characterization sections (Secs. 2.7, 2.8) – the cleaning and first-look steps are part of describing the dataset, not of the planned analysis in Chapter 3.

Table I2.1 collects the implementation status of every step discussed below; the sections point to it instead of repeating status remarks.

Pipeline step	**Status**	**Where**
Data acquisition (fetch, download, verify)	implemented	Sec. I2.1
IGC parsing and format validity	implemented, unit-tested	Sec. I2.3
Catalog and season index (`build-catalog`)	implemented	Sec. I2.4
Census scan, diagnostic figures and macros	implemented	Sec. I2.7, Table I1.1
Glider-label normalisation	implemented, unit-tested	Sec. I2.5
Altitude-channel choice (`alt_source`)	implemented, unit-tested	Sec. I2.7.1
Fix-level cleaning (three detectors)	implemented, unit-tested	Sec. I2.7.2
Validation battery for the cleaning	planned, follows the cleaning	Sec. I2.7.2
Trimming of ground phases, interior guard	implemented, unit-tested	Sec. I2.7.3
Flight-level filtering	implemented, unit-tested	Sec. I2.7.4
ENU conversion	implemented, unit-tested	Sec. I2.7.5
Uniform sampling and segmentation	implemented, unit-tested	Sec. I2.7.6
Smoothing and differentiation	implemented; timescales measured	Sec. I2.7.7
Preliminary characterization	implemented	Sec. I2.8
MSD audit (residuals, controls, isotropy)	implemented	Sec. I3.1
Ensemble MSD (Sec. 3.1)	implemented, unit-tested	Sec. I3.1

Table I2.1. Implementation status of the pipeline steps, collecting the status remarks of this appendix in one place. *Implemented* code is committed and in use; *designed* steps have their rules and thresholds fixed (and, where noted, their parameters already in the configuration) but no running routine yet; *planned* steps follow the designed ones.

I2.1 Data acquisition

Mirrors Sec. 2.1.

I2.1.1 Pipeline and tooling

Package `soaring.acquisition.ffvl`, exposed as two CLI entry points, `soaring-para` and `soaring-delta`, sharing subcommands: `fetch-xml` (archive season XMLs), `download` (fetch IGC files), `build-catalog` (rebuild catalog/index), `status` (per-season coverage), `verify` (post-hoc integrity scan, Sec. I2.1.3). One package serves both disciplines because the two FFVL sites expose an identical URL structure and the same machine-readable XML export format; only the host and the data root differ between the two CLIs.

Resumable downloads. Files already present on disk are skipped, so an interrupted job resumes freely without re-fetching.

Atomic writes. Each file is written to a temporary `.part` and then renamed, so a file under its final name is always complete – important on the exFAT external disk, which has no journaling.

Content validation. A response is accepted only if it looks like a real IGC file (an `A` header record followed by at least one `B` fix record); HTML error pages and truncated responses are rejected and retried.

Rate-limited networking. A small bounded thread pool, per-request timeouts, exponential backoff, and a minimum delay between requests keep the load on the server low; a browser TLS fingerprint is impersonated to pass the site's Cloudflare challenge.

I2.1.2 Reproducibility

Destination directory (`data_root`) set per machine via an environment variable – `SOARING_PARA_DATA_ROOT` for paragliders, `SOARING_DELTA_DATA_ROOT` for hang gliders – or a config file shipping a placeholder (`configs/para_download.yaml/ configs/delta_download.yaml`); every command validates the setting at start-upand aborts with an explicit message if the target is missing.

I2.1.3 Integrity

The `verify` subcommand re-scans every downloaded file on disk, reusing the same IGC validation applied at download time, and flags truncated or invalid files, leftover `.part` fragments, and read errors.

I2.2 Where the data lives

Everything is on one external disk and nothing is in the repository, so a reader who has the disk can locate any number this thesis quotes. There is one root per discipline, identically laid out, found through an environment variable rather than a hard-coded path (`SOARING_PARA_DATA_ROOT`, `SOARING_DELTA_DATA_ROOT`); the repository keeps only the small versioned `data/<discipline>/seasons_index.csv`, so the season tables of Sec. 2.5 build without the disk mounted.

Path	Size	What it is
raw/ — immutable, never modified after download		
`raw/igc/<season>/`	186,052 files	the track files, one directory per season, named `<date>_<flight_id>.igc`
`raw/raw_xml/<year>.xml`	27 files	the season listings exactly as the site served them: the provenance of the catalog
catalog/ — regenerable from raw/, metadata only		
`catalog/catalog.csv`	87 MB	one row per listed flight, 23 columns (Sec. 2.3)
`catalog/seasons_index.csv`	2 kB	one row per season: listed, with a track, downloaded
logs/ — what the acquisition did		
`logs/download.log`, `logs/failures.csv`	80 kB	the run log and the retry registry, one row per track that could not be fetched
derived/ — regenerable from raw/, everything the analysis makes		
`derived/track_scan.parquet`	8.5 MB	the census cache: one scalar row per parsed flight (Sec. I2.7)
`derived/fixes.parquet`	43 GB	the processed trajectories: 1.36×10^9 rows for paragliders
`derived/segments.parquet`	4 MB	one row per segment the splitting produced, retained or not
`derived/flights_meta.parquet`	10 MB	one row per flight *attempted*, dropped ones included

The three levels are a rule and not a habit: `raw/` is never written to after the download, `catalog/` is a function of `raw/`, and `derived/` is a function of both. Deleting `derived/` costs computation and never data. It also means the two census families cannot be confused: `track_scan.parquet` describes *parsed* tracks and is what the `\StatScan*` macros query, because a cut must be audited on the population it acts on; `flights_meta.parquet` describes what the pipeline *did*, and is what the `\StatPipe*` macros query.

The trajectory table. `fixes.parquet` carries one row per grid point of every retained segment, keyed `(source,flight_id,segment_id)`. Printed as it is on disk (the hang-glider root; `scripts/reporting/tools/show_dataset.py` produces this view):

```
34,389,245 rows in 43 row groups, 1,156.0 MB on disk, ZSTD

dtypes:
  source   str      t  float32      v_E  float32      interpolated     bool
  flight_id str     E  float32      v_N  float32      edge             bool
  segment_id int16 N  float32      v_z  float32      hampel_flagged   bool
                    z  float32      a_*  float32      alt_invalidated  bool

head(4), columns 1-9:
    source flight_id  segment_id    t          E           N           z        v_E       v_N
hangglider       975           0  0.0  -2.644085   -4.948205 1458.585693 13.275144  1.215108
hangglider       975           0 10.0 111.172691  -33.921207 1460.657104  9.778760 -5.975531
hangglider       975           0 20.0 198.742096 -103.775276 1468.514282  8.025670 -6.961107
hangglider       975           0 30.0 279.220947 -138.911942 1463.914307  7.008325 -2.623670

head(4), columns 10-17:
      v_z       a_E       a_N       a_z  interpolated  edge  hampel_flagged  alt_invalidated
-0.532143 -0.436803 -1.029317  0.192857         False  True           False            False
 0.721429 -0.262474 -0.408811  0.057857         False  True           False            False
 0.625000 -0.088144  0.211696 -0.077143         False False           False            False
-1.466667  0.072790  0.463077 -0.047143         False False            True            False
```

The four rows are the first four grid points of a 10 s logger: t starts at zero, E and N within metres of it, z at its measured 1,458 m. Coordinates are `float32`, a part in 10^7 on a coordinate of tens of kilometres and orders below the GPS noise on it. The file is written in batches of 400 flights, and an analysis streams it rather than loading 43 GB—which is how the mean-squared displacement of Sec. 3.1 is computed.

A row group is not a batch, and the difference cost a measurement. The writer hands Parquet 400 flights at a time; Parquet then cuts what it is handed into row groups of its own default size, which is why the header above reads 43 row groups for an archive written in 16 batches. A reader that takes a row group for a unit of flights therefore meets, at each boundary, one flight arriving as two fragments—and processes it as two flights. The ensemble MSD counted 157,134 paraglider flights where the pipeline had written 156,017, 0.7 % too many; worse, the time-averaged MSD is computed *within* a segment and cut at half its length, so a straddling segment stopped answering at half of its longer fragment instead of half of itself, and the longest segments—the ones that carry the long-lag tail, and the ones likeliest to straddle—were the ones truncated. The reading was thus biased by the writer's buffering rather than by the data. Every consumer now goes through one reader, `soaring.analysis.derived.stream_flights`, which holds back the last flight of each row group and prepends it to the next, so what it yields is always a whole flight; it raises rather than continues if the writer's ordering ever stops holding. The dataset verifier reads through it too, since it had the same blind spot in a worse place: the step *joining* two fragments was never differenced, so it was never checked, and a check with a blind spot reports success over it.

The three timing conventions a reader of this table has to know are argued elsewhere in the chapter and worth restating together: t is zero at the first fix of free flight *of the parent flight*, so a segment never restarts it (Sec. 2.7.6); E and N are zero at that same fix (Sec. 2.7.5); and z is the adopted altitude at its measured value, never re-zeroed. The only times in another clock are `ground_phase_start_s` and `ground_phase_end_s` in `flights_meta`, which are in the recorded clock of the IGC file because the integrity gate counts its removals inside that window.

The segment table. One row per segment the splitting produced, retained or not, and the reason for each drop:

source	flight_id	segment_id	t_start	t_end	n_fix	n_fix_raw	frac_interpolated
hangglider	975	0	0.0	20640.0	2065	2064	0.001453
hangglider	1032	0	0.0	14175.0	0	691	0.001479

censored_start	censored_end	kept	drop_reason
False	False	True	<NA>
False	False	False	channel_not_reconstructable

`n_fix` counts the rows the segment contributed to `fixes` (zero when dropped) against `n_fix_raw`, its measured fixes. `segment_id` is assigned at split time and stays stable, so a gap in the numbering is itself the record of a drop.

The flight table. One row per flight *attempted*, forty-one columns, read down rather than across—one retained flight beside one the flight-level filter rejected:

	a retained flight	a dropped one
source	hangglider	hangglider
flight_id	975	830
pipeline_version	1.1.0	1.1.0
drop_stage	NaN	flight_filter
drop_reason	NaN	duration_below_minimum
alt_source	baro	gnss
n_fix_raw	2099	349

n_fix_clean	2069	348
n_removed_frozen	30	0
n_flagged_kept	40	8
n_boundaried	0	0
integrity_fraction	0.0	0.004587
ground_phase_start_s	300.0	485.0
duration_flight_s	20645.0	1085.0
path_km	283.408578	6.044969
lat0	43.812717	NaN
dt_native_s	10.0	NaN
n_segments_kept	1.0	NaN
savgol_window_horiz	5.0	NaN

(abridged: the table carries 47 columns in all.) A dropped flight keeps everything the stages *before* the verdict had already measured—flight 830 was judged on a duration of 1,085 s, and its cleaning counters remain auditable even though none of its fixes reached the trajectory table. That is what makes the removals a census rather than an absence.

What is not stored. Anything that is pure algebra of the stored columns—the speed $|\mathbf{v}|$, the heading, the angular velocity, the turn radius, the glide ratio, the mechanical energy—is computed at analysis time and never materialised. Only the outputs of a non-trivial parametric step are kept, which is what makes the table's size a function of the archive rather than of the questions asked of it.

Checking it. `scripts/verify_dataset.py` reads the three tables back and tests, on the data itself, the properties this chapter claims: completeness within a segment, a strictly increasing and uniform time base, no impossible step between two interior samples of a segment (Sec. 2.7.2), the origin at $r(0) = 0$, reachability—$|\mathbf{r}(t)| \le v_{xy}^{\max} t$ for every sample—and referential integrity between the tables. Each check names the section it comes from, so a failure says which claim is false rather than only that something is wrong.

Reachability is the one check stated without reference to anything the pipeline did: it follows from the frame and the speed bound alone, because the flight began at the origin. That is what makes it able to catch a defect no per-step test can see. A flight whose *first* fix is corrupt has every step plausible—it simply sits, in the archive, 4,500 km from where it says it began, with the whole trajectory referred to that point (Sec. 2.7.2).

The qualification in the third check is not a hedge but the statement being precise. The bound is a claim about *measured* motion, and two kinds of sample in the table are not measurements. At a segment edge the smoothing polynomial is evaluated off-centre (Sec. 2.7.7), so its position carries more variance than a centred one and on a slow logger can imply a step over the bound where the underlying fixes do not. Across a bridged gap the monotone cubic is a reconstruction, and its slope may exceed the secant it spans—a hole crossed at $42\,\mathrm{m\,s^{-1}}$, itself under the bound and so legitimately bridged, leaves interpolated points implying nearly $60\,\mathrm{m\,s^{-1}}$. Both kinds are flagged in the table for exactly this reason (`edge`, `interpolated`), so the check reads the invariant between samples that are measured and centred, and *reports* the excess on the others. That reported number is one of the acceptance criteria Sec. 2.7.7 asks for: measured, rather than assumed away.

The script found two defects this text would otherwise have asserted without warrant. A randomised robustness suite (`test_pipeline_robustness.py`, which combines injected spikes, null-island runs, frozen locks, time defects, gaps and offsets at random and asserts only these invariants) found two more, and both were faults in the *statement* rather than in the pipeline: the invariant had been written without either qualification above. A third, the corrupt opening block of Sec. 2.7.2, was found by neither: it was found by the ensemble MSD of Chapter 3 being wrong, which is the expensive way to find a bug, and the reachability check above is what makes it cheap next time. It also found one in the cleaning—a track crossing the antimeridian is read as motionless, because the great-circle distance is correctly aware of the wrap and therefore reads the two numerical extremes of a straddling bounding box as metres apart. No flight in the present archive reaches within ten degrees of that meridian; the further sources of Sec. 2.1 may.

I2.3 The IGC tracklog format

The parser (`soaring.analysis.igc.parse_igc`) checks, at decode time and independently of any flight-dynamics judgement: that the time is a valid 24-hour `hh:mm:ss`; that the arc-minutes field of latitude and longitude is < 60 (the `DDMM.mmm` encoding is undefined otherwise); that the hemisphere letter is one of the two valid characters; and that the decoded latitude/longitude fall within $[-90, 90]/[-180, 180]$ degrees. A record failing any of these is dropped as unparseable, on the same footing as a record with the wrong length or a non-numeric field. This format-validity check and the fix-level plausibility checks are distinct stages, and both are implemented and unit-tested (Sec. I2.7.2, Table I2.1).

I2.4 Flight metadata and the catalog

Both `catalog.csv` and `seasons_index.csv` are regenerated on demand (Sec. 2.3) by the acquisition CLI's `build-catalog` subcommand (Sec. I2.1.1) – never hand-edited.

I2.5 Glider categories

`aile\_class` is the FFVL's own code ("C ou 2", "non homologuée"), not the EN/FAI label of Table 2.2, so it is mapped rather than merely read: `soaring.reporting.glider\_class.canonical\_wing\_class` carries the map, one dictionary per discipline, and raises if a flight's raw code is not a key of it rather than letting an unrecognised value enter a stratum under its own spelling. The placeholder class, the literal "0" in `aile\_class`, and a blank entry both come back as `NaN` – unclassified and excluded from a class stratum, not recovered from the wing-model field, since that would need a wing-model-to-class lookup this repository does not have. The mapping runs where the class is joined onto a discipline's flights for a class-stratified figure or macro (Sec. 2.4), not at the fix-level stage, since `aile\_class` is catalog metadata (Sec. I2.4) rather than a per-fix quantity. Status: Table I2.1.

Provenance of the EN/FAI class ladders. The two systems of Table 2.2 are not documented on equal terms. The FAI classes come from the Sporting Code [11], which the FAI publishes for free: anyone can download the current edition and check a definition against it. The EN classes come from a *European standard*, which is a different kind of document. Standards are drafted by the European Committee for Standardization (CEN) and then published and *sold* by the national standards bodies — AFNOR in France, DIN in Germany, BSI in the United Kingdom — as copyrighted documents, priced per copy at a few hundred euros. There is no public version: unlike a law or a sporting code, the authoritative text of EN 926-2 cannot be consulted without buying it, and it cannot be quoted here.

The EN entries of Table 2.2 are therefore condensed from Table 1 of the *final draft* of the standard, FprEN 926-2:2012, the version submitted to formal vote, which the CEN working group circulated publicly [24]. That draft is the closest thing to the source text that is openly available, and the class descriptions in it are unlikely to have moved in the published EN 926-2:2013, though that has not been checked against the published text.

I2.6 Coverage and statistics

The season tables of Sec. 2.5 are typeset from the versioned `seasons_index.csv` (Sec. I2.4); the trajectory-level census numbers quoted from Sec. 2.7 onward come from a different pipeline, the cached track scan described at the head of Sec. I2.7.

I2.7 Trajectory pre-processing

Mirrors Sec. 2.7 and its subsections below. Parsing: each IGC file is decoded by the `soaring.analysis.igc` parser, which reads the `B` records at their fixed character positions (Sec. I2.3) and returns both altitude channels alongside time and geodetic coordinates. The Welch/PSD procedure used to compare altitude channels is stated once, in Sec. I2.7.1; the smoothing calibration (Sec. I2.7.7) reuses it, applied to the horizontal components and to the flight's adopted altitude channel.

The census scan and its cache. One full parse of the archive backs every trajectory-census number and figure in this appendix and in Chapter 2: the `load_or_scan_tracks` scan reads every parsed track (186,025 paraglider, 6,716 hang-glider) once and caches a per-flight summary table at `<data_root>/derived/track_scan.parquet`, one cache per discipline.

That cache is worth describing, since every census number in the chapter is a query against it. It is a table with *one row per parsed flight* and one column per summary quantity, all produced by `track_stats` in `soaring.analysis.census` as it walks a flight's fixes once:

- `duration_s`, `n_fix`, `dt_s` — the recorded span, the number of `B` records, and the median inter-fix interval;
- `path_km`, `extent_km` — the summed great-circle steps and the farthest any fix lies from the first;
- `max_gap_ratio`, `missing_fraction` — the largest inter-fix gap in units of `dt_s`, and the share of a uniform grid at that interval with no fix;
- `max_vxy_mps`, `max_vz_mps` — the extreme inter-fix speeds;
- `baro_alt_min_m`, `baro_alt_max_m`, `baro_present_frac` — the barometric range and how much of the flight the channel covers.

Nothing in it is a trajectory: it is deliberately one scalar row per flight, which is what makes it small enough to hold in memory for the whole archive and to re-query in seconds. Its purpose is exactly that—to decouple the cost of *reading* the archive (hours, once) from the cost of *asking* it a question (instant, as often as a threshold changes). Every `\StatScan*` macro, and the flight-level and sampling diagnostics of Figs. 2.3 and 2.6, are functions of these columns alone.

What the census is not. It is measured on the **raw** archive, one parse of the `B` records with no cleaning, no trimming and no filtering—which is the only thing it could be, since its purpose is to choose the thresholds that cleaning, trimming and filtering then apply, and a threshold chosen on the population that survives it is chosen circularly. So every `\StatScan*` number in this thesis is a statement about the archive *as recorded*: a duration is the recorded span and not the airborne one, a path includes whatever a corrupt fix contributed, and a retained-fraction curve is what a cut would keep if applied alone to the raw population. The retained ensemble has its own numbers and its own family, `\StatPipe*`, computed from `flights_meta.parquet` after all seven stages have run (Table 2.6); where a quantity appears in both, the two are not expected to agree, and the difference is precisely what the pipeline did. The two families are never mixed in one statement.

`ch2_dataset/generate_census_stats.py` recomputes the quoted macros from that cache against the thresholds currently in `configs/preprocessing.yaml`, so a threshold change propagates to every quoted count by re-running one script, with no rescan. The cache is refreshed when the archive changes—which for the FFVL collections means once a season closes, and more substantially when a new source is added (the sailplanes of Sec. 2.1 above all); between refreshes the cached counts can trail the season tables' download counts, and the two are not expected to agree to the flight. The same scan feeds Figures 2.3, 2.6 and 2.7, regenerated together by one run of `ch2_dataset/generate_preproc_figure.py` (Table I1.1); the fix-level diagnostics of Figure 2.2 are the one exception, drawn from a separate sampled pass (Sec. I2.7.2).

I2.7.1 Choice of altitude channel

The `alt_source` field. The channel a flight uses is stored as an `alt_source` column (`baro` or `gnss`)— two-valued and categorical, so its storage cost is negligible —in the processed dataset's per-flight table, alongside the duration and the raw fix count—the number of `B` records the IGC file yields at parse time, before any cleaning, since the channel choice precedes the cleaning in the pipeline order.

Welch/PSD procedure and its sample (Figure 2.1a–c). The sample is a seeded random subsample of 3,000 flights per discipline.The size is set by convergence of the ensemble spectrum, not by the spectral shape: with a few hundred flights the high-frequency part of the median barometric curve is still visibly ragged (under-averaged), and it smooths out only with a few thousand flights, so 3,000 buys a stable band at modest cost. This is an assertion the reader cannot presently check, and it should not stay one: the convergence figure—the same median spectrum drawn at a few hundred, a thousand and 3,000 flights, on one pair of axes—is to be produced and shown here, so that the choice of sample size rests on a picture rather than on a claim. Panel (a)/(b) use one representative flight (the one with the most fixes among the 3,000 sampled flights of that discipline); its measured offset (panel b) is annotated on the panel itself, not restated here.

IGC loggers record one fix per second nominally, but timestamps are not guaranteed to be perfectly regular (a missed fix, a logger writing at 1.05 s instead of 1.00 s); Welch's method (Appendix 2.A) requires a uniformly sampled signal, so the following runs on every flight before the transform.

(i) **Modal sampling period.** For each flight the median inter-fix interval is computed and rounded to the nearest integer second. The value that appears most often across all flights in the subsample (the *mode*) is taken as the common target Δt (in practice $\Delta t = 1\,\mathrm{s}$ for both disciplines, giving $f_s = 1\,\mathrm{Hz}$ and $f_{\mathrm{Nyq}} = 0.5\,\mathrm{Hz}$).

(ii) **Flight selection.** Only flights whose own median period lies within 0.25 s of the modal Δt are retained, so pooled spectra share one frequency axis. Mixing cadences here would not be admissible: a spectrum is defined on a frequency axis that runs to $1/(2\Delta t)$, so a 1 s and a 9 s logger do not even share a Nyquist frequency, and averaging their periodograms per frequency bin would compare different bands. That is the whole reason for this cut—and it is a cut on the *diagnostic* sample only. The pipeline proper keeps every flight at its native cadence (Sec. 2.7.6); it is only this spectral estimate, which pools flights, that needs them commensurate. Flights with fewer than $N_{\mathrm{seg}} = 256$ fixes ($\sim$4 min at $\Delta t = 1\,\mathrm{s}$) are also excluded (a series shorter than one Welch block of $N_{\mathrm{seg}} = 256$ samples cannot contribute a periodogram, Appendix 2.A). Of the 3,000 sampled flights per discipline, this leaves 1,641 paragliders but 727 hang gliders: paragliders are predominantly 1 s loggers (69 %, Sec. I2.7.6), so the population mode is a good match for most of them, whereas hang gliders split across several native rates (1, 2, 5, $\geq$8 s, Figure 2.7) and only their 1 s-logger minority survives this cut.

(iii) **Uniform resampling.** The altitude series $z(t)$ is linearly interpolated onto a grid $t_0, t_0 + \Delta t, t_0 + 2\Delta t, \ldots$ spanning the flight. Linear and not cubic, because the resampling must not disturb the band the estimate is about to read. Interpolation acts on a spectrum as a filter: linear interpolation attenuates high frequencies smoothly and monotonically, whereas a higher-order interpolant has a transfer function that can exceed unity near the Nyquist frequency, adding power that was never in the signal. Since the quantity being measured *is* the flat floor at high frequency, an interpolant that can lift it would bias the very number in question; one that only attenuates cannot manufacture a floor, and the timestamps are near-uniform to begin with, so the attenuation it does apply is slight.

(iv) **Welch on the resampled series.** `scipy.signal.welch` is called with $f_s = 1/\Delta t$, $N_{\mathrm{seg}} = 256$, 50% overlap, a Hann window of the same $N_{\mathrm{seg}} = 256$ samples (the window spans one Welch segment by construction—256 s at the 1 s cadence, giving a frequency resolution of about 4 mHz), and linear detrending per segment.

(v) **Robust ensemble estimator.** The per-flight PSDs are kept individually (per discipline and channel) and reduced across flights by the per-frequency *median*, with a shaded 10 to 90 percentile band; panel (c) then pools the two disciplines, whose medians coincide. The median, not the arithmetic mean, because the per-flight spectra are heavy-tailed: a mean swings by one to two orders of magnitude between samples – for the hang-glider barometric floor, from 0.8 to $93\,\mathrm{m}^2\,\mathrm{Hz}^{-1}$ between an $n = 114$ and an $n = 727$ draw – whereas the median is stable. Because Δt is the same for every retained flight by construction, the figure's Nyquist dotted line is a single fixed value.

Reading the band (panel c).

Why the medians coincide. For the typical flight both channels are dominated at high frequency by the metre resolution to which the IGC format rounds every altitude: the white-noise floor of a 1 m quantiser, $\frac{1}{12}/f_{\mathrm{Nyq}} \approx 0.17\,\mathrm{m}^2\,\mathrm{Hz}^{-1}$,[1] sits just below where the measured medians level off ($\approx 0.3\,\mathrm{m}^2\,\mathrm{Hz}^{-1}$).

Where the channels part. In the tail: at the 90th percentile the high-frequency GNSS floor is 4× the barometric one for paragliders and 108× for hang gliders, and 13 % to 20 % of flights carry a GNSS floor above three times their own barometric one. What sits in that tail is receiver tracking noise scaled by the dilution of precision, since a raised *floor* is by definition a white contribution; multipath, being correlated over minutes (Sec. 2.7.1), does not show up there. Why the hang-glider fleet is worse in this respect is not settled by these spectra alone—older or lower-grade receivers are the obvious candidate—and the figure is not claimed to answer it.

Consequence. This heterogeneity, rather than a uniform gap, motivates the choice of the barometric channel (Sec. 2.7.1).

Census (panel d). Panel (d) is a full census, not a sample: the barometric-presence fraction is read off the shared track scan (Sec. I2.7), already paid for by the flight-level diagnostics.

"Present" needs a number, and the number is a decision. The scan records `baro_present_frac`, the share of a flight's fixes carrying a non-zero pressure altitude, and a flight counts as barometric when that share reaches `BARO_PRESENT_MIN` = 95 % (defined once, in `soaring.`

[1] Rounding to a step q adds an error that is well modelled as uniform on $[-q/2, q/2]$ and independent between samples, of variance $q^2/12$. A white sequence of variance σ^2 has a flat one-sided power spectral density $\sigma^2/f_{\mathrm{Nyq}}$, since integrating it over $[0, f_{\mathrm{Nyq}}]$ must return the variance. With $q = 1\,\mathrm{m}$ and $f_{\mathrm{Nyq}} = 0.5\,\mathrm{Hz}$ this is $(1/12)/0.5 \approx 0.17\,\mathrm{m}^2\,\mathrm{Hz}^{-1}$.

`analysis.altitude_noise`, and imported by both the cleaning and the census so the three cannot drift). The threshold is high deliberately. Presence is bimodal—the census below shows the channel is either essentially absent or essentially complete—so any value in the middle of the range separates the same two populations; but a mid-scale cut would also admit a flight missing, say, a third of its pressure altitudes, and such a flight is not one the vertical analysis should treat as barometric: its v_z would rest on a channel with holes, reconstructed by interpolation over stretches long enough to erase transitions. Setting the cut at 95 % sends those flights to the GNSS fallback, where their altitude is at least uniformly sourced. The number quoted in the body (Sec. 2.7.1) is the same one.

Presence is whole-file, checked on the same census: when the channel is absent it reads exactly zero for 99.3 % (paragliders) / 100 % (hang gliders) of those flights, and when present it covers every fix for 96 % / 98 % of them; the rest miss individual fixes, dropped one by one (Sec. 2.7.2), and no more than 5 % of the flight by construction, since below that the channel is not counted as present at all.

The bimodality asserted above is what these numbers measure. Where the channel is absent it is *exactly* zero for 99.3 % of those flights, and where it is present it is complete for 96 %; the population in between is thin. Raising the threshold from 0.5 to 0.95 reclassified 241 paraglider flights and no hang-glider ones—0.13 % of the archive—which is the direct measurement of how thin. Those flights are also where the worst partial coverage sat: the largest number of missing barometric fixes on a flight still counted as barometric fell from 14,878 to 1,397 when the threshold moved, so most of what looked like a pathological tail was a classification artefact rather than a sensor one.

What the presence test does not check. It decides *that* a flight falls back to GNSS, not *whether* GNSS is the better channel there — an assumption the pipeline never tests, since no cache used by this scan records how complete the GNSS altitude is. It is safe wherever the channel is absent outright, 99.3 % of the fallback population: a barometer that never recorded implies nothing about the GNSS solution, which is computed independently. It is untested on the rest, 107 paraglider flights (0.06 % of the archive) present on 90 % to 95 % of their fixes, close enough to the cut that a real defect rather than a missing sensor could plausibly explain the shortfall; no hang-glider flight falls in that band. Confirming the fallback also holds there would need the same presence statistic computed on the GNSS channel, which this scan does not carry. Whatever that comparison would show, the population is too small to move an archive-wide result: 0.06 % of the paraglider archive is 107 flights, and even a worst case where every one of them lands on a noisier GNSS record than the barometric channel it left is 107 flights out of 186,025 carrying a degraded vertical channel — not a share any statistic pooled over the archive would register. The liveness threshold is `baro_min_range_m` (key `alt_channel`; value in Table 2.7); rule and rationale in Sec. 2.7.1.

Why GNSS altitude is vertically noisier. The asymmetry behind the vertical dilution of precision is geometric: a receiver sees satellites only above the horizon, so the ranging directions bracket it in the horizontal plane but all arrive from the same side vertically, leaving the vertical coordinate the worst-determined one. The GPS performance standard commits to 8 m horizontal against 13 m vertical error at 95 % globally, and 15 m against 33 m at the worst site [25]. Multipath — the direct signal arriving together with replicas delayed by reflection off terrain or the pilot's own equipment — biases the range on timescales of minutes rather than adding independent noise from fix to fix [12], which is why it raises the low-frequency end of the spectrum rather than the high-frequency floor read in panel (c) above.

What the liveness test does not catch. The statistic `baro_min_range_m` is a single whole-flight max – min, so a barometer that records normally for the first ten minutes and then freezes keeps a range of hundreds of metres, is judged alive, is adopted, and delivers a vertical velocity of identically zero over most of the flight — precisely the outcome the test exists to prevent. Nothing downstream closes the gap: the frozen-lock rule of Sec. 2.7.2 needs the *position* to be frozen, and a cruising wing never trips it; the flight-level altitude-activity cut consumes the same extremum difference and inherits the same blindness. A test that did close it would be a *local* one — a moving-window range, or a longest-constant-run bound — and that is a new threshold with its own failure mode against a wing genuinely holding altitude, so it is not adopted here on the strength of an argument alone. What the present rule guarantees is exactly this: a channel constant throughout is refused, one that dies partway is not.

I2.7.2 Fix-level cleaning

This section gives the working definitions the narrative (Sec. 2.7.2) states in words: the three detectors, their parameters, the order they run in, and how the whole step is validated. In the order of Table 2.5, the three families map onto a robust local test (a short window of context), a structural rule (a whole run), and an absolute bound (no context). Status: Table I2.1.

The detectors run in a fixed order, which the subsections below follow:

(1) format validity, at decode time (Sec. I2.3);

(2) time defects: backward steps, then duplicates;

(3) position outliers (the impossibility gate; the Hampel flag is recorded, not required—Sec. 2.7.2);

(4) frozen-lock runs;

(5) altitude: absolute bounds and altitude outliers.

The order matters because a speed or a residual is only meaningful once the time base is monotone and free of duplicates. Merging duplicates is componentwise and unweighted: fixes sharing one UTC second collapse to a single fix at that second, carrying the mean of their latitudes and longitudes and the mean of each altitude channel over the members that have one; the `V` flag of the merged fix is the logical *or* of the members', so a degraded member cannot be averaged away. The position-outlier and frozen-lock passes are one left-to-right scan that carries the last accepted fix as an anchor: it never re-walks the whole track, and it absorbs a run by extending the anchor across the flagged block – tracking the block's running bounding box to test its bounding diameter against δ_{xy} – so the cost is $O(N)$ in the number of fixes.

Time-base defects

The parser unwraps the UTC-midnight roll-over — a drop of ∼86,400 s advances the day count — and *leaves* residual backward jitter in place (Sec. I2.3). That is deliberate, and it is the one behaviour the redesign changed: the parser used to flatten the jitter with a running maximum, which is a repair, and repairing it there destroys the evidence this stage has to act on. A backward step that is *not* a roll-over is now removed rather than absorbed – the fix with the corrupt time is deleted as a node, just as a fix with a corrupt position is. A corrupted timestamp is recorded as a defect rather than silently repaired. Which fix carries the corrupt time is decided by minimal removal: delete the smallest set whose removal leaves a strictly increasing clock (the complement of the longest increasing subsequence), so a single fix with a forward-jumped clock is itself removed, rather than condemning every later fix against a running maximum. This requires moving the clamp out of the parser: `parse_igc` keeps the roll-over unwrap (a format concern, needing the raw time of day) but stops flattening backward jitter with the running maximum – flattened, the evidence the cleaning must act on no longer exists downstream.

Fixes sharing one UTC second are collapsed to a single representative at that second's centroid position. This replaces the provisional "keep the first" of the earlier design: keeping the first privileges an arbitrary member and, for a genuine $\geq 2\,\mathrm{Hz}$ logger, would silently halve the cadence, whereas the centroid is order-independent and the sub-second resolution it discards is discarded anyway once the flight is resampled to a common Δt (Sec. I2.7.6). One dilution case is accepted: if one member of a duplicate pair is itself a position spike, the centroid halves its amplitude before the outlier pass (which runs after the time defects) sees it; a large spike is still flagged at half size, and one small enough to slip under $k\sigma_k$ after halving is within what the smoother absorbs.

Robust local-outlier test and action

The test is Eqs. (2.2)–(2.3) of Sec. 2.7.2, evaluated on a *time* window of half-width w (not a sample count), so it adapts to the native cadence and to irregular sampling; working defaults in Table I2.2. The window must hold at least a handful of fixes (order five); a sparser neighbourhood falls back to the absolute bounds alone.[2]

Three properties distinguish the test from the bare speed bound:

- *The floor $\varepsilon_{\min}$ is not cosmetic.* On a very smooth stretch $\sigma_k \to 0$, so without it any sub-metre wiggle would exceed $k\sigma_k$ and the test would chase GPS noise where there is no outlier.
- *A burst is one block.* A short burst of consecutive spikes inflates the windowed MAD only mildly, so it is still flagged, and it is removed as the block whose bracketing neighbours rejoin within scale (the attribution argument is the removal-policy paragraph of Sec. 2.7.2).
- *It does not clip real manoeuvres.* A genuine sharp turn is drawn by consecutive, mutually consistent fixes, each well predicted by its neighbours and so of small residual; only an isolated point off the local trend scores large. The one failure mode, the under-sampled tight turn (Sec. 2.7.2), is measured directly by the injected-defect test (check (1) below).

The flag itself is context-sensitive – a $30\,\mathrm{m\,s^{-1}}$ step out of a $12\,\mathrm{m\,s^{-1}}$ glide is flagged while the same step passes unremarked where it is ordinary – but a flag alone does not delete (next paragraph): a flagged-but-physically-possible fix is *kept*, its flag recorded as a per-flight anomaly count alongside the integrity-gate counters. The test runs on the horizontal position *alone*. An earlier draft of this appendix had it running on the adopted altitude channel as well, which the three-part argument of Sec. 2.7.2 supersedes: on z an absolute bound is sufficient where on xy it is not, a local test on z would flag the thermal entries the segmentation is built on, and a false vertical flag is expensive exactly where it is most likely. The vertical is therefore cleaned by its absolute bounds and by nothing else.

From flag to action. The identifier only *flags*; it never repairs. A flagged horizontal fix is *deleted* as a node – not overwritten with the local median – with all reconstruction deferred to the single resampling step (Sec. I2.7.6; the one-policy rationale is Sec. 2.7.2). This is the

[2] The residual and run-diameter distances are the haversine great-circle distance of Eq. (2.1); a vectorised off-the-shelf implementation exists as `sklearn.metrics.pairwise.haversine_distances`, should the hand-rolled one need cross-checking.

Hampel identifier, the detection half of the eponymous filter, not the median-substituting filter itself: the local median enters only as the reference for the residual r_k, never as an output value.

Why the flag is not required. The rule was originally a conjunction — flagged *and* impossible — and running it over the archive exposed a limit in the *flag*. The Hampel identifier has a breakdown point of 50 %: past that, the median moves onto the corruption instead of resisting it. Some loggers write the null position $(0, 0)$ for scattered runs of a few seconds each; where several such runs fall inside one 20 s window they exceed half of it, the median follows them, the corrupt fixes score a residual of *zero* and go unflagged, and it is the good fixes around them that are flagged instead. On one such flight the identifier caught 17 of 99 null fixes, and every survivor left a step of some 5,000 km in the record. The rule adopted is therefore the impossibility gate on its own: a genuine tight circle is still never a candidate, because it flies at an ordinary speed, and the attribution the flag supplied is instead supplied by the rejoining test, which names the block whose removal restores the trend. The change bites only where the identifier is outside its own stated range of validity, which is precisely where the conjunction failed.

Position. Deletion is the impossibility gate: a fix goes when it is unreachable from the last accepted fix at the absolute v_{xy} bound *and* the removal of the block it belongs to lets the track rejoin within that bound. The flag is neither necessary nor sufficient—why it cannot be required is Sec. 2.7.2—and the genuine under-sampled tight turn is kept because it flies at an ordinary speed, not because anything exempts it.

Byte-identical, on the parsed table. The frozen-lock witness of Eq. (2.4) is specified on the raw `B` records, but the parser returns decoded floats and keeps no raw text. No raw text is needed: the `DDMM.mmm` field is a fixed-width integer count of thousandths of an arc-minute, so the decode is injective and two records are byte-identical in latitude and longitude *iff* their decoded values compare exactly equal. The test is therefore `lat[i]==lat[i+1]andlon[i]==lon[i+1]` on the parsed columns, with exact equality and no tolerance — a tolerance would turn the conclusive signature into the jittering-freeze case it is meant to exclude.

Discontinuity. When no bounded block restores continuity – a position discontinuity, such as a re-acquisition offset, where every removal still leaves an impossible step – the scan stops deleting and marks a split at the step, handled like a long gap (Sec. I2.7.6): both sides are genuine, only the transition is unknown.

How long a block may run. "Bounded" needs a number, and it is the Hampel half-width w: a block that has not rejoined within 20 s is called a discontinuity instead. The value is reused rather than introduced, and it also fixes the scale at which the two cases part—an offset smaller than $v_{xy}^{\max} w \approx 900\,\mathrm{m}$ becomes reachable from the anchor before the block expires and is removed as an excursion rather than split. That is a real limit of the rule and a bounded one: at most one block of genuine fixes, at a discontinuity, against the alternative of splitting a flight in two for every isolated spike.

The postcondition, and its counter. After the scan, any step between two *surviving* fixes that still breaks the bound is made a segment boundary (Sec. 2.7.2). The number of boundaries this adds is recorded per flight as `n_boundaried` in `flights_meta`, alongside the other cleaning counters: it is the audit of a rule that would otherwise repair without saying so, and it is expected to be small—a large value would indict the scan rather than the data.

Altitude. No flag reaches it: the vertical is cleaned by the absolute bounds of Table 2.5 alone, which mark the value missing and never touch the fix(a deferral: Sec. 2.7.2). Both bounds act on the channel the flight *adopted*, whichever it is. The $|v_z|$ bound was placed on barometric distributions, but what it expresses is a climb/sink envelope of the aircraft, which does not change with the instrument reading it; leaving a GNSS-fallback flight with no vertical cleaner at all — the consequence of dropping the local test — would be the worse reading. It fires more often there, and the rate is recorded per `alt_source` so the audit can see by how much.

Isolated, not sustained. The $|v_z|$ bound marks an altitude missing when the record jumps away from it and back — both adjoining steps past the bound, with opposite signs. This is check (5) below made operational: a *run* of super-threshold steps of one sign is a sustained manoeuvre, a spiral dive holding $15\,\mathrm{m\,s^{-1}}$ to $20\,\mathrm{m\,s^{-1}}$ of real sink, and it is counted and left alone rather than censored.

Why not a Hampel test on z too. The obvious symmetric design — a local-outlier test on the altitude, corroborated by the climb-rate bound, on the same footing as the horizontal rule — is wrong here, for three reasons. First, an absolute bound is *sufficient* on z and insufficient on xy: the vertical envelope is narrow and known (no wing in this dataset sustains $|v_z| > 13\,\mathrm{m\,s^{-1}}$, no altitude outside $[-100, 6000]$ m is sensor-plausible), whereas a 50 m horizontal spike is an impossible $50\,\mathrm{m\,s^{-1}}$ at a 1 s cadence but an unremarkable $5\,\mathrm{m\,s^{-1}}$ at 10 s, so the same defect passes or fails the bound depending on the logger and needs a local, scale-free test the vertical case does not. Second, a Hampel test on z would flag the physics: the residual from a local median is largest where the signal turns fastest, and on the altitude channel that means thermal entries and exits, where v_z swings by several $\mathrm{m\,s^{-1}}$ within seconds (Sec. 2.7.6) — exactly the transitions the phase segmentation is built on. A horizontal circle, by contrast, can be told from a spike by its speed, a corroborating witness the vertical channel lacks: the quantity that would have to arbitrate a suspected altitude spike *is* the climb rate, the quantity under suspicion. Third, the costs are not symmetric: a false vertical flag is interpolated back cheaply, except across a thermal entry, where the interpolation erases the very transition it should preserve, so a cheap-looking false positive is expensive exactly where it is most likely. Keeping the vertical rule to a bound that only fires on the physically impossible avoids that failure mode altogether.

The first fix. An anchor-carrying scan never questions its own first anchor, and no rule here supplies one: the discontinuity is left as a discontinuity, for the reason Sec. 2.7.2 gives. The guarantee is enforced one stage later, by the reach criterion of Sec. 2.7.4, and checked on the written table (Sec. I2.2).

Two rules tried, both wrong. An impossible step is a statement about a *pair* of fixes and says nothing about which endpoint is wrong; away from the ends the bounded-removal test settles it by evidence (which removal restores the trend), but at the very first fix there is no trend behind the step to ask. Two attribution rules were tried regardless, and both failed, in opposite directions. One fired on the first impossible step, which accuses the earlier end always: applied to an ordinary position spike three seconds into a record, it deleted the good fixes and promoted the spike to origin. The other fired on a discontinuity the scan itself declares near the start — a discontinuity survives only where no bounded removal after the step restored the trend — but the split is placed at the first fix of the block that could not be rejoined, and that block lies *after* the step whenever the corruption simply outlasts w: a 50 s corrupt run five seconds in therefore deleted five good fixes, kept fifty bad ones, and left the origin 5,039 km out. What such a rule would have bought is the recovery of two flights in 156,017, against a failure mode demonstrated twice.

This is the *default to keep* rule of Sec. 2.7.2: removal requires positive, corroborated evidence.

Frozen-lock detection

The rule, its witnesses and its margins are Eq. (2.4) and Sec. 2.7.2; parameter values in Table I2.2. Three points are implementation-level.

- *Diameter, not step.* The run's bounding diameter is the discriminating statistic because a step-based test would fire on every 1 Hz climb: the fix-to-fix step of a circling wing, $v_{xy}\Delta t \approx 10\,\mathrm{m}$, is of the order of δ_{xy} itself.
- *Robust witness.* Δz_{baro} is evaluated between medians of the run's ends, so an altitude spike inside the run – not yet cleaned, since the altitude pass comes last – cannot fake a climb.
- *Bookkeeping.* On a barometric flight the `V` flag and the zeroed GNSS altitude are recorded as corroborating diagnostics only; they fill the witness slot on GNSS-fallback flights alone (Eq. (2.4)).
- *A declaration must speak for the run.* Where the recorder's declarations are the witness, they are required on more than half the run's fixes rather than on any one of them. A single `V` flag inside an otherwise healthy stretch is a momentary loss of a satellite, which every flight has; it is the sustained declaration that testifies to a frozen receiver.

A run the rule condemns is marked as a data gap; the split it induces is realised at the uniform-sampling stage (Sec. I2.7.6), on the same footing as a native long gap.

The margins, quantified. A circling wing turns at $R = v^2/(g\tan\phi)$[3], so the tightest paraglider circles ($v \approx 10\,\mathrm{m\,s^{-1}}$ at a bank of $\phi \approx 35°$, ordinary values for tight thermalling) give $2R \approx 30\,\mathrm{m}$, and hang gliders, which fly faster, turn wider ($v \approx 13\,\mathrm{m\,s^{-1}}$ at the same bank gives $2R \gtrsim 50\,\mathrm{m}$). Against the working value of δ_{xy}, the diameter test misses by a factor of at least 2 for the tightest paraglider circles and $\gtrsim 3$ for hang gliders, more for anything wider — ratios to be rescaled if δ_{xy} moves. Barometrically, a climb of order $1\,\mathrm{m\,s^{-1}}$ sustained over τ_{freeze} moves the barometer by $\gtrsim 60\,\mathrm{m}$ against the flatness tolerance δ_z, an order of magnitude margin. Both would have to fail well outside their designed range at once, on the same run, for the conjunction to cut a genuine climb: on a barometric flight the rule is, in this sense, conservative by construction, and errs towards keeping.

Why the duration condition abstains. A run that is spatially collapsed and witnessed but lasts less than τ_{freeze} is kept, deliberately: at durations of a few tens of seconds a collapsed run and a genuine slow, horizontally localized stretch — a search, the core of a tight climb — are not distinguishable by these signatures, and cutting on them would remove the very events the analysis measures. The cost is bounded by the threshold itself: a kept short freeze lengthens one waiting time by at most τ_{freeze}, small against the tail of $\psi(\tau)$ where the anomalous exponent is read, whereas a wrongly cut climb removes a waiting time outright. Short freezes that do get through leave a run of interpolated-looking, near-identical positions,

[3] The coordinated-turn relation: in a steady banked turn the lift L balances gravity vertically, $L\cos\phi = mg$, while its horizontal component provides the centripetal force, $L\sin\phi = mv^2/R$; dividing the two gives $\tan\phi = v^2/(gR)$, i.e. $R = v^2/(g\tan\phi)$.

and the $\psi(\tau)$ sensitivity checks sweep τ_{freeze} precisely to bound how much of the tail depends on this choice.

Absolute bounds and gates

The six numbers of Table 2.5 are the `FixLevelThresholds` dataclass, which `load_preproc_config` reads from `configs/preprocessing.yaml` under the key `fix_level`; nothing is hard-coded in Python. They are the context-free floor of the design, and are placed on the real per-fix distributions (Figure 2.2) rather than guessed. The horizontal-speed bound is set where the bin-to-bin ratio of a finely binned ($2.5\,\mathrm{m\,s^{-1}}$) histogram settles at ≈ 1, i.e. where the decay has actually stopped, not where it first visibly slows ($\approx 40\,\mathrm{m\,s^{-1}}$ for hang gliders, versus the adopted $55\,\mathrm{m\,s^{-1}}$), since a shallow but real decreasing stretch can still be genuine rare dynamics. The Hampel and frozen-lock parameters live under the same key; Table I2.2 collects them.

Flight-level integrity gate. The pass records how much it changed each flight (or segment) – `n_fix_raw`, `n_fix_clean` and `frac_interpolated` in `flights_meta`. A flight whose deleted-plus-reconstructed fraction exceeds f (10 %, Table I2.2) is dropped whole; the counting window (airborne only) and the rationale are Sec. 2.7.2, whose threshold sweep validates the adopted value rather than setting it. This is the cleaning counterpart of the sampling-regularity cut (Sec. I2.7.6).

Parameter	**Role, and the scale that sets it**
Robust local-outlier test (Hampel)	
w	half-width of the local time window; a few native intervals
k	flag threshold, in robust standard deviations σ_k
$\varepsilon_{\min}$	absolute residual floor; a few times the horizontal GPS noise, so a smooth stretch ($\sigma_k \to 0$) is not chased
Frozen-lock rule (Eq. 2.4)	
δ_{xy}	run-diameter tolerance; a few times the GPS noise, below the $2R \approx 30\,\mathrm{m}$ to $100\,\mathrm{m}$ of a circling climb
δ_z	barometric-flatness witness; a few times the sensor noise and quantisation, $\ll$ the $\gtrsim 60\,\mathrm{m}$ a real climb gains over τ_{freeze}
τ_{freeze}	minimum span of a cut run; two thermalling periods, so a slow, tight climb is never mistaken for a freeze
Flight-level integrity gate	
f	maximum deleted-plus-reconstructed fraction; past it the flight is being rebuilt, not cleaned, and is dropped whole

Table I2.2. Fix-level cleaning: what each parameter of the robust and structural detectors does, and the physical scale that fixes it. The *values* are not repeated here: they are stated with the rules that use them in Sec. 2.7.2, and gathered with the rest of the pipeline's working values in Table 2.7. They are fixed a priori and checked a posteriori on the per-rule removal fractions (Sec. 2.7.2), with the injected-defect calibration as the direct measurement of the error rates. All live as config keys under `fix_level` in `configs/preprocessing.yaml`, alongside the absolute bounds of Table 2.5.

Diagnostics and validation

Diagnostic sample (Figure 2.2). Computed by `fix_level_distributions` and rendered by `make_fixlevel_diagnostics_figure` (Table I1.1). Panels (a)–(c) pool a seeded random sample of 15,000 paraglider flights (126,723,499 consecutive-fix pairs, ~73 s to compute) and the complete 6,716-flight hang-glider population (a full census at this size, 37,915,334 pairs,

$\sim$25 s) – under two minutes total. The large sample is deliberate: resolving the sparse artefact tail well enough to tell it from real dynamics needs far more points than a headline statistic would. The x-range in (a)/(b) extends to each discipline's own 99.99th percentile rather than cropping just past the bound; panel (c) keeps a narrower range, its cut sitting far out in an otherwise tight, unimodal distribution. Panel (b) bins one integer $\mathrm{m\,s^{-1}}$ per bin because barometric altitude is logged to the metre: differencing at a 1 s cadence returns integer rates, and a finer grid would alias them into a spurious comb.[4]

Validating the cleaning. The primary validation is the two-stage procedure of Sec. 2.7.2: freeze the working values a priori, then audit the per-rule removal fractions over the archive and sweep each threshold around its working value. The checks below are the fuller battery behind it, run as the implementation matures.

(1) *Injected defects.* The synthetic-defect procedure of Sec. 2.7.2, item (d), gives the detection floor – the smallest spike caught – and, where the removal-fraction audit flags a rule, recalibrates k, w, δ_{xy} and τ_{freeze} to a target rate: the only direct measurement of the cleaning's false negatives.

(2) *Downstream invariance.* Recompute the transport estimators – the displacement-scaling exponent and the waiting-time-tail exponent – over a grid of cleaning strengths, i.e. re-running the cleaning with all thresholds tightened or loosened together by a common factor around the working point, and take the operating point on the plateau where they stop moving; movement under stricter cleaning is the signature of a cut biting into signal.

(3) *Detector nesting.* The fixes flagged by the absolute bound should nest inside those flagged by the local test (the impossible is a subset of the locally anomalous); a large disagreement points to a defect class neither was designed for – a cheap guard.

(4) *Per-phase removal rates.* The error this pipeline most wants to avoid is losing a slow-phase fix. Once the segmentation is in place (Ch. 3), the removal and split rates are measured *per phase* and required to be flat across transition, search and climb rather than enriched in the two slow phases; and synthetic tight climbs – circles at flying speed, over a range of radii and wind drift – planted in otherwise clean tracks confirm directly that the frozen-lock detector never splits a real climb. This is the false-*positive* complement of check (1), and its natural readout is the tail of $\psi(\tau_w)$: a cut biting into climb or search would thin that tail before it moved anything else.

(5) *Isolation of the vertical cuts.* The fixes cut by $|v_z| > 13\,\mathrm{m\,s^{-1}}$ must be *isolated*, not consecutive: a sustained spiral dive holds $15\,\mathrm{m\,s^{-1}}$ to $20\,\mathrm{m\,s^{-1}}$ of real sink, so a run-shaped excess is the signature of a genuine manoeuvre – calling for a raised bound or a coherent-run exemption rather than censoring – whereas an isolated super-threshold difference, bracketed by ordinary rates, is a sensor error and safe to cut.

I2.7.3 Trimming of ground phases

The two thresholds of Eq. (2.5) (v_0, T_0) are the `TrimmingThresholds` dataclass, loaded from `configs/preprocessing.yaml` (key `trimming`); values in Table 2.7.

The interior-ground guard of Sec. 2.7.3 keeps `interior_ground_s` (T_{ground}, 600 s) and `ground_flatness_m` (5 m) in the same group. The flatness is tested on the *detrended* altitude, as Sec. 2.7.3 requires: fit a straight line to the adopted channel over the candidate stint by

[4] Not every flight samples at 1 s (Figure 2.7), and differencing over a longer native step returns finer values (multiples of $0.2\,\mathrm{m\,s^{-1}}$ at 5 s) that partly fill the gaps once cadences are pooled: among 1 s-native fixes about 99% land within $0.05\,\mathrm{m\,s^{-1}}$ of an integer, against only 34–42% at every other cadence.

least squares, subtract it, and compare the residual's central span—its 5th to 95th percentile—against `ground_flatness_m`. The linear term is the barometric drift, which over a stint of T_{ground} is already of the order of the tolerance; the residual is what is left of genuine vertical motion, and the central span rather than the full range because a range grows with the length of the stint and a central span does not (Sec. 2.7.3). No new config key is needed.

Building it made one thing explicit that the rule as stated leaves open. Detrending cannot by itself separate a pressure drift from a steady climb, because both are linear and the fit removes either completely: a wing thermalling at $1\,\mathrm{m\,s^{-1}}$ in still air holds a ground speed near zero, gains 600 m over a T_{ground} stint, and leaves a detrended residual of *zero* — so the flatness condition alone would excise it as a mid-flight landing, which is precisely the failure this guard exists to avoid. What separates the two is the *rate*: drift runs at tens of metres per hour, a soaring climb two orders of magnitude faster. The fitted slope is therefore bounded as well, at $\delta_z/\tau_{\text{freeze}}$ — the frozen-lock rule's own statement of what counts as motionless, read as a rate, so the guard borrows a number the configuration already fixes instead of introducing one. Below T_{ground} the same predicate is stored as a list of suspect intervals per flight, which the segmentation later maps onto the phases they touch (consumption of the flag: Sec. 2.7.3).

Why the central span, not the full range. A full range is a maximum statistic: over zero-mean noise it grows without bound with the number of samples, as $2\sigma\sqrt{2\ln n}$. At the metre-scale barometric noise of this archive it reads 4.6 m over a 60-sample stint and 7.1 m over 3,000, so against a 5 m tolerance the same motionless pilot would pass the test on a one-minute stint and fail it on a ten-minute one, on noise alone — and since an interior stint must last T_{ground} to be considered at all, it would fail on every stint the rule is ever offered. The central span carries the same units and reading and sits at 3.3σ whatever the stint's length, so δ_z^{gr} means the same thing on a stint of any duration; a genuine climb of hundreds of metres exceeds it either way.

The GNSS-fallback abstention. The intended rule widens the tolerance on a GNSS-fallback flight to match that channel's noise; the widening is not implemented, because the factor it needs is a measured ratio between the two channels' short-timescale noise, and a first measurement on the archive did not reproduce the ordering Fig. 2.1 reports. Until that is settled, the single barometric tolerance is applied to both channels, which on a noisier one simply means the flatness condition is not met and the stint is kept — the conservative half of the intended rule, since the error this guard exists to avoid is excising a genuine climb. The cost is that a mid-flight landing on a GNSS-fallback flight is neither excised nor flagged.

I2.7.4 Flight-level filtering

The three cuts of Sec. 2.7.4 are the `FlightLevelThresholds` dataclass (key `flight_level` in `configs/preprocessing.yaml`); values in Table 2.7. The census measurements behind the path floor and the upper duration bound, and the totals of Figure 2.3, are quoted through macros computed from the shared track scan (Sec. I2.7); the headline duration and path medians are quoted in Sec. 2.8.

The plausibility and reach bounds of Sec. 2.7.4 are not further `flight_level` keys: both read `max_horizontal_speed_mps` from the `fix_level` block, so the same number governs a single step, a whole-flight average and a whole-flight displacement, and the three cannot drift apart. Both are evaluated on the trimmed track, after the per-block sums of the duration and the path, and they are the last of the whole-flight cuts because each is a statement about a record the earlier ones have already accepted. The extent they need is returned by `flight_quantities` alongside the duration and the path, from the same haversine call, so it costs nothing: the maximum of a vector the function already forms. Unlike the duration and the path it is *not* summed per block—a displacement from a fixed origin has nothing to sum—which is why a split does not weaken it.

The chain behind every number quoted in Sec. 2.7.4 is short enough to name in full, so that a count can be traced to the code that produced it:

- `soaring.analysis.igc.parse_igc` decodes one IGC file into a table of fixes;
- `soaring.analysis.census.track_stats` reduces those fixes to the one summary row described in Sec. I2.7, and `scan_tracks` / `load_or_scan_tracks` do that for every flight of a discipline and cache the result;
- `fraction_retained` and `retention_curve` in the same module evaluate a single cut against a column of that table—the first at one threshold, the second along a grid of them, which is what the bottom row of Fig. 2.3 plots;
- `scripts/reporting/ch2_dataset/generate_census_stats.py` calls the same functions with the thresholds read from `configs/preprocessing.yaml` and writes the `\StatScan*` macros the text quotes.

Two consequences follow, and both matter for reading the numbers. The counts are *marginal*: each is the effect of one criterion applied alone to the whole parsed population, not of the cascade. And they are pre-cleaning: the scan runs on parsed tracks, so a count in Sec. 2.7.4 is what the criterion removes from the parsed archive and not from the cleaned one. That is deliberate and is argued for above (*What the census is not*): a threshold chosen on the population the cleaning has already reshaped would be chosen against its own effect. The post-cleaning counts are the cascade of Sec. 2.9, and the two answer different questions.

I2.7.5 From geographic to local Cartesian coordinates

No implementation content: this step is pure coordinate geometry with no configurable parameters, and both of its figures (Figure 2.4, Figure 2.5) are hand-drawn TikZ diagrams, not data plots. The mathematics behind Eqs. (2.6) and the ENU rotation is derived in Appendix 2.B.

I2.7.6 Uniform sampling within a flight

The sampling thresholds (key `sampling`, the `SamplingThresholds` dataclass; values in Table 2.7) act as follows.

Split bound. $g_{\max} = \min\left(\texttt{max\_gap\_factor} \times \Delta t,\ \max(\texttt{max\_gap\_seconds},\ 2\,\Delta t)\right)$, pinned by the constraint chain of Sec. 2.7.6. Its audit is a per-flight count – the census scan records `max_gap_ratio` and `dt_s`, so the absolute gap is their product – and joins Figure 2.6 when regenerated.

Missing fraction. Acts per *segment* (rule: Sec. 2.7.6); the per-flight value in the census scan is the pre-split diagnostic, not the operational gate.

Segment gate. `min_segment_duration_s` (90 s). It only needs to support the smoothing window and the within-segment statistics: a few smoothing timescales, never less than $w\,\Delta t$; the only gate applied at segment level (Sec. 2.7.6).

Measured vs interpolated. A grid point counts as measured when a fix lies within half a native step of it, interpolated otherwise; the same tolerance decides whether a flight is uniform as recorded (no interpolated points) or was resampled (`was_resampled`).

The bound is on the *time base* and is the same for every channel: an order-of-magnitude smaller v_z buys no wider gap for the altitude, since the interpolation error scales with curvature, not speed, and a wide altitude gap would erase the climb–glide transitions that v_z is there to resolve (Sec. 2.7.6).

Segment handling.

Origin and clock. A split assigns a `segment_id` and nothing else (semantics: Sec. 2.7.6).

Tables. One row per fix in `fixes`, keyed (`flight_id`, `segment_id`); per-segment records (start/end, fix count, `frac_interpolated`, censoring flags) in a `segments` table beside `flights_meta`. No separate files; the ensemble MSD at lag t reads $\mathbf{r}(t)$ wherever some segment covers t.

Censoring. Any phase duration truncated at a segment boundary is flagged censored, for the $\psi(\tau)$ fits to exclude.

Per-channel fill. Across a short gap the horizontal position is interpolated by monotone piecewise cubics (`scipy.interpolate.PchipInterpolator`) and the adopted altitude channel linearly (its within-phase smoothness makes the choice among low-order methods immaterial), realising the completeness invariant of Secs. 2.7.2 and 2.7.6. Each filled grid point is flagged in an `interpolated` boolean column of `fixes`, and `frac_interpolated` records the per-segment fraction for the integrity gate.

Why linear for the altitude, cubic for the horizontal. The asymmetry follows from what each channel is used for. The altitude is differentiated once more than the horizontal coordinates before it is read — v_z is the segmentation's discriminant — so an interpolation artefact in z propagates directly into a phase assignment, whereas the same artefact in E or N is absorbed by the smoothing. Linear interpolation cannot overshoot, and its error over a hole of width g is bounded by $\max|\ddot{z}|\, g^2/8$[5], whereas a cubic is more accurate where the second derivative is smooth and worse exactly where it is not — the transitions a phase assignment turns on.

The test flight that forced the `z_reconstructed` flag. $g_{\max}$ bounds the interval between two *fixes*, not an interval over which a fix exists but carries no altitude value; that case is not rare, since the vertical channel can be absent while the horizontal record is perfect (a logger writing no barometric value, or one the fix-level cleaning removed). No time gap opens there, so no split is declared, and the linear fill spans the hole whatever its length. On a test flight missing 500 s of altitude the filled z departed from the truth by up to 540 m, the smoothing differentiated the straight line into a vertical velocity that was pure invention, and the record read `frac_interpolated` $= 0$ with `was_resampled` false — the flight declaring itself uniform as recorded, because the `interpolated` flag is a statement about the time base alone. The vertical therefore carries its own per-fix flag, `z_reconstructed`, with the per-segment share `frac_z_reconstructed` and, per flight, the longest unbroken run `z_gap_max_s`, reported next to the fraction because a fraction alone cannot separate a hundred two-second holes, which a straight line bridges well, from one twelve-minute hole, which it does not. Any analysis of the

[5] On an interval of width g the difference between a C^2 function z and the chord through its endpoints is $z(t) - p(t) = \frac{1}{2}\,\ddot{z}(\xi)\,(t - t_0)(t - t_0 - g)$ for some ξ in the interval; the quadratic factor is largest at the midpoint, where it equals $g^2/4$, giving $|z - p| \le \max|\ddot{z}|\, g^2/8$.

vertical excludes on that flag; the horizontal analyses are untouched by it, which is why the flag is per channel rather than one for all three.

Figures 2.6 and 2.7 are rendered by `make_gap_diagnostics_figure` and `make_sampling_figure` from the shared track scan (Sec. I2.7).

Gap-ratio discreteness (Figure 2.6). Checked directly: only 86 %/63 % (para/hang) of largest-gap-ratio values are exact integers, the rest a genuinely continuous scatter; the integer peaks visible for hang gliders sit around $k = 3, 6, 10, 15$. Panels (a)/(b) of the figure therefore use linear axes: a logarithmic axis would compress precisely this small-integer structure, on which the adopted cut sits.

Native-rate discreteness (Figure 2.7). Checked directly: Δt lands on an exact whole second for 99.9 % of paragliders and 100 % of hang gliders. Per-rate breakdown: paragliders sit mostly (69 %) at 1 s; hang gliders spread across the rates below.

Native rate	1 s	2 s	5 s	10 s	other
Hang gliders	30 %	13 %	27 %	15 %	remainder

I2.7.7 Smoothing and differentiation

Three decisions are pinned for the implementation (status: Table I2.1):

- the filter runs per segment (never across a boundary) with `scipy.signal.savgol_filter` in `mode='interp'`, so the terminal half-windows are fitted on the edge window rather than padded, and edge samples inherit the segment-boundary caution of Sec. I2.7.6;
- the window is floored at $w = 5$, the smallest a cubic fit accepts (slow-logger consequences: Sec. 2.7.7, item iii);
- the polynomial order and the three smoothing timescales are the `SavgolParams` dataclass in `configs/preprocessing.yaml` (key `savgol`): `tau_c_horizontal_s` for (E, N), `tau_c_vertical_baro_s` and `tau_c_vertical_gnss_s` for the vertical channel (the flight's adopted altitude, Sec. 2.7.1); values in Table 2.7.

The measured timescales. The knees were measured with `tools/estimate_savgol_timescales.py`, which applies the Welch procedure of Sec. I2.7.1 to E, N (local equirectangular metres) and to the altitude channel each flight actually uses, on a seeded sample (908 horizontal flight-components, 719 barometric and 189 GNSS-fallback flights at 1 Hz). The knee rule: the floor is the median PSD over 0.35 Hz to 0.5 Hz; f_c is the lowest frequency whose median PSD falls below three times it. The measured floors sit at ≈ 0.60 and $\approx 0.26\,\mathrm{m^2\,Hz^{-1}}$ against the theoretical quantisation floors $q^2/12/f_{\mathrm{Nyq}}$ of 0.57 and 0.17; the three knees coincide at $f_c \approx 0.2$ Hz, hence $\tau_c = 5$ s for every channel (Sec. 2.7.7, item (iv), for the reading), and even the 90th-percentile spectra move the knee by less than a second. The two vertical keys stay, conditioned on `alt_source`, for the noisy-minority refinement; the validation of Sec. 2.7.7 (flat residual spectrum, stability under one window step) is what confirms the values. The filter runs on the resampled grid, so the small interpolated fraction (`frac_interpolated`) is already in place when it reads its window.

I2.8 Preliminary characterization

The reduction `scripts/reporting/ch2_dataset/generate_prelim_figure.py` produces the figures of Sec. 2.8 and of Sec. 3.3 (Chapter 3) and the `\StatPrelim*` macros. It reads three things: the per-flight summary and the per-lag positions written by `ch3_global_transport/audit_msd.py` (Sec. I3.1), `flights_meta.parquet` for the take-off coordinates, and the season catalogue for `wing_class` and `season`. The join on `flight_id` resolves for 99.99 % of retained flights.

Reading the audit arrays rather than the fix table is what makes the stratification affordable: a stratified MSD is then a row selection on a matrix already in memory, so the three stratifications and the isotropy check together cost one pass that has already been paid, instead of one 43 GB scan each.

The basemap. The take-off panels are drawn on Natural Earth admin-0 country polygons (public domain), which give the coastline and the borders in one layer and carry the overseas departments inside the France feature—which is what puts La Réunion on the map at all. The geometry is not fetched at draw time and Cartopy is not a dependency: `scripts/reporting/tools/build_basemap.py` downloads it once, keeps the rings that intersect each panel, decimates them (0.01° for France, 0.001° for La Réunion, 0.25° for the world frame) and writes `data/basemap.json`, which is committed at 0.76 MB. The figure then needs nothing but Matplotlib and regenerates from a clean checkout with no network. Rings are kept whole rather than clipped to the frame—Matplotlib clips to the axes anyway, and clipping a polygon properly would need the geometry library the arrangement exists to avoid. Re-run the builder only to change a panel's extent.

The three frames. Metropolitan France $\lambda \in [-5.5, 10]$, $\varphi \in [41, 51.5]$; La Réunion $\lambda \in [55.15, 55.92]$, $\varphi \in [-21.42, -20.82]$; the world panel for everything outside both. The split is the archive's, not a choice: 4,939 paraglider flights launch on La Réunion and 3,359 outside France altogether, and a single frame that held them would render metropolitan France four pixels wide. Each panel sets its aspect to $1/\cos\varphi$ at its own mid-latitude, so the coastline has its true shape rather than a stretched one.

The orographic boxes. The strata are longitude/latitude boxes rather than polygons traced from the map: a box that is written down can be checked against a gazetteer, and the test the strata support is whether the curves separate, not where exactly the Alps end. Alps $\lambda \in [5.4, 10.0]$, $\varphi \in [43.8, 46.6]$; Pyrenees $\lambda \in [-1.9, 3.3]$, $\varphi \in [42.0, 43.5]$; Massif Central $\lambda \in [1.8, 4.6]$, $\varphi \in [43.6, 46.2]$; anything else inside $\lambda \in [-6, 10]$, $\varphi \in [41, 52]$ is "lowland" and anything outside it is "abroad". The boxes are coarse and overlap along their edges, with the first match winning in the order listed; they are a stratification, not a geography, and the test they support is whether the curves separate rather than where exactly the Alps end.

The stratum floor. A stratum below 2,000 flights is pooled into the residual group rather than drawn. Below that the curve's own sampling error at the long-lag end is comparable to the separations being tested for, and a panel that draws it invites the reader to compare noise. The floor is why the hang-glider archive, at 6,132 flights, contributes only to the isotropy panel: one wing class and one orographic group clear it, so the stratified panels would compare a stratum against itself.

The isotropy test. The variance ratio is computed on the same per-lag position matrix, column by column, over the flights covering that lag. The Kolmogorov–Smirnov comparison runs at four lags spanning the fit range, on the signed components rather than their magnitudes, so that a difference in *spread* and a difference in *shape* both register. Only the distance is quoted. At 155,788 flights the two-sample p-value underflows to zero for any departure worth naming, so it separates nothing; the distance is the effect size and is what the text reads.

Stratum departure. Each stratum's curve is divided by the pooled curve over the fit range and the largest $|\cdot - 1|$ recorded. Two summaries are generated: the maximum over strata (`\StatPrelim*MaxDeparturePct`) and the median over strata (`*TypicalDeparturePct`). The maximum alone reads as a statement about the smallest and noisiest stratum, which is what it usually is; the median is what distinguishes an axis along which the ensemble genuinely splits from one along which a single small group wanders. Both are quoted in Sec. 2.8.

Chapter I3

Global transport

This chapter mirrors Chapter 3. Built and recorded here: the two mean-squared-displacement estimators and their audit, the filtered-variation family that replaces them as a source of exponents, the moment spectrum, the non-Gaussian parameter, and the velocity autocorrelation and its tail. One observable the body catalogues is defined and not measured, and Sec. 3.10 says which.

I3.1 Observables on the whole trajectory

The mean-squared displacement of Sec. 3.1 is implemented and unit-tested, as is every other observable Chapter 3 defines except the return probability, which Sec. 3.10 names as the one gap. The estimators live in `soaring.analysis.observables.transport` and the figure, the curve table and the `\StatMsd*` macros come from one pass of `generate_msd_figure.py` (Table I1.1). A second pass, `audit_msd.py`, keeps what the first one averages away, and `audit_msd_report.py` reduces it into the `\StatAudit*` macros; both are described at the end of this section.

I3.1.1 The two estimators

`MSDAccumulator` builds the ensemble average and `TAMSDAccumulator` the time average. Both accumulate by streaming rather than by loading: the `fixes` table runs to $\sim 1.4 \times 10^9$ rows over the two archives, and neither estimator needs more than one flight in memory at a time.

Coverage. A flight answers a lag t with the fix nearest to it, and only if two conditions hold: that fix lies within $\max(0.5\,\mathrm{s}, \Delta t/2)$ of t, and $t \geq \Delta t$, where Δt is the median inter-fix time of the flight's own concatenated segments. The second condition is the one that is easy to omit and expensive to omit. Without it a 5 s logger asked for a lag of 1 s answers with its fix at $t = 0$—a displacement of exactly zero—and a slow-cadence minority drags the short-lag MSD down until the grid crosses their step. The same rule appears in the time-averaged estimator, which refuses any lag below its segment's own step. A lag falling inside a gap between two retained segments is answered by neither side, since the split bound (Sec. 2.7.6) makes every gap wider than the tolerance.

The lag grid. `log_lag_grid` spaces 90 points geometrically from 1 s to 43,200 s, the duration bound of the flight-level filter, and snaps them to the second. The snapping is what makes the short end usable: unsnapped, lags of 1.00, 1.13 and 1.27 seconds all resolve to the same fix at a 1 Hz cadence, and the curve there is a staircase of repeated values whose local slope alternates between zero and a spurious spike. Snapping merges them instead, which is why the grid delivers 80 distinct lags rather than 90, and why the lags below 8 s are consecutive integers. At long lags the rounding is a part in 10^4. The consecutive-integer run is also what lets the coverage artefact of Sec. 3.1 have a period; that is the cost of the snapping, it is measured in Sec. 3.1 by splitting the archive by cadence, and it sits an order of magnitude below the lower end of every fit range.

The time average. `time_averaged_msd` is computed within a segment, never across the gap that ends it. The direct double loop is $O(n^2)$ and hopeless at 10^4 samples per segment; writing $D_j = |\mathbf{r}_j|^2$, the identity $\overline{\delta^2}(k) = \langle D_j + D_{j+k}\rangle_j - 2\langle \mathbf{r}_j \cdot \mathbf{r}_{j+k}\rangle_j$ turns the second term into an autocorrelation, which the FFT gives in $O(n \log n)$. The curve is cut at half a segment's length, beyond which the average over starting times has too few of them to be one.

I3.1.2 The fit, and what is quoted from it

The range. The lower end is an argument, not a consequence: at 120 s it sits above the thermalling period, so what the slope measures is transport rather than the circling superimposed on it, and above the whole region the short-lag artefacts occupy. The upper end is a consequence. `coverage_limited_range` walks the count-per-lag curve and cuts where it has fallen to `min_coverage` $= 0.25$ of its value at the lower end; past that the curve is an average over a survivorship-selected minority. Applied to the four curves this gives the ranges quoted with each exponent.

The two errors. `fit_msd_exponent` fits unweighted in log space, which gives every decade the same say; a fit weighted by the much larger absolute errors at long lags would be dominated by the last decade alone. It returns the least-squares error on the slope, which is written out as `\StatMsd*AlphaErrOls` and is *not* the quoted uncertainty. `bootstrap_alpha_error` resamples 200 ensembles of flights with replacement, refits each over the same range, and reports the standard deviation as `\StatMsd*AlphaErr`. Neither figure is attached to an exponent in the body: both MSD exponents are quoted bare, because the bootstrap treats clustered flights as independent records and because the range dependence is fifty times larger where both have been measured. Sec. 3.1 makes that argument and gives the synthetic check that establishes what each of the two errors measures; the exponents that do carry an interval are those of Sec. 3.4.

The cohorts. Three fixed-duration cohorts at $T = 1$, 2 and 4 h. Within a cohort every member is present at every lag up to T, so the population is constant by construction and any remaining growth is motion; each is therefore fitted from the range's lower end up to its own T, which is as far as the guarantee reaches. The time-averaged cohorts are keyed on segment span and capped at $T/2$, for the same reason the time-averaged curve is. A cohort is compared against the *pooled curve refitted over the cohort's own lags*, never against the pooled exponent over the pooled range: the local slope is not constant across those two spans, so the naive comparison charges the shape of the curve to the change of population. Both readings are generated (`\StatMsd*CohortSpread` and `*CohortSpreadNaive`), and on the ensemble estimator they differ by a factor of five, which is the size of the mistake avoided.

The local slope, and why it is a regression. The panel that carries the most interpretation is the one whose estimator is easiest to get wrong. The local slope is $\mathrm{d}\log \mathrm{MSD}/\mathrm{d}\log t$, and the obvious way to compute it—a centred difference between neighbouring lags—divides by the grid spacing, which on the logarithmic grid used here is about 0.05 decades. The division multiplies whatever scatter the curve carries by $1/(2 \times 0.05) \approx 4$: a point-to-point scatter of a few tenths of a per cent in the MSD, which is about what the ensemble's own standard error allows, becomes a slope that jitters by 0.05 and reads as noise. The panel then testifies against a curve that is not in fact noisy.

`soaring.analysis.observables.transport.local_slope` therefore works on a window: at each lag it takes the points within 0.15 decades on either side—five to six at this grid spacing—and reports the *median of the slopes of all pairs* of them, a Theil–Sen estimator. The choice is for resistance to a single displaced lag rather than to a heavy-tailed residual, which is the failure this curve actually has: a coverage artefact of a few per cent at one lag tilts a least-squares window and moves a median by nothing. The uncertainty reported with it, and drawn as a band, is the interquartile range of those pairwise slopes divided by 1.35 and by the square root of the window's size—the window's own scatter about its line, expressed as an error on the slope. Over the fitted range it is 0.017, against a point-to-point variation of 0.052: the curve moves more between neighbouring lags than the scatter within a window accounts for, which is the quantitative statement that what the panel shows is structure and not noise. The half-width is a stated compromise rather than a tuned one: narrower and the amplification returns, wider and a genuine bend is flattened into the straight line the panel exists to test for; 0.15 decades resolves a crossover of half a decade. It is not a smoother in disguise. The residual wiggle of this archive is correlated across some four grid points, so it survives the window—which is the right outcome for something that is not noise, and the reason the window is reported with the figure rather than chosen to make the curve look better.

I3.1.3 The audit pass

The ensemble curve turned out not to be a power law (Sec. 3.1), and none of the questions that decides—is the shape the coverage rule, the survivorship, a shared heading, or the motion—can be asked of an averaged curve after the fact. Each is a different reduction of the same per-flight quantity, and that quantity is gone by the time the average exists.

`audit_msd.py` therefore makes a second streaming pass and writes what the first one discards: one row per flight and one column per lag, holding each flight's E and N at that lag under the estimator's own coverage rule, `NaN` where it does not cover it, plus a per-flight summary (duration, path, native step, mean-velocity components, largest in-segment step speed, coordinate extent). The coverage rule is *copied* into that script rather than imported, so that a change to the estimator surfaces as a disagreement between the figure and the audit instead of being tracked silently. The arrays run to a few hundred megabytes per discipline and are analysis products, not thesis ones: they live outside the repository, under `AUDIT_DIR`.

`audit_msd_report.py` reduces them, together with the curve table `msd_curve.csv` and the segment and flight tables, into `\StatAudit*`. Everything the audit asserts in the body reaches it through those macros. What it computes: the rms residual each estimator leaves against the power law fitted to it, in dex and as a percentage; the range of the local slope over the fitted decades; the mean vector's share of the mean square, $|\langle \mathbf{r} \rangle|^2 / \langle |\mathbf{r}|^2 \rangle$; the local slope recomputed on sub-ensembles of a single native cadence, where the population cannot change, and on fixed-duration cohorts, where it cannot be selected; the first-crossing time of 5 km from take-off, per flight; the per-component variance ratio and the Kolmogorov–Smirnov distance between the E and N marginals; and the pre-processing losses of Table 2.6 decomposed by the rule that caused them.

Two of its numbers are worth naming here because they are checks on the pipeline rather than on the estimator. The retained tables carry 0 non-finite coordinates and 0 non-monotone clocks, which is what Sec. I2.2 asserts and this pass confirms independently. And the flights whose first retained segment did not survive resampling start their record at $t > 0$ rather than at the origin—1.4 % of paraglider and 11.7 % of hang-glider flights. Their position is still measured from their own take-off, since the ENU frame is anchored there and a dropped segment does not move it; they simply do not answer the lags before their record starts. For the rest, the position at $t = 0$ is the origin to within 0.09 m in the median, the smoothing's displacement of the first sample.

I3.1.4 The shape observables, and what calibrates them

The non-Gaussian parameter. Computed from the moment spectrum already accumulated, as $\alpha_2^{\mathrm{NG}} = \langle|\Delta\mathbf{r}|^4\rangle/2\langle|\Delta\mathbf{r}|^2\rangle^2 - 1$ at each lag, on order-1 increments. Two calibrations make the archive's value readable and both are pinned by `tests/analysis/observables/test_moments.py`, at the sample size the test uses: an exact fractional Brownian motion returns 0.02 in the median over a ten-point grid spanning the quoted range, and a Lévy walk of tail index 1.5 returns 0.55. The Gaussian 0.02 is estimation bias and not shape — a fourth moment over 10^4 increments is not yet its own expectation — and sixteen times the data collapses it to within a hundredth of zero, sign included. The same test pins that too, since it is what makes the archive's own value a departure rather than an offset. It is quoted over 60 s to 2,000 s rather than over the whole grid; above that the declared task governs the trajectory and the fourth moment rests on the fewest flights, and the curve climbs to +0.47 there.

The tail of the velocity autocorrelation. Green–Kubo is used in its scaling form, $C(\tau) \sim \tau^{-\gamma}$ with $\gamma < 1$ giving $\langle|\Delta\mathbf{r}|^2\rangle \sim \Delta^{2-\gamma}$, and not as the integral. Two reasons, both properties of the archive: the integral runs from zero and C is estimable only above the smoothing scale of the slowest logger, so its first decade is missing; and C is estimated per flight with that flight's own mean velocity removed, which pulls every lag down by about the record-mean of C and so steepens the tail. Both push γ up, so $2 - \gamma$ is a floor and `vacf_tail_exponent` documents it as one. The one-sidedness is itself tested, on an Ornstein–Uhlenbeck velocity whose true correlation is known: the estimate with the mean removed is always the steeper.

The filter kernel, against a closed form. For any process with stationary increments and $S(\Delta) = \sigma^2\Delta^{2H}$, the two orders are related exactly by $V_2 = 4S(\Delta) - S(2\Delta) = S(\Delta)\,(4 - 2^{2H})$. That identity pins the kernel in a way a second convolution cannot: the order-2 kernel is symmetric, so an independent `np.convolve` route performs the same three products in the same order and agrees to the last bit whether or not the kernel is right — which is what a recomputation of the paraglider exponent by that route in fact found. The estimator also has to be unbiased where the archive sits, at H near 1, which is the boundary of the family: on exact fractional Brownian motion of $H = 0.99$ it returns 1.98.

When a knee is declared. A bilinear fit to $q\nu(q)$ has two more parameters than a straight line through the origin, always reduces the residual, and on a grid of ten moment orders buys the reduction cheaply — fitted to a fractional Brownian motion, which has no knee, BIC prefers the bilinear form while the straight line already fits to an rms of 0.003. So BIC alone is not the rule. `bilinear_fit` declares a knee only when three conditions hold together: BIC prefers the bilinear form, *and* the straight line leaves a residual above `min_departure` $= 0.02$, *and* the knee sits inside the q grid rather than on its edge. On the synthetic pair those three give no knee for fractional Brownian motion and a knee at 1.57 with a right-hand slope of 0.96 for a Lévy walk of index 1.5. A Lévy walk of index 1.5 leaves a departure of about 0.084 on this grid; the archive leaves 0.0109 and 0.0057, which is roughly eight and fourteen times smaller, and about half and a third of the declaration threshold. The margin against a knee is therefore an order of magnitude; the margin against the gate is a factor of two, and the two should not be confused.

Why the drift is filtered and never subtracted. `centring_bias` measures the alternative rather than arguing against it: fractional Brownian motion of known exponent is generated, the mean-squared displacement is computed raw and after subtracting each realisation's own net velocity, and the two are compared lag by lag. Subtracting a velocity estimated from the record's endpoint turns the series into a bridge whose increments must sum to zero, so the centred curve is pulled down at every lag and hardest at the long ones — 2 % at a lag of 0.05 % of the record and 76 % at half of it, with no range in between where the distortion is negligible. On $\alpha = 1.50$ the operation returns 1.35. The filtered variation of `soaring.analysis.observables.variations` estimates no drift at all, so none can be estimated badly, and the order scan is what reads off how much drift there was.

The variation pass, and why stratifying is free. `measure_variations.py` traverses the fix table once and keeps, per flight and per filter order, one curve over the lag grid together with the flight's cadence, wing class, season and declared task. Every stratification of Sec. 3.4 is then a row selection on that table rather than another pass over 1.4×10^9 rows, which is what makes four cuts affordable where one would otherwise be a day's work.

First-passage times, and why they are not reported. The audit arrays keep each flight's position at every lag, so the first-passage time $T(L)$ — the first time a flight is L from take-off — costs no traversal, and $\langle T(L)\rangle \sim L^{1/H}$ would give the exponent in the space domain, independently of any moment fitted over lags. It was computed and is not reported, for a reason worth recording so it is not attempted again. Read naively it returns $\alpha = 2.46$, which no process with stationary increments can give. The cause is right censoring: at 34 km only a quarter of paraglider flights arrive and the missing three quarters are the slow ones. Treating non-arrival as censored and taking a Kaplan–Meier median instead returns 1.95 for paragliders and 2.03 for hang gliders, both inside the paraglider interval — but refitting over sub-ranges of radius moves the paraglider value between 1.51 and 2.03, a spread five times the uncertainty it was meant to check. The reason is the one Sec. 3.1 gives for the ensemble MSD. $T(L)$ is measured from take-off, so it shares that estimator's origin: at small L the wing is still climbing out inside its launch area, and at large L the censored median extrapolates past the duration of most flights. It is the same crossover seen along the space axis instead of the time axis, and it is not an independent check on anything.

Why the runs test is not read against a normal table. The Wald–Wolfowitz statistic is asymptotically normal under a null of *independent* signs, and on a curve every lag of which averages the same flights the signs are nothing of the kind: the measured lag-1 residual autocorrelation is 0.89 to 0.95 on all four curves. Simulating an exact power law observed with AR(1) noise of that correlation on the same forty-lag grid, the rule $|z| > 2$ rejects it in 84 % of trials at $\rho = 0.8$ and 90 % at $\rho = 0.9$, and the true five-per-cent threshold is $|z|$ near 5.1. The z values this archive returns, -4.9 to -5.8, are therefore not significances, and `residual_runs` now says so in its own docstring.

Calibrating the null on the *observed* residual autocorrelation would be circular in the way this project has been caught by once already, since a genuinely bent curve has a high residual autocorrelation because of the bend. So the conclusion rests on a different comparison, `residual_against_noise`: the residual against the standard error of each lag's own mean, taken from the stored 10 % to 90 % band and count on a normal equivalence, which understates a right-skewed spread and so returns a lower bound. It gives 22 to 51 across the four curves — a departure that many times the error on each point cannot be scatter, whatever its arrangement.

I3.1.5 Known limitations, with their size

Four, each bounded rather than repaired.

Edge samples enter. The smoothing evaluates its polynomial off-centre at the ends of a segment and the table flags those samples `edge` (Sec. 2.7.7). `verify_dataset.py` qualifies its invariants by that flag—which is why its speed invariant holds—and the MSD estimators do not, and read every fix. Those samples are under 1 % of the table and account for 26,368 of the 26,368 + 32 in-segment steps above the fix-level speed bound, 0.001,9 % of all steps. The worst step between two *measured* samples is 49 $\mathrm{m\,s^{-1}}$ against a bound of 45 $\mathrm{m\,s^{-1}}$: an excess of under a tenth, which the verifier reports and does not fail on, its threshold for a data defect being twice the bound. Excluding the flagged samples moves the ensemble MSD by up to 13 % below 10 s and by under 0.2 % above 60 s, so the fit ranges are unaffected; excluding them would also cost every flight its shortest lags, since the first samples of a record are exactly the flagged ones.

Absolute position carries re-acquisition offsets. Bounded in Sec. 3.1 at under a per cent of the mean over the fitted range, on 0.029 % of paraglider flights.

The archive holds a few duplicate flights. The same physical flight uploaded twice, detected by take-off coordinates, duration and path length agreeing to four decimal places, one second and ten metres: 186 paraglider flights in 93 groups, 0.119 % of the archive, and 29 hang-glider flights in 14 groups, 0.473 %. Ensemble averaging assumes independent draws, and at this rate the violation is immaterial; it is reported so that it is not assumed to be zero.

The last lag of the grid is empty. 43,200 s is the flight-level duration bound, so no retained flight reaches it and the row is written with a null MSD and a zero count. It is dropped by every fit and every plot through the finiteness mask, and appears in `msd_curve.csv` for grid completeness.

Reading the table. Through `soaring.analysis.derived.stream_flights`, never by iterating Parquet row groups: a row group is a unit of storage and cuts across flights, so the direct reading counted 0.7 % of flights twice and truncated the longest segments, which are precisely the ones the long-lag end of the time-averaged curve rests on.

Chapter I4

Flight phases

This chapter mirrors Chapter 4. None of that code exists yet, so each entry is a placeholder rather than a populated record; each will gain its own implementation content — module names, sample sizes, reproduction commands — as the code is written.

I4.1 Segmentation into flight phases

Not yet implemented; see Sec. 4.1.

I4.2 Phase-resolved observables

Not yet implemented; see Sec. 4.2.

I4.3 Reproducing the reference results

Not yet implemented; see Sec. 4.3.

I4.4 Modelling and simulation

Not yet implemented; see Sec. 4.4.

I4.5 Tooling

Largely underway already: see Sec. 4.5 and the reproducibility map, Table I1.1.